Moral Constructivism

▶ Modern Moral Relativism

Moral Conventionalism

▶ Modern Moral Relativism

Moral Development

▶ Theory of Moral Development

Moral Dispositions

▶ Moral Instincts

Moral Domains

▶ Evolved Moral Foundations

Moral Emotions

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Synonyms

Other-focused emotions; Pro-social emotions

Definition

"Moral" emotions are those thought to relate to the capacity for human morality. Examples of such emotion types include disgust, shame, pride, anger, guilt, compassion, and gratitude.

Introduction

Emotion can be defined with reference to its input conditions, its characteristic output, and the internal mechanisms that mediate input and output. It is generally accepted that at least some emotion kinds are adaptive products of evolution. An appraisal that the environment affords an opportunity or threat is the input condition for such emotion kinds, and the output is some behavior that (if successful) constitutes an adaptive response to the situation. The internal mechanism prompting the adaptive action is a coordinated set of changes in appraisal, expression (facial, vocal), action tendencies (fleeing, threat displays), feeling, and physiological arousal that together explain how the adaptive response takes place and the characteristic internal sensations associated with the emotion.

Moral emotions are those emotion types linked with our capacity for moral thinking or moral action. They are thought to either (i) enable a capacity for moral action or (ii) contribute to the project of grounding or justifying moral norms (i.e., normative statements that define moral obligations). Section "Grounding Moral Norms" discusses the relationship between moral emotions and the project of grounding moral norms. Section "Evolved Emotions" introduces the idea that (at least some emotion types) evolved for their adaptive significance. Section "Moral Emotion Types" discusses some putative moral emotion types that have been studied in recent years.

Grounding Moral Norms

Finding some sort of nonarbitrary ground for moral norms is the traditional "holy grail" of moral philosophy. Virtue ethics, divine command theory, rationalism, and sentimentalism are four broad traditions of thought that engage in this project. Divine command theory takes moral norms to be grounded by a heteronomous force, namely, the Christian conception of God. Virtue ethics, a tradition initiated by Aristotle in the *Nicomachean Ethics*, grounds the virtuous (good) action of a thing in its ability to perform its "natural" or "proper" function (in the human case, this capacity is the exercise of rationality; see Korsgaard 2008). The rationalist tradition, exemplified by Kant's *Groundwork for the Metaphysics of Morals* (1785/2003), sees moral norms grounded in autonomous reason – individuals rationally reflecting on the norms appropriate to the situation at hand.

None of the accounts just described support the idea that emotion can, on its own, ground moral norms. It is within sentimentalist moral theories, however, that emotions perhaps play the most central role in a more modest project, namely, explaining our *capacity* for moral behavior. Sentimentalism is perhaps best understood in contrast with rationalism. Rationalism claims that it is our capacity to reason, not our capacity to feel (i.e., to experience emotion), that undergirds the human capacity for moral behavior. The rationalist approach has a long and venerable philosophical heritage. Plato's *Timaeus* (360 BCE) presents a myth in which gods invent human heads whose distinctive faculty is reason. Recognizing the heads requires bodies – full of unbridled desires and passions – to help them move through the world. The struggle to live a moral human life is expressed as the struggle of the head to control the body through the sublimation of the body's passions toward virtuous ends.

In the seventeenth and eighteenth centuries, the sentimentalists Hobbes and Hume instrumentalized practical reason. Hume, for instance, demoted reason, describing it as a "slave of the passions" that can "never pretend to any other office than to serve and obey them" (Hume 1738, p. 462). Hume's point was that the passions determined which ends were desirable, while reason acted as a conscientious servant, formulating plans to deliver those ends. Moral knowledge was attained by an "immediate feeling and finer internal sense," rather than a "chain of argument and induction" (Hume 1777, p. 2). For the sentimentalists, reason alone can enable the inference that a particular action will lead to an avoidable loss of human life, but it cannot tell us whether this loss ought to be avoided.

The focus of Hume (and Hobbes before him) was on casting emotion as (i) a causal antecedent to moral judgment and (ii) a sufficient ground for the legitimacy of moral claims. But this is not to say that sentimentalism offers succor to the moral realist who wants to ground morality in facts extraneous to our nature as embodied beings. We now set aside this latter project and return to the more modest project of explaining our capacity for moral behavior and decision-making and how we foster conditions conducive to its exercise.

Since the Enlightenment, rationalism has exerted an influential hold within moral philosophy. The rationalist tradition takes cool-headed reason to represent the ideal conditions for moral decision-making. Kant (1785) is the canonical representative of this view and has exerted a tremendous impact on modern moral and political philosophers (e.g., Rawls 1971), many of whom have attempted to follow Kant by providing a foundation for ethics derived from the meaning of rationality itself. However, rationalism has also dominated much of twentieth-century psychology. Kohlberg's work was a sustained attack on "irrational emotive theories" (1971, p. 188), and his cognitive-developmental theory was an important part of the cognitive revolution. More recently, neo-sentimentalists have claimed that moral judgments are produced by automatic intuitive or emotional processes. Cushman et al. (2010) describe two related ideas associated with neo-sentimentalism in contemporary literature on moral psychology. The first "claims that moral judgment takes the form of intuition, accomplished by rapid, automatic and unconscious psychological processes," while the second "claims that moral judgment is driven primarily by affective responses" (p. 47). Empirical evidence cited in favor of the first claim focuses on demonstrations of people's inability to articulate a rational basis for many of their most strongly held moral beliefs. Evidence favoring the second challenge draws on neuroscientific data relating to localized brain regions (Ciaramelli et al. 2007) but also includes behavioral studies of moral judgment that invoke affective manipulations.

Two influential neo-sentimentalist proposals have come from Damasio (2006) and Haidt (2001) in recent years. Both figures emphasize the importance of emotional and/or intuitive processes in moral decision-making. In particular, both claim that moral judgments are produced by automatic intuitive or emotional processes. They also claim that, when we think we are deliberating about the rightness of a given moral principle, we are not so much reasoning from first principles as we are engaging in post hoc rationalization. That is, we are simply reasoning in whatever way will support the emotionally driven judgment we have already pledged allegiance to. Haidt's social intuitionist model aims to explain how this post hoc rationalization occurs. Haidt claims that "moral dumbfounding" occurs when a judgment, solution, or other conclusion appears suddenly and effortlessly in consciousness, without any awareness by the person of the mental processes that led to the outcome.

Proponents of neo-sentimentalism need not be read as claiming that emotional and/or intuitive processes actually ground or justify specific moral claims. Rather they can be seen as showing that the capacity to feel certain emotions (i.e., moral emotions) is tied up with our *capacity* to formulate moral norms (see Kitcher 1994 for discussion). Despite its popularity, neo-sentimentalism has met with skepticism, for instance, by champions of "universal moral grammar" (Hauser 2007). Hauser and others have claimed that moral judgment requires computations performed over representations of agents, intentions, and causal relationships in order to output judgments of right and wrong. Information processing of this sort must precede an emotional response, so the argument goes, and thus it must be that moral judgment leads to emotion, not the other way around. One promising way to make progress in this debate about the role emotions play in our capacity to be moral is to explain the adaptive function of particular emotion types in relation as facilitating conformity to moral norms. The remainder of the article focuses on this project. First, we motivate the claim that emotions evolved.

Evolved Emotions

Evolutionary approaches to emotion describe them in terms of the function they currently perform or performed in our ancestral past. The claim that at least some human emotions evolved and are shared with lower mammalian species is now uncontroversial and began with Darwin's *Expression of the Emotions in Man and Animals* (1872). Darwin himself defined emotions as subjective feelings. Following Herbert Spencer, Darwin distinguished emotions as feelings caused by external states, in contrast to feelings caused by bodily states, such as hunger and pain. Later evolutionary theorists have preferred behavioral and physiological definitions of emotion, and so consequently Darwin's theory of emotional expression has been seen as a theory of the emotions themselves. The central principle Darwin proposed to explain the evolution of expression was the "principle of serviceable associated habits." Darwin believed that instinctive behaviors derive from habits. The consistent acquisition of the same habit for many generations causes it to become hereditary, or instinctive, as a result of the inheritance of acquired characteristics.

The founders of ethology saw themselves as the direct heirs of Darwin's work on mental evolution (Lorenz 1965). They retained Darwin's principles but reinterpreted them to fit the theory of evolution as it emerged in the 1930s in the "modern synthesis" of Darwinism and Mendelian genetics. The principle of serviceable associated habits was transformed into the ethological concepts of "ritualization" and "derived activity" (Tinbergen 1952). Derived activities are behaviors that originally evolved for one purpose but were later selected for another purpose. Ritualized behaviors are derived activities that originally evolved to fulfill some practical function but were later selected to function as signals. For example, grimacing in anger may have originally evolved to enable our ancestors to prepare to attack an opponent using their sharp canine teeth. We have evolved such that we no longer have fang-like canines, and yet we still grimace in anger. The derived activities account sees grimacing serving a derivative, signalling function of indicating that one is angry and that the possibility of attack is imminent. Grimacing was originally selected to enable direct attack, and through this association came to serve a secondary function of signalling to conspecifics the possibility of attack. The concept of ritualization allowed ethology to reconstruct Darwin's principle of serviceable associated habits while avoiding his commitment to the inheritance of acquired characteristics. Darwin's descriptions of the psychological rewards that led to the reinforcement of emotional behaviors are equally plausible as descriptions of the original selective advantage of those behaviors.

Darwin's detailed accounts of human facial expressions were confirmed and extended during the 1960s and early 1970s, and many of his suggested homologies between human and primate facial expressions were confirmed. It became generally accepted that some emotional expressions, the so-called basic emotions, are found in all cultures and have an evolutionary history at least as long as the history of the human species (Ekman 1973). The basic emotions are usually referred to as fear, anger, disgust, contempt, joy, sadness, and surprise, although in normal usage each of these words has a much wider meaning. Each basic emotion corresponds to an "affect program" stored somewhere in the brain. When activated, this program coordinates a complex of actions that include facial expression, autonomic nervous system changes, expressive vocal changes, and muscular-skeletal responses such as flinching or orienting. The concept of an affect program inherits many of the features of the ethological concept of a "fixed action pattern." Both concepts suggest that certain apparently complex behaviors are, in reality, atomic units of behavior that unfold in the same stereotyped sequence whenever they are triggered. The affect programs are controlled by an "automatic appraisal mechanism." This is a specialized neural system that applies its own distinctive rules for stimulus evaluation to a limited set of data derived from the earliest stages of the processing of perceptual information. Considered together, the appraisal mechanisms and affect programs form a cognitive "module" in the sense favored by more recent evolutionary psychologists (Barkow et al. 1992).

Evolutionary accounts of basic emotions have inspired work on whether "nonbasic" emotions (such as gratitude) may have also evolved (▶ Emotion). This work is discussed in the section below with a focus on emotion types for which an adaptive function related to conformity with moral norms is available.

Moral Emotion Types

Considering the unique adaptive functions of different moral emotion types is the basis of a more complex account of how the "suite" of moral emotions has, as a whole, evolved to support the capacity for moral behavior. Below, we consider a number of putative moral emotion types (disgust, shame, pride, indignation, guilt, compassion, and gratitude). We focus on adaptationist hypotheses that relate to obeying rules or socially sanctioned behaviors, or acting so as to benefit others.

Disgust. Kelly's (2011) influential synthesis of work on disgust suggests that the emotion combines a taste-aversion mechanism with a mechanism for avoiding infection and parasitism. Kelly argues that disgust is uniquely human. It is an example of the co-option and developmental entanglement that occurs as an evolving lineage confronts new adaptive challenges and must solve them by modifying existing features. Both elements of the disgust response have homologues or analogs in other species. It is their entanglement that creates a uniquely human emotion. On this account, moral disgust represents a further piece of evolutionary tinkering in which the emotion of disgust was co-opted to help enforce conformity with in-group norms. More recent work accords with this view, with moral disgust described as disgust that evolved to "respond to sociomoral elicitors [. . .] such as immoral acts" (Russell and Giner-Sorolla 2013, p. 330).

Pride and shame. Although competing explanations of the evolution of pride and shame are extant in the literature, a consensus is emerging. Behaviors homologous to those we associate with pride and shame are components of displays employed by nonhuman primates (and other mammals) during the negotiation or affirmation of relative rank. We can label the emotions that underlie such displays as “protopride” and “protoshame.” In a competitive environment, the expression of protoshame signals the actor’s acceptance of a subordinate position in a hierarchy. It functions to deflect aggression from the dominant party. The expression of protopride, in contrast, signals the actor’s dominance. Importantly, protopride is a pleasant feeling, while protoshame is unpleasant. Fessler (1999) argues that protopride and protoshame were selected for in primates because they promoted dominance-seeking behaviors, which increased the chances of reproductive success. These emotions are the phylogenetic ancestors of pride and shame in human beings (Fessler 1999, 2004).

However, Clark (2012) has argued that humans retain the ability to experience protopride and protoshame alongside their more sophisticated descendants, pride and shame. Pride and shame are more cognitively complex than protopride and protoshame. They require that the organism has “theory of mind” – the capacity to attribute mental states to other individuals. They also serve a different adaptive function in the relatively cooperative, nonhostile environments human beings inhabit. In such contexts, pride and shame are experienced in response to the actual or imagined assessment of an actor’s behavior by an audience. If the assessment inspires the contempt or pity of the audience, the actor feels shame. If the assessment inspires the envy or admiration of the audience, the actor feels pride. In the cooperative context, then, pride and shame encourage the actor to seek the admiration and envy of others, which in turn encourages them to act in ways that support the well-being of the group (Fessler 1999). In this way, these emotions inspire adherence to socially sanctioned norm-governed behavior.

Anger. Philosophers sometimes draw a distinction between simple or “basic” anger and more cognitively complex varieties such as resentment or moral indignation. Say that a simple form of anger arises in response to feeling a sharp jolt in the back from an unknown cause while standing in line at the supermarket. The reflex-like expression of basic anger, consistent with a basic appraisal (however implicit) of goal obstruction, is one we likely share with nonhuman species (Ekman 1973). In contrast, resentment or moral indignation is generally understood as accompanied by a more complex appraisal, namely, the judgment another party or parties are worthy of blame (Wallace 2008). In the supermarket example, resentment might only arise upon turning around and seeing that the cause of the jolt in the back was a person indelicately attempting to queue-jump. This case requires making an appraisal that has a more specific content, namely, that the goal obstruction arousing the initial anger has been caused by a third party and that it was wrong of that party to cause it.

An appraisal of this sort is consistent with Rozin et al.’s (1999) association of anger with violations of the standard of autonomy, which refers to concerns about fairness, reciprocity, and the prevention of harm. Anger elicited in this way could have served three distinct adaptive functions. The first is encouraging behavior that is reciprocal and fair (and therefore likely to lead to the survival of the population at large). The second is motivating an aggressive response to hostility, which would have increased the likelihood of the emoter’s survival. The third is that an angry expression would serve as an important social signal of imminent threat, encouraging the opponent to flee the situation.

Guilt. The case of guilt provides a useful example of the role evolutionary game-theoretic scenarios play in establishing adaptationist hypotheses for emotions and thereby connecting them to the moral domain. Modeling such scenarios is especially useful when considering the role of moral emotions in promoting the survival of the group at the cost of the individual. Unlike the case of anger, guilt-prone individuals act so as to benefit the group at a cost to themselves. That is to say that they behave altruistically, even in cases when they do not expect to be caught. Those who feel guilty after a transgression accept punishment readily and even punish themselves. For this reason, one might think that guilt evolved for its benefits to groups, rather than individuals.

Recent evolutionary game-theoretic modeling (Rosenstock and O'Connor 2018) shows that when actors play an iterated prisoner's dilemma and sometimes err, guilt can evolve to facilitate forgiveness and social reparation. Guilt functions in this setting by either ensuring the trustworthiness of apology or by leading apologizers to pay a cost. This model adds to previous literature on the evolution of apology by showing how both emotion reading and costly apology can interact to stabilize guilt, thereby describing the environmental conditions that would allow guilt to evolve.

Compassion and gratitude. Accounts of the evolution of compassion and gratitude build on the literature on biological altruism. In evolutionary biology, altruism refers to an action that benefits another at a cost to the actor (such as sharing food). The evolution of altruism is explained either by "kin selection" (▶ Kin Selection) or "group selection" (▶ Group Selection). Kin selection proposes that an altruistic gene will be favored by natural selection when it motivates action that confers a fitness advantage to a recipient of the same kin (Hamilton 1964). Group selection proposes an alternative mechanism by which altruism spreads as a trait in a population: altruistic actions (like helping conspecifics on a hunting trip) increase the fitness of the group even though it decreases fitness for individuals. Groups with larger numbers of altruists will be more likely to survive than those with lower numbers, and so altruism will spread as a trait through the mechanism of group selection.

The relation between altruistic behavior and pro-social emotions is a matter of some contest. However, one influential argument by Trivers (1971) claims that compassion (used synonymously with "sympathy") evolved within a complex system of other emotional states (including gratitude, guilt, liking, and anger) to enable nonkin to initiate, maintain, and regulate reciprocal, altruistic relationships (see also Nesse 1990). Within this system of emotions, compassion emerged as a state that motivates altruism in mutually beneficial relationships and contexts. It is an emotional reaction that involves feelings of concern and sorrow for the other party, based on an apprehension of that party's emotional state.

Gratitude is relatively understudied compared with other moral emotions. Like compassion, gratitude motivates altruistic behavior but stems from the appraisal that one has just received help or aid from another, which creates a motivational state to either "return the favor" (call this "direct reciprocity") or "pay it forward" (by helping someone else, also known as "upstream reciprocity"). Experiments show that gratitude in humans fosters pro-social behavior. In particular, the recipient of a favor is more likely to engage in acts of both direct and indirect reciprocity. Recent modeling scenarios by Nowak and Roch (2007) suggest both that "spontaneous" cooperators are outcompeted by gratitude-related upstream cooperators (i.e., that gratitude is a better motivator of cooperation) and, moreover, that upstream reciprocity greatly enhances the level of altruism in a population.

Conclusion

The surge of literature on the moral emotions over the past 30 years (e.g., Haidt 2003) represents a shift away from the (normative) question of what grounds or justifies moral norms and toward the (descriptive) question of what capacity or capacities in human beings enable moral behavior. While these two projects should be undertaken in tandem, the focus on the latter question holds the promise of explaining much of moral behavior by recourse to "instinct-like" emotional capacities that evolved to encourage norm conformity, especially in relation to norms often described using moral language.

While the hypothesis that basic emotions are adaptations is well-established, the hypothesis that "moral" emotions evolved for their role in enforcing norm conformity is more contested. This is so for multiple reasons. One is that the latter hypothesis requires mapping readily observable behaviors (such as cooperating by sharing food) to normative moral concepts (e.g., fairness, generosity, etc.). Another is that establishing that certain emotion-inspired behaviors increase fitness at the level of kin or group selection (e.g., in the case of compassion and gratitude) requires game-theoretic modeling of change in frequencies of a trait (e.g., cooperation) in a population over time. Finally, establishing how a "suite" of moral emotions may have evolved over phylogenetic time as part of a multilevel selection strategy would require modeling scenarios of multiple behavioral traits that act as proxies for the suite of emotion types under investigation. The next step would require moral psychologists and moral theorists to determine how much of moral behavior can be explained by exclusive appeal to the behavioral traits under consideration. This is an ambitious and valuable task in and of itself but one quite distinct from the project of justifying or grounding moral norms.

Cross-References

▶ Altruism
▶ Emotion
▶ Evolution of Morality
▶ Evolutionary Game Theory
▶ Group Selection
▶ Kin Selection

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Moral Foundation

► Sacredness/Purity/Disgust

Moral Foundations

► Function of Morality (Haidt)

Moral Foundations Theory

► Evolved Moral Foundations

Moral Instincts

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Synonyms

Moral dispositions; Moral senses; Moral sentiments

Definition

Inborn tendencies to act morally and to experience a sense of morality.

Introduction

Over the years, the concept of instinct has drifted in and out of favor in the natural and social sciences and has been defined in many different ways. Here, instincts are defined as natural inborn tendencies inherited by all members of a species to feel, think, and act in particular ways in response to particular internal and external triggers. Evolutionary theorists such as Pinker (2008) have invoked the concept of moral instincts. Moral instincts can be defined in two qualitatively different, but related, ways – (a) as instincts that induce individuals to *behave* in ways that people consider moral and to refrain from behaving in ways that people consider immoral and (b) as instincts that induce individuals to *categorize* certain phenomena as moral and immoral and to make moral judgments. People experience the second kind of moral instinct as positive and negative feelings, intuitions, and ideas that engender a sense of approval and disapproval about certain forms of conduct, desires, motives, character traits, and people. Darwin (1874) referred to them as the moral senses. The challenges faced by evolutionary theorists are to identify the mental mechanisms that dispose individuals to behave in moral ways and to make moral judgments and to explain how these mechanisms evolved. These challenges entail explaining why people feel that they and others ought and ought not to behave in certain ways; why people view certain phenomena as right or wrong, good or bad, fair or unfair, and virtuous or vicious; and why people pass judgment on themselves and others for behaving in ways that uphold or violate moral standards.

Culturally Universal Moral Norms

Anthropologists and other social scientists have concluded that although people from different cultures may develop different, culturally relative, moral norms, people from all known cultures view particular types of behavior as right and their opposites as wrong. For example, Boehm (2012) found that members of all the hunter-gatherer groups he surveyed considered generosity, group harmony, and cooperation good and excessive selfishness, bullying, murder, lying, cheating, stealing, incest, and adultery bad. In addition to these forms of conduct, other investigators have adduced evidence that people from all cultures view deference to legitimate authority and purity as moral, and their opposites as immoral. In summary, all cultures possess moral norms upholding (a) altruism, (b) support for one's group, (c) fairness, (d) respect for legitimate authority, and (e) purity.

Culturally Relative Moral Norms

Although people from all cultures consider these five forms of conduct moral, cultures differ in the salience of moral norms upholding these forms of conduct, the relative value they attach to each of them, and the particular kinds of social behavior that instantiate them. As Pinker (2008) has pointed out, Western cultures tend to consider moral norms opposing harming others and upholding care and justice as regnant; Japanese cultures tend to elevate norms pertaining to group solidarity; and Hindu and Orthodox Jewish cultures tend to place a high value on purity (e.g., as reflected in dietary restrictions) and authority (e.g., not insulting Mohammad). Other investigators have found that different groups within Western cultures also tend to place different values on different types of moral behavior. For example, liberals tend to value altruism and fairness more than conservatives do, whereas conservatives tend to value group loyalty, authority, and purity equally highly.

Different cultures and subcultures also may classify particular behaviors in different categories, producing cultural and individual differences in the instantiation of moral norms. For example, people may have different conceptions of what it means to help and to harm others and what it means to treat others justly and unjustly; and they may harbor different ideas about what makes these forms of conduct right and wrong. Investigators have documented notable differences among cultures with respect to moral norms pertaining to sexual conduct, marital unions, preservation of life, and dietary restrictions. In addition, the threshold for the activation of moral instincts tends to be lower for in-group members than for out-group members (Greene 2013), and cultures may differ in the extent to which moral instincts are activated by nonhuman animals.

Theorists such as Haidt (2001) and Pinker (2008) have accounted for the evolution of culturally universal and culturally specific moral norms by suggesting that all people inherit mental mechanisms that endow them with a set of universal, species-specific moral intuitions, just as people from all cultures inherit mechanisms that endow them with the capacity to produce and to understand language. However, in the same way that people from different cultures acquire different languages, people from different cultures instantiate universal moral norms in different ways.

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Origins of Instincts to Behave in Moral Ways

Evolutionary theory generates the expectation that instincts to behave morally originated either because they gave early humans an adaptive advantage or as by-products of other adaptations. Evolutionary theorists such as Alexander (1987), Krebs (2011), and Richerson and Boyd (2005) have suggested that the overriding adaptive function of moral instincts is to motivate members of groups to advance their interests in cooperative ways. Moral instincts induce members of groups to resist the temptation to advance their short-term personal interests in ways that diminish the long-term biological and genetic benefits they can reap from upholding the cooperative social orders of their groups. Cooperation can pay off by enabling members of groups to maximize the acquisition of resources, to engage in superior forms of defense, and to outcompete selfish groups. “Inasmuch as every member benefits from a unified, cooperative group, one expects them to care about the society they live in, and to make an effort to improve and strengthen it similar to the way the spider repairs her web, and the beaver maintains the integrity of his dam. Each and every individual has a stake in the quality of the social environment on which its survival depends” (Flack and de Waal 2000, p. 14). However, in spite of the potential adaptive benefits of cooperation, there is a significant obstacle to the evolution of cooperative dispositions, namely, the conflicts of interest that inevitably occur within cooperative groups.

As expressed by the evolutionary biologist, Alexander (1987), “agreement seems to be universal. . .that moral (and ethical) questions and problems arise because of conflicts of interest; I have never found an author who disagrees. If there were not conflicts of interest among people and societies it is difficult to see how concepts of right and wrong, ethics and morality, and selfishness and altruism could ever have arisen” (p. 3). Conflicts of interest give rise to moral dilemmas that, if not resolved in cooperative ways, can diminish the reproductive success of all members of a group.

The anthropologist Boehm (2000) has argued that the central adaptive problems that moral instincts evolved to solve stemmed from “competitive dispositions to dominance” and the social conflicts they produced (p. 97). “Morality is the human invention that addresses such problems, and it is based very heavily upon ancestral dispositions. These were the raw materials, out of which moral communities were forged” (p. 97). According to Boehm, “morality is based heavily on social pressure, punishment, and other kinds of direct social manipulation” (p. 82), and the most significant sources of morality are instincts that induce individuals to obey the social norms of their groups and resist the temptation to dominate others. When members of groups succumb to the temptation to resolve conflicts of interest in selfish ways, it disrupts the cooperative order of the group and prevents everyone, including the cheaters, from reaping the benefits of cooperative exchanges.

Origins of Instincts to Behave in Altruistic and Group-Upholding Ways

It is much easier for evolutionary theorists to explain how dispositions to induce others to behave altruistically could have evolved than to explain how individuals acquired instincts to behave altruistically themselves. How could altruistic instincts have been selected if they decreased altruists’ chances of surviving and reproducing? This problem plagued Darwin (1874), who considered evidence of altruism potentially fatal to his theory. Modern evolutionary theorists have offered three main answers to this question: that altruism can evolve through kin selection, group selection, and sexual selection.

Kin Selection

Although Darwin (1874) focused most extensively on the possibility that altruistic instincts could evolve through group selection (see below), he briefly entertained the idea that even if altruistic members of groups were less likely to survive and to reproduce than selfish members of their groups were, their “tribe would still include their blood-relations” who could pass their altruistic traits on to future generations. Darwin did not fully appreciate the significance of this insight because he did not understand that blood relatives may share genes that code for altruism.

Armed with knowledge that genes are the primary unit of inheritance, Hamilton (1964) recognized that individuals can propagate their genes in two ways: (a) directly, by behaving in ways that increase their reproductive success, and (b) indirectly, by behaving in ways that increase the reproductive success of kin who possess replicas of their genes. This insight enabled Hamilton to predict that mental mechanisms that induce individuals to behave altruistically could evolve when the genetic costs to them (C) were less than the genetic benefits to recipients (B), multiplied by their degree of relatedness (r) – conditions captured in what came to be called Hamilton’s rule: $C < Br$ (also formulated at $Br > C$). The forms of altruism evoked by kin-selected mental mechanisms are most clearly exemplified by the sacrifices parents make for their offspring (r = 0.5), but they also extend to other blood relatives, such as brothers and sisters (r = 0.5), nephews, and nieces (r = 0.25).

Hamilton's rule implies that, all else equal, individuals will favor themselves (r = 1) and their relatives such as sons, daughters, and siblings who are most likely to share the genes that code for altruism; it does not imply that individuals will allocate their altruism to others in proportion to their degree of relatedness. Hamilton's rule implies that individuals will be most strongly disposed to help their relatives when the costs of helping them are low and the benefits the helpers receive are high and that helpers will be more prone to make sacrifices for relatives who possess high reproductive potential than for those who possess low reproductive potential. Researchers have amassed a great deal of evidence that humans and other animals behave in accordance with Hamilton's rule.

Triggering Kin-Selected Altruistic Instincts

In order for animals to propagate replicas of their genes in accordance with Hamilton's rule, they must be able to distinguish kin from non-kin and estimate the degree of to which they are related to them. Animals are not able to perceive their degree of relatedness to others directly, so they must derive inferences about genetic relatedness from kinship cues such as how similar others are to them (e.g., in appearance and odor), how familiar they are, and how close to them they reside (proximity). Because non-related members of groups may possess such cues, animals may end up helping those who look and act like their kin, thus contributing little or nothing to the propagation of their genes. Altruistic instincts probably originated as maternal (and in some species paternal) instincts, then were generalized in species such as our own to siblings, more distant relatives, and members of in-groups who looked and acted like kin.

Mental Mechanisms and Emotions Supporting Kin-Selected Altruistic Instincts

Triggers that induce humans and other animals to behave altruistically activate mediating mental mechanisms that affect their emotional and cognitive states. Although the mental mechanisms that induce animals to help their kin (and those who resemble their kin) are designed in different ways in different species, research suggests that there are significant commonalities across mammals in the ways in which they operate. For example, these mental mechanisms appear to be regulated by hormones such as oxytocin, vasopressin, prolactin, and endogenous opioids, and they appear to generate psychological experiences of attachment, commitment, and love (Brown and Brown 2006).

Sympathy and Empathy

Because kinship equates to genetic similarity, it is not surprising from the perspective of evolutionary theory that people tend to identify with their relatives, to empathize with them, and to sympathize with them when they are in distress. de Waal (2008) has reviewed research demonstrating that empathy is evoked by kinship cues such as "similarity, familiarity, social closeness, and positive experience with the other" and that human "subjects empathize with a confederate's pleasure or distress if they perceive the relationship as cooperative" (p. 16). de Waal cites research showing that "seeing the pain of a cooperative confederate activates pain-related brain areas, but seeing the pain of an unfair confederate activates reward-related areas, at least in men." de Waal concludes that "the empathy mechanism is biased the way evolutionary theory would predict. Empathy is (a) activated in relation to those with whom one has a close or positive relationship, and (b) suppressed, or even turned into Schadenfreude, in relation to strangers and defectors."

de Waal summarizes his account of the evolution of empathy as follows:

> Empathy covers all the ways in which one individual's emotional state affects another's, with simple mechanisms at its core and more complex mechanisms and perspective-taking abilities at its outer layers. Because of the layered nature of the capacities involved, we speak of the Russian doll model, in which higher cognitive levels of empathy build upon a firm, hard-wired basis.... The claim is not that [this basis] by itself explains [all forms of empathy], but that it underpins ... cognitively more advanced forms ..., and serves to motivate behavioral outcomes. (p. 11)

According to Moll et al. (2008), "preliminary functional imaging results in normal subjects point to the involvement of the anterior PFC [prefrontal cortex], the dorsolateral PFC, the OFC [orbitofrontal cortices], the anterior insula, and the anterior temporal cortex" (pp. 15–16) in the production of feelings of sympathy, pity, and compassion.

Group Selection

Contemplating the problem that altruism presented for his theory, Darwin (1874) considered the possibility that altruistic instincts that reduced the fitness of individuals could evolve if they increased the fitness of other altruistic members of their groups. If altruistic groups outcompeted selfish groups, then altruism could evolve through group selection – a process that selects traits that benefit groups even though they may be costly to individual members of the groups. However, Darwin immediately saw a problem with this idea, namely, that even if altruistic groups outcompeted selfish groups, the selfish members of altruistic groups would fare better than their altruistic cohorts, eventually driving them and their altruistic traits into extinction.

Although most contemporary evolutionary theorists are wary of the idea that altruism could have evolved through group selection, they acknowledge that it is possible under ideal conditions, which, according to some theorists were met during the evolution of the human species. In a prominent defense of group selection, Sober and Wilson (1998) identified five conditions under which altruism could evolve in a population even though it decreased within altruistic groups: (a) there are many groups in the population, (b) the groups contain different proportions of altruistic and selfish individuals, (c) the groups containing high proportions of altruists are more fit than selfish groups are, (d) altruistic and selfish groups compete against one another and form new groups, and (e) the fitness benefits to altruistic groups are greater than the fitness benefits to selfish individuals within the groups. Sober and Wilson suggested that when humans evolved, altruistic groups outcompeted more selfish groups, grew in size, and reconstituted themselves before within-group section for selfishness transformed the groups. In recent years, several eminent evolutionary biologists have endorsed the idea that altruism can evolve through group selection.

Many studies have found that people are willing to sacrifice their own interests to promote the collective interests of their groups. Perhaps the most striking examples stem from the behavior of soldiers during battle. As pointed out by Brown and Brown (2006), the US Army Field Manual contains the following passage, "while patriotism and sense of purpose will get American soldiers to the battlefield, the soldiers' own accounts (and many systematic studies) testify that what keep them there amid the fear of death and mutilation is, above all, their loyalty to their fellow soldiers" (p. 38).

A group of evolutionary theorists has suggested that instincts to uphold large culturally defined groups evolved later than, and through a different form of group selection from instincts to help friends, relatives, and members of smaller in-groups. For example, Richerson and Boyd (2005) suggested that "tribal instincts" evolved relatively recently in the human species through cultural group selection. According to Richerson and Boyd (2005), tribal instincts "are laid on top of more ancient social instincts rooted in kin selection and reciprocal altruism" (p. 191), which causes humans to be "simultaneously committed to tribes, family, and self, even though our simultaneous and conflicting commitments very often cause us…great anguish" (p. 191). Richerson and Boyd view tribal instincts as analogous to Chomskyan linguists' "principles and parameters . . . of language. . .. The innate principles furnish people with basic predispositions, emotional capacities, and social skills that are implemented in practice through highly variable cultural institutions, the parameters" (p. 191).

The sometimes acrimonious exchanges between evolutionary theorists who have argued for and against group selection are fueled by the different perspectives from which they view evolution. Several biologists have demonstrated that although models of group selection and models of kin selection are formulated in different ways, they are mathematically equivalent. The essence of both models lies in the idea that genes that dispose a unit of selection to behave altruistically – whether this unit be individuals, families, or groups – can evolve when the altruistic behaviors have the ultimate effect of increasing the frequency of altruistic genes in the population.

Triggering Group-Selected Altruistic Instincts

The basic triggers for group-upholding altruistic instincts are cues to shared group membership. Although it might seem that humans would be evolved to identify with groups that are defined in terms of features such as race and physical appearance, and to sustain such group identifications throughout their lives, social psychologists have found that humans tend to be more flexible. For example, studies have found that individuals who are assigned to groups on an arbitrary basis such as whether they had blue eyes or brown eyes or whether they preferred one form of abstract art or another quickly identify with such groups and favor them when they distribute resources.

Mental Mechanisms and Emotions Supporting Group-Upholding Altruistic Instincts

According to Richerson and Boyd (2001), "we are adapted to living in tribes, and the social institutions of tribes elicit strong – sometimes fanatical – commitment" (p. 215). Social psychologists have attributed the feelings of solidarity that emerge in groups to people's tendency to identify with in-groups and incorporate "social identities" into their self-concepts.

Sexual Selection

The third way in which altruistic instincts (and other moral instincts) can evolve is through sexual selection. Darwin opened the door to understanding the sexual selection of altruism by recognizing that adaptations that affect reproduction contribute as much – indeed in sexually reproducing species, often more – to the evolution of traits than adaptations that affect survival and that one of the ways in which individuals can increase their reproductive success is by sacrificing their interests for the sake of others. Altruistic traits that are a burden to survival can evolve through sexual selection when they help individuals increase their reproductive success by attracting mates.

It is in the genetic interest of members of sexually reproducing species to select mates who (a) possess "good genes" that they can pass on to their offspring and (b) are disposed to help them and their offspring. One of the ways in which animals advertise that they possess good genes is by demonstrating that they are able to survive even though they possess a biologically altruistic "handicap" that induces them to make sacrifices for the sake of others. Altruistic behaviors constitute costly signals that advertise that the individuals who posses them are fit enough to survive in spite of them. Although studies on mate choice in humans have featured sex differences in the qualities that appeal to potential mates, members of both sexes rank altruistic qualities at or near the top of their lists.

Triggering Sexually Selected Altruistic Instincts

The primary triggers for sexually selected altruistic instincts are cues that signal the availability of attractive mates with high reproductive potential. Although we might expect the individuals doing the choosing to favor potential mates who signal that they are willing and able to help them and their offspring, individuals also may select mates who display a willingness to help a wider range of recipients, because such behaviors may signal that potential mates possess good genes and high social potential.

Mental Mechanisms and Emotions Supporting Sexually Selected Altruistic Instincts

It seems plausible that sexual selection played some role in the design of mental mechanisms that induce humans to experience feelings of romantic love. According to Fessler and Haley (2003), "a number of investigators. . .have suggested that some emotions can be understood as mechanisms designed to commit people to behavior that yields long-term payoffs, thus overcoming the temptation for short-term defection. Romantic love, a universal human emotion that underpins pair bonding. . .appears to be such a mechanism" (p. 10). When individuals are in love, they are motivated to impress their lovers, to assist them, and to bring them gifts. Neuroscientists have found that feelings of love are correlated with the activation of the ventral tegmental and caudate nuclei of the brain (areas associated with experiences of pleasure), the production of neurochemicals such as dopamine, and the inhibition of neurochemicals such as serotonin.

An Integrative Account of the Evolution of Emotional Sources of Altruism

Brown and Brown (2006) have advanced an account of the evolution of altruism called selective investment theory that integrates other theoretical approaches. The central principle of selective investment theory is that individuals are instinctually disposed to help those with whom they share fitness interdependence. Brown and Brown (2006) review a large number of studies that support their claim that mental mechanisms have evolved in many species that induce them to form emotionally based social bonds with mates, offspring, friends, members of in-groups, and other individuals or groups on whom their fitness is dependent and that these bonds dispose them to treat these recipients altruistically. Selective investment theory postulates that the key to understanding the evolution of altruism lies less in determining whether altruists and recipients share genes or whether they pay one another back than in determining "the direction and magnitude of the correlation between the reproductive success of potential altruists and their recipients" (p. 9).

Origins of Instincts to Respect Authority

In addition to believing that it is right to behave altruistically and to uphold their groups, people from all cultures believe that it is right to respect legitimate authority and to obey rules that uphold the social orders of their groups. Instincts to respect authority probably originated in social species as elaborations of deferential instincts that upheld dominance hierarchies. Deferential instincts induce members of groups to avoid punishment from those who are more powerful than they are and to accept subordinate positions until they are able to garner enough power to assume more dominant positions in their groups.

In addition to preserving subordinate group members' safety and physical well-being, deferential instincts induce subordinate members to uphold social orders that ultimately contribute to their fitness: "Low-ranking individuals are not simply dominated or exploited; they typically benefit from protection, advice, leadership, and intervention in disputes" (Haslam 1997, p. 300). Dominance hierarchies structure social orders in ways that benefit all members of groups, even if the benefits are distributed unevenly. Well-organized, cohesive groups tend to fare better than poorly organized groups in conflicts with other groups.

According to Haslam (1997), "there appear to be clear continuities between humans and nonhuman primates regarding the realization and representation of dominance" (p. 304) and, by implication, the realization and representation of instincts to respect authority. Deferential instincts evolved in humans in essentially the same ways in which they evolved in other primates, even though humans define dominance and authority in significantly more complex ways than other primates do and belong to a larger number of groups that are organized in a greater variety of ways. Primitive brain structures similar to those possessed by other primates dispose humans to behave in primitively deferential ways in conditions in which these forms of conduct increased the fitness of early humans. However, in contrast to other primates, humans also possess more sophisticated brain structures that endow them with the capacity to engage in complex forms of abstract and reflective thought and to communicate through symbolic language. The primitive and later-evolved brain structures communicate with each other, which affects the ways in which humans defer to and respect authority.

Triggering Deferential Instincts

Like other animals, humans who are faced with conflicts of interest evaluate the power and status of their opponents and select their strategies accordingly, respecting the authority of those who have the power to defeat them and those they admire and dominating those who are weaker and of lower status than they are. However, humans are exposed to many more types of authority than other primates are. Although deferential instincts may be triggered by anyone identified as an authority, and even by the rules and norms of groups, humans tend to be more discriminating than other primates in deciding which authorities to obey.

Mental Mechanisms and Emotions Supporting Deferential Instincts

Evidence suggests that instincts to obey authority may be mediated by two types of qualitatively distinct emotions – (a) fear and insecurity and (b) admiration and awe. Fear is evoked by threatening stimuli such as powerful adversaries, whereas feelings of admiration and awe are evoked by accomplished people of high status. Fear is mediated by the autonomic nervous system and primitive brain mechanisms, such as the amygdale, interacting with interpretive mechanisms in the prefrontal cortex. According to Moll et al. (2008), "the neuroanatomy of awe is still obscure" (p. 16). Fear disposes individuals to respect authority in order to avoid punishment, whereas awe and admiration dispose people to make sacrifices for those they "worship" and to curry their favor. Different mixtures of fear and awe may engender different kinds of moral instincts. A spate of research has found that hormones such as androgen and serotonin play an important role in the activation of deferential instincts (Cummins 2005).

Origins of Instincts to Behave Fairly

Social animals induce members of their groups to treat them in fitness-increasing ways by rewarding those who benefit them (and/or their relatives) and punishing those who harm them. Animals that behave in these ways often are motivated to "get even," which entails behaving fairly. With respect to primates, de Waal and Brosnan (2006) have suggested that "the squaring of accounts in the negative domain. . .may represent a precursor to human justice, since justice can be viewed as a transformation of the urge for revenge, euphemized as retribution, in order to control and regulate behavior" (p. 88). In the positive domain, "the memory of a received service, such as grooming, induces a positive attitude toward the same individual, a psychological mechanism described as "gratitude" by Trivers" (p. 93).

Game theory research on the evolution of social strategies has demonstrated that individuals who invoke unfair strategies against one another in social exchanges tend to fare more poorly than individuals who invoke fairer strategies. Investigators have found that, in appropriate conditions, fair strategies such as tit for tat and firm but fair outcompete selfish strategies such as All-D [always defect] and therefore evolve. Fair strategies tend to pay off better than unfair strategies because it is in the adaptive interest of members of groups to ensure that cheaters do not prosper. Group members may reinforce fairness by getting even, rewarding those who behave fairly and punishing those who behave unfairly, selecting fair individuals as exchange partners and rejecting or ostracizing unfair individuals, and enhancing the reputation and social status of fair partners and turning others against unfair partners.

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Triggering Instincts to Behave Fairly

Evolutionary theorists expect instincts to behave fairly to be triggered in contexts in which behaving fairly paid off better biologically than behaving unfairly did in the environments in which the animals in question evolved, such as contexts that involve repeated social exchanges between members of cooperative groups. These instincts are expected to be sensitive to cues that signal positive and negative consequences of behaving fairly, such as the probability that one will encounter an exchange partner in the future, exchange partners' willingness and ability to reward fair behaviors and punish unfair behaviors, and the presence of an audience. Individuals are expected to be more strongly disposed to treat members of their in-groups fairly than they are to treat strangers, partners in one-shot exchanges, and members of out-groups fairly. In contrast to deferential instincts, which tend to be activated in relations between dominant and subordinate members of groups, and in contrast to altruistic instincts, which tend to be activated in relations among friends, relatives, and loved ones, instincts to behave fairly tend to be activated in economic exchanges among equals. Evolutionary psychologists have pointed out that friends and loved ones rarely engage in tit-for-tat-like forms of reciprocity and that they normally are willing to support one another over long periods of time without any expectation of immediate returns. Indeed, people often feel offended when their friends adopt an "exchange orientation" and offer to pay them back.

Mental Mechanisms and Emotions Supporting Instincts to Behave Fairly

Some evolutionary psychologists have asserted that calculating the value of the goods and services that exchange partners give and receive, comparing them, and evaluating them in terms of an appropriate standard of fairness require sophisticated mental abilities that only humans may possess. However, other evolutionary theorists have disagreed, arguing that a sense of fairness may be mediated by less sophisticated mental abilities. In support of the latter position, Brosnan and de Waal (2003) found that capuchin monkeys that were offered a relatively undesirable reward for performing a task (a cucumber) after observing other monkeys receiving a more highly valued reward (a grape) got angry, refused to accept the less desirable reward, and even threw it away. The observers' anger was intensified when the other monkeys did not have to work for the reward. Brosnan and de Waal (2003) concluded that "capuchin monkeys thus seem to measure reward in relative terms, comparing their own rewards with those available and their own efforts with those of others" (p. 48).

A suite of emotions has evolved to support instincts to behave fairly and through such behaviors to uphold the cooperative orders of groups (Fessler and Haley 2003; Trivers 2006). One set of emotions, which includes feelings of gratitude and a sense of indebtedness and guilt, motivates individuals to even the score by paying their debts and righting the wrongs they have perpetrated. Another set of emotions, which includes anger and righteous indignation, motivates individuals to induce others to behave fairly. Forgiveness motivates individuals to renew damaged social exchanges. Researchers have begun to identify triggers for these emotions, to map the design of the mental mechanisms that support them, and to identify the areas in the brain that induce them.

Gratitude

In a review of the research on gratitude, McCullough et al. (2008) suggested that gratitude evolved "to stimulate not only direct reciprocal altruism, but also "upstream reciprocity:" passing benefits on to third parties instead of returning benefits to one's benefactors" (p. 283). According to Moll et al. (2008), people feel gratitude when they experience positive outcomes that they attribute to the agency of a person who intended to improve their well-being. Gratitude engenders attachment to benefactors and the motivation to pay them back. "Activated brain regions include the ventral striatum, the OFC, and the anterior cingulate cortex" (p. 16). McCullough et al. (2008) have found that feelings of gratitude are affected by the value of benefits, the costs of giving, the perceived intentions of the benefactor, and the extent to which the beneficial behaviors are viewed as voluntary, expected, and normative. In contrast to feelings of indebtedness, which people experience as a negative state that motivates them to redress perceived inequities, people experience gratitude as a positive state that motivates them not only to help those who benefited them but also to help third parties.

Guilt

Members of cooperative groups inevitably make mistakes and treat others unfairly. Feelings of guilt evolved to motivate people to uphold the social norms of their groups and avoid punishment from others by making amends, righting the wrongs they perpetrated, repairing damaged social relations, and getting mutually beneficial cooperative relations back on track. Social psychologists have found that people feel guilty when they feel responsible for causing those to whom they feel attached to experience bad outcomes, especially when their behaviors violate moral norms. Theorists disagree about the emotional origins of guilt. Some assert that it is rooted in fear, whereas others assert it is rooted in sadness. According to Moll et al. (2008), "recent neuroimaging data showed the involvement of the anterior PFC, the anterior temporal cortex, the insula, the anterior cingulated cortex, and the STS [superior temporal sulcus] regions in guilt experience" (p. 13).

Indignation and Vindictiveness

Being treated unfairly and observing others being treated unfairly may trigger feelings of indignation and vindictiveness, which may motivate individuals to right the wrong by punishing those who have behaved in unfair ways. Inasmuch as getting even entails righting a wrong, it entails behaving fairly. In addition, the anticipation of punishment for behaving unfairly motivates individuals to treat others fairly. According to Moll et al. (2008), people feel indignant when they observe others intentionally violating norms: "Indignation relies on . . .engagement of aggressiveness, following an observation of . . .bad outcomes to the self or a third party. . . and "evokes activation of the OFC. . .the anterior PFC, anterior insula, and anterior cingulate cortex" (p. 15).

Research has revealed that feelings of vindictiveness and associated motives to take revenge stem from different mental mechanisms from those that produce feelings of moral indignation (McCullough 2008). Areas of the brain such as the left prefrontal cortex and caudate nucleus that are associated with the pursuit of rewards light up when people feel vengeful, but areas of the brain such as the nucleus accumbens that are associated with feelings of satisfaction light up when people take revenge or learn that those who have treated them unfairly have been punished. According to McCullough (2008), "revenge pays neurochemical dividends. . . Natural selection's logic here seems pretty easy to comprehend: by paying us back with pleasure, our brains ensure that we'll go to the trouble of seeking the social advantages that come from returning harm for harm."

Forgiveness

Although members of all human groups right wrongs and get even by punishing those who treat them and other members of their groups unfairly, this strategy runs the danger of disrupting mutually beneficial cooperative relations and inviting reciprocally punitive tit-for-tat types of reactions that lead to fitness-reducing blood feuds. To counteract this danger, dispositions have evolved that dispose members of groups to repair damaged social relations by forgiving those who have trespassed against them. Strictly speaking, forgiveness does not support the disposition to treat others fairly unless it is invoked to even the score with those who have forgiven one in the past; however, forgiveness supports fair systems of cooperative exchange by helping to get them back on track after they have been derailed.

McCullough (2008) has reviewed research demonstrating that forgiveness constitutes a culturally universal social instinct that mediates reconciliation in humans, chimpanzees, and other primates. According to McCullough (2008), forgiveness is activated when individuals (a) feel close to and empathize with those who have wronged them, (b) are motivated to guard against the loss of beneficial relationships, (c) are confident that the wrongdoers will not harm them again, and (d) believe that forgiving the wrongdoers will increase their security. When making decisions about whether or not to forgive those who have trespassed against them, people attempt to determine whether the wrongdoers intended to harm them, could have avoided harming them, feel regretful about the harm they perpetrated, and are likely to repeat the offense. Being wronged activates stress hormones such as cortisol that generate unpleasant feelings of anger and anxiety that victims can relieve by forgiving those who wronged them or seeking revenge against them. The neurochemistry of forgiveness involves mechanisms that reduce the stress engendered by ongoing social conflicts.

Origins of Instincts to Uphold Purity

It is in the adaptive interest of group members to ensure that their environments, including other members of their groups, do not contaminate them and cause them to get sick. Rozin et al. (2000) have found that stimuli associated with the risk of harm due to germs, such as bodily fluids, dead animals, garbage, and rancid smells, evoke feelings of disgust and that this emotion tends to activate negative moral instincts in humans. Rozin et al. (2000) have pointed out that sexual acts involve the exchange of bodily fluids and that sexual prohibitions such as those associated with menstruation, anal sex, and bestiality are often based on ideas of pollution and feelings of disgust. According to Moll et al. (2008), "the neural representations of disgust . . . include the anterior insula, the anterior cingulated and temporal cortices, the basal ganglia, the amygdale, and the OFC" (p. 15).

Rozin et al. (2000) have suggested that in humans, moral disgust is an "embodiment" of nonmoral disgust that originated in digestive repulsion. As with other forms of disgust, the adaptive function of moral disgust is to motivate people to avoid potentially noxious stimuli, whether physical or social in nature. It is in the adaptive interest of members of groups to steer clear of those who emit signs of impurity and by implication to behave in pure ways themselves. "Cleanliness is next to godliness." Rozin et al. (2000) also have suggested that instinctive reactions to physical impurity and purity may be generalized by humans into ideas about spiritual impurity and purity. People tend to view acts that corrupt people as immoral and pureness of spirit as moral. In support of these ideas, research has revealed that disgusting events that have little to do with morality can intensify moral instincts. For example, investigators have found that the harshness of people's moral judgments is intensified when they are in a filthy, messy room.

Imperfections in the Design of Psychological Mechanisms and the Role of Manipulation in the Activation and Evolution of Moral Instincts

The mental mechanisms that give rise to and regulate moral instincts are not perfect, and imperfections in their design may induce individuals to behave in ways that render their conduct more or less moral than the adaptive forms of conduct that mediated the evolution of the mechanisms. For example, instincts to avoid contamination may induce individuals to purify themselves in religious rituals and to uphold food prohibitions that foster or diminish their fitness. Instincts to defer to authority may induce people to follow the orders of cult leaders, abide by the dictates of gods, and obey arbitrary rules. Instincts to help kin may be triggered by individuals who look and act like kin, which may contribute little or nothing to the propagation of their genes.

Members of most social species engage in strategic forms of interaction in which they attempt to manipulate one another into behaving in ways that increase their fitness by preying on imperfections in the mental mechanisms that regulate their moral instincts. Like cuckoo birds that trick neighboring birds into suffering the costs of sitting on their eggs, humans attempt to manipulate others into behaving morally by acting like they are their kin, by misrepresenting their value as mates, by acting more dominant and pure than they really are, by exaggerating the value of the assistance they have proffered to them, by persuading others that altruism pays off, and so on. However, because it is in the biological interest of recipients of these overtures to detect and counteract such manipulations, countervailing mechanisms have evolved. In effect, manipulative mental mechanisms have selected increasingly effective countervailing mechanisms, which have selected increasingly effective manipulative mechanisms, and so on in an arm race manner that has caused members of many social species to develop highly sophisticated social strategies. Mutual manipulation probably played a key role in the evolution of the human brain as well as the evolution of moral judgments, moral beliefs, and moral norms (Krebs 2011).

Summing up, people from all cultures are naturally inclined to behave in altruistic, group-upholding, deferential, fair, and pure ways, and people from all cultures consider these forms of conduct right and their opposites wrong. Instincts to behave in ways that uphold universal moral norms are triggered by particular stimuli and are mediated by mental mechanisms that produce emotional experiences such as love, sympathy, gratitude, and indignation. Although such emotions have been called "moral emotions" because they give rise to forms of conduct that people consider moral, they may be evoked by nonmoral triggers and fail to be accompanied by attributions of right and wrong. For example, people may sympathize with a friend who has been diagnosed with cancer or killed by lightning without feeling that anyone has committed a moral transgression, and people may feel gratitude when they experience a stroke of good luck without feeling that they deserve the positive outcome.

Explaining how humans acquired instincts to behave in ways that they consider moral does not equate to explaining how humans acquired the mental mechanisms that induce them to feel or to sense that these forms of conduct are moral. To fully account for the evolution of moral instincts, evolutionary theorists have attempted to explain how the mental mechanisms that give rise to a sense of morality, or more exactly, to the moral senses, evolved (Krebs 2011).

Origins of a Sense of Morality

All normal human beings possess a sense of morality that induces them to think and feel that certain phenomena are right and wrong, good and bad, and virtuous and vicious. All people make prescriptive and prohibitive moral judgments about how they and others ought and ought not to behave; and all people pass judgment on the moral and immoral behaviors of others. Research has indicated that children distinguish between rules that uphold moral orders and other kinds of rules very early in their lives and that they believe it is significantly more important to uphold the former than to uphold the latter. As expressed by Cummins (2005), "children are significantly better at detecting violations of rules in the social domain, such as those dealing with permissions, prohibitions, promises, obligations, and warnings, than they are at detecting violations of rules in other domains" (p. 698). Children distinguish between moral rules and rules that uphold social conventions by the time they are 3 years old. "When asked to test compliance with social rules, 3-year-olds have been found to spontaneously seek out potential rule violations just as adults do…., readily distinguish rule-violating behavior from compliant behavior…The average 3-year-old appears to have as firm a grasp on the implications of socially prescriptive rules as the average adult" (Cummins 2005, p. 689).

Emotional and Rational Aspects of a Sense of Morality

There is a long-standing debate among scholars about the extent to which people derive moral judgments from emotional and rational mental processes (Denton and Krebs 2016). In one camp, philosophers such as Kant and cognitive-developmental psychologists such as Lawrence Kohlberg have argued that all moral judgments stem from reason and that judgments that stem from emotional sources do not qualify as moral. In the other camp, philosophers such as Hume and social psychologists such as Haidt (2001) have asserted that reason is the "slave of the passions" and that moral judgments are primarily manifestations of emotional states.

Some theorists have attempted to resolve this dispute by demonstrating that two sets of mental mechanisms have evolved in humans – one that induces them to make moral judgments in emotional ways and the other that induces them to make moral judgments in more rational ways. According to Haidt (2001), "The affective system has primacy in every sense: It came first in phylogeny, it emerges first in ontogeny, it is triggered more quickly in real-time judgments, and it is more powerful and irrevocable when the two systems yield conflicting judgments" (p. 819).

In one line of research that has supported dual-process models of moral judgment, neuroscientists have used techniques such as fMRI to assess brain activity as participants make decisions about hypothetical moral problems such as those that are contained in trolley dilemmas. For example, Greene and his colleagues (see Greene 2013) asked participants to make moral judgments about two versions of a dilemma that involved a trolley heading down a track containing five people whom it is destined to kill. In the first version, called the "switch" dilemma, participants are asked to decide whether they should pull a switch that would cause the trolley to be redirected onto a different track containing only one person. In the second version, called the "footbridge" dilemma, participants are asked to decide whether to push a large man off a footbridge onto the track to stop the trolley. Consistent with the findings of other investigators, Greene and his colleagues found that most participants said that it would be right to pull the switch but wrong to push the large man onto the tracks. In support of the idea that different mental mechanisms mediated this difference in moral judgment, Greene and his colleagues observed an increase in activity in regions of the brain mediating rational moral judgments when participants responded to the "switch" dilemma and an increase in activity in regions of the brain mediating emotional judgments when participants responded to the "footbridge" dilemma. Greene (2013) argued that in contrast to the position advanced by philosophers such as Kant, these findings indicated that emotional reactions induce people to make categorical, deontological moral judgments (e.g., it is never right to violate people's right to life) and rational reactions induce people to make utilitarian judgments (e.g., it is right to sacrifice the life of one person to save five). However, some philosophers have disagreed with this conclusion.

In another line of research supporting dual-process models of moral judgment, investigators have examined the effects of damage to areas of the brain that mediate emotional reactions to moral issues. In a classic set of studies, the neuroscientist Damasio (1994) found that although patients with damage to the ventromedial area of the prefrontal cortex – an area that mediates emotional processes – made normal kinds of moral judgments to hypothetical moral dilemmas, they behaved in highly immoral ways in their everyday lives. Damasio (1994) concluded that the deficiencies in real-life moral judgment displayed by these patients stem from problems with the activation of moral emotions. People with injuries to their ventromedial prefrontal cortex are able to understand the difference between right and wrong in principle, but they do not *feel* the difference, because they do not experience moral instincts that give rise to emotions such as shame, guilt, gratitude, and moral outrage in the same way that normal people do.

Denton and Krebs (2016) advanced an "evolutionary-developmental" explanation for the findings from research supporting dual-process models of moral judgment, suggesting that early-evolved brain mechanisms that humans share with other primates give rise to moral emotions; later-evolved brain mechanisms give rise to sophisticated forms of moral reasoning; and the latter increase in power in humans as they develop throughout their life span. According to Denton, "The basic moral emotions produced by primitive brain structures work quite well in enabling social species such as chimpanzees that live in relatively small groups and form relatively simple societies to coordinate their efforts in mutually beneficial ways . . . [and] they also work reasonably well for modern humans in social contexts that are similar to those experienced by chimpanzees and early humans, such as those that include family, friends, and small in-groups. However, because humans are the only species that evolved neocortices capable of performing sophisticated forms of moral reasoning we can safely infer that, early humans experienced adaptive social problems different from those experienced by other species. The high correlation between brain size and group size in social animals. . . suggests that adaptive problems related to complex systems of cooperation and social relations among members of large groups may have played an important role in the evolution of the neocortex."

The Moral Senses

Although all aspects of a sense of morality induce people to think and feel that certain phenomena are right and wrong, different aspects induce people to experience different kinds of moral sentiments and to make different kinds of moral judgments. Some moral sentiments pertain to one's self and one's behavior (e.g., guilt), whereas other moral sentiments pertain to other people and their behavior (e.g., righteous indignation). Some moral sentiments are imbued with positive thoughts and feelings (e.g., moral approbation), whereas others are imbued with negative thoughts and feelings (e.g., moral disapprobation). Some moral sentiments are activated by thoughts and feelings about how people should and should not behave in the future (e.g., a sense of moral obligation), whereas other moral sentiments are activated by events that already have occurred (e.g., moral outrage). Moll et al. (2008) have suggested that "moral emotions might prove to be a key venue for understanding how phylogenetically old neural systems, such as the limbic system, were integrated with brain regions more recently shaped by evolution, such as the anterior PFC [prefrontal cortex], to produce moral judgment, moral reasoning, and behavior" (p. 17).

Origins of the Moral Senses

Although the basic adaptive function of all moral instincts is to induce individuals to uphold moral norms that support the social orders of their groups, different moral sentiments accomplish this in different ways. The function of some moral sentiments is to induce others to behave in moral ways (e.g., sentiments that induce individuals to approve of group-upholding, cooperative, and altruistic behaviors, to feel vengeful when members of their groups exploit them, and to experience feelings of righteous indignation when they observe third parties behaving unfairly). The function of other moral sentiments is to dispose the individuals who experience them to behave in moral ways (e.g., sentiments that engender feelings of moral obligation, sentiments that induce people to resist temptation, and sentiments that evoke feelings of guilt).

Origins of Moral Sentiments About Others

Moral sentiments triggered by the behavior of others probably originated in positive and negative emotional reactions to social behaviors that had good and bad effects on recipients' fitness and the fitness of their relatives, friends, and members of their in-groups. It is safe to assume that early humans liked being treated in kind, loyal, respectful, fair, and uncontaminated ways by others because these forms of conduct helped them survive, reproduce, and propagate their genes. On the other hand, early humans surely disliked being treated in cruel, disloyal, disrespectful, unfair, and unclean ways, just as modern humans do. It also is safe to assume that positive reactions to being treated in fitness-increasing ways and observing one's relatives and friends being treated in these ways evoked a primitive evaluative sense that these forms of conduct were right and that the people who emitted them were good, whereas negative reactions to being treated in fitness-decreasing ways and observing one's friends and relatives being treated in these ways would have engendered the sense that these behaviors were wrong and that those responsible for perpetrating them were bad.

Origins of Sense of Moral Obligation

Darwin (1874) believed that the essence of morality lies in a sense of duty and that this sense is engendered by the activation of social instincts: "The imperious word ought seems merely to imply the consciousness of the existence of a rule of conduct, however it may have originated" (p. 112). In contrast to nonmoral feelings of duty, feelings of moral obligation pertain to shared standards of conduct that are prescribed by moral norms, supported by sanctions, and affect the welfare of members of one's group.

Origins of Conscience and a Sense of Guilt

Whereas a sense of moral obligation involves forward-looking motives to behave in moral ways, a guilty conscience involves backward-looking reactions to misdeeds that people have committed. When people's consciences bother them, they pass judgment on themselves. (Anticipatory feeling of guilt also may be evoked when people entertain the idea of committing immoral deeds.)

Darwin (1874) advanced an account of the evolution of conscience in which he suggested that satisfying a pressing but transient antisocial need that leaves a more enduring prosocial need unsatisfied induces people to feel that they made the wrong choice. Supporting such feelings, Darwin suggested that "man would be influenced in the highest degree by the wishes, approbation, and blame of his fellow-men," and the "deep regard" people feel for the "good opinion" of their fellows would cause them to regret their misdeeds and feel guilty about them (p. 106).

Findings from research on the development of conscience in children are consistent with expectations derived from a Darwinian framework. For example, in a review of the literature, Thompson et al. (2006) concluded that "one of the strong motivators for morally compliant behavior is the salient emotion that arises from cooperative and uncooperative conduct" (p. 277). "Conscience development has its origins in infancy, where the sanctions (and rewards) of adults in response to the child's actions have emotional and behavioral consequences. . . .experiences of approval and disapproval are important" (pp. 269–270). "Young children are motivated to cooperate with the expectations of parents to enjoy the positive affectionate relationship that they enjoy. Viewed in this light, the parent-child relationship in early childhood can be conceived of as the young child's introduction into a relational system of reciprocity that supports moral conduct by sensitizing the child to the mutual obligations of close relationships" (p. 282). Moll et al. (2008) have found that guilt stems from "the blending of elementary subjective emotional experiences, which are ubiquitous in mammals, with emotional and cognitive mechanisms that are typically human" (p. 4).

Origins of a Sense of Justice

Trivers (1985) has asserted that "a sense of fairness has evolved in the human species as the standard against which to measure the behavior of other people, so as to guard against cheating in reciprocal relationships" (p. 388), and "such cheating is expected to generate strong emotional reactions, because unfair arrangements, repeated often, may exact a very strong cost in inclusive fitness" (Trivers 2006, p. 77). Evolutionary theory has no difficulty explaining why members of groups would feel that it is right for others to treat them fairly and wrong for others to treat them unfairly, but it is more challenging to explain how the sense that it is right for them to treat others fairly evolved. As expressed by Trivers (2006), "an attachment to fairness or justice is self-interested and we repeatedly see in life. . .that victims of injustice feel the pain more strongly than do disinterested bystanders and far more than do the perpetrators" (p. 77).

Psychologists have amassed a great deal of evidence in support of self-serving biases in people's sense of justice. For example, studies have found that people hold themselves to lower standards of fairness than they apply to others, underestimate how much they owe others and overestimate how much others owe them, overvalue the costs of giving to others and undervalue the benefits of receiving, and react less strongly to treating others unfairly than to being treated unfairly by others. Such biases notwithstanding, evolutionary theorists also expect constraints on self-serving biases in people's sense of justice to evolve for much the same reasons that evolutionary game theorists expect unconditional selfishness to be a nonoptimal strategy in repeated interactions among members of groups. Indeed, some evolutionary psychologists have found that people's sense of justice may, under some conditions, be biased in favor of others. For example, researchers have found that people may feel worse when they fail to repay their friends than they do when their friends fail to repay them.

The precursors to a sense of justice in other primates observed by investigators such as de Waal appear to be limited to positive and negative reactions to being treated fairly and unfairly by other members of their groups, and humans also display such reactions. However, in addition, humans may react strongly to the ways in which third parties are treated – a phenomenon that has been labeled "strong reciprocity." As expressed by Wilson (1993), "our sense of justice. . .involves a desire to punish wrongdoers, even when we are not the victims, and that sense is a 'spontaneous' and 'natural' sentiment" (p. 40).

Conclusion

Although it may seem on the surface that evidence of moral instincts that induce humans and other animals to behave altruistically, to uphold their groups, to respect authority, to behave fairly, and to purify themselves is inconsistent with evolutionary theory, on close examination it becomes clear that it is not. Instincts to behave in moral ways and instincts to make moral judgments have evolved because they help social animals survive, reproduce, and propagate their genes. Moral instincts induce social animals to sacrifice their short-term interests in order to achieve the long-term biological benefits they can obtain by upholding their groups and supporting those on whom their fitness is dependent. Moral instincts enable members of groups to reap the benefits of sociality and cooperation and to foster their fitness by resolving their conflicts of interest in mutually beneficial ways.

Cross-References

- ► Altruism
- ► Cooperation
- ► Group Selection
- ► Kin Selection
- ► Moral Emotions
- ► Moral Foundations
- ► Sexual Selection

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Moral Instincts and Morality

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Synonyms

Gut feelings; Moral intuitionist theory; Moral intuitions

Definition

Moral instincts, or otherwise social intuitions, are unconscious abilities that differentiate between right and wrong actions.

Introduction

Since the advent of psychology, Bargh (1989) stated that people access substantial types of knowledge without phenomenological cognizance. According to Reber (1993), awareness and phenomenological cognizance are comparatively new concepts in comparison to other functions of the brain considered to be unconscious. Lately, some scientific disciplines such as evolutionary psychology and cognitive sciences proposed intuitionist approaches to human functioning (Klein 2003).

Long ago, gut feelings were considered unreasonable. Now, however, things have changed significantly, and often reasoning is considered needless (Cummins 2003). Certain theories, like the moral intuitionist theories, emphasize people's sudden instinct reactions that they exhibit toward others. As per this view, the intuitive moral judgments are vital part of human moral behavior. These theories further vie that moral intuitive responses result from a person's inherent computational elements obtained from the evolutionary history of the human species and are shaped by the way people were raised in their culture.

Haidt (2001) proposed the "social intuitionist model" of moral functioning. The model suggests that social instincts are a fundamental and quick response that emerges to differentiate between rightness and wrongness without any cognizant of their source. Therefore, moral instincts may or may not be followed by rationale. The "social intuitionist model" was initiated from the data of specific set of scenarios that were designed to provoke repulsive behavior (Haidt et al. 1993). For instance, one scenario details consensual sexual intercourse between adult siblings who, although they both seek pleasure from their experience, decided that they will not do it again. Another scenario describes eating the family dog after it got hit by a car. A third scenario describes a man who bought chicken from the supermarket and cooked and ate it after masturbating on it. After participants were presented with these provoking, repulsive scenarios, they were asked whether the actions in the scenarios were right or wrong and indicated why. Participants were able to decide rightness or wrongness immediately. In contrast, after confronted with an opposite opinion claiming that although the scenarios seem disgusting, no harm could result from people's actions, the participants were unable to give a reason for their judgment. Haidt's "social intuitionist model" had been influential to other approaches followed afterward to support or reject the importance of instincts in moral judgments (Gigerenzer 2008).

Conclusion

Moral instincts are considered important in the study of morality in several aspects. Firstly, people can differentiate between rightness and wrongness without any cognizant of their source and therefore can shape some moral judgments. Second, moral instincts revolve around the concept of unconscious inherent processes that are necessary for human functioning, and thus these processes lead to moral actions.

Cross-References

▶ Morality

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Moral Intuitionist Theory

▶ Moral Instincts and Morality

Moral Intuitions

▶ Moral Instincts and Morality

Moral Language Regulation

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Synonyms

Altruism; Evolution of morality; Rule of law

Definition

Recursive language evolved as a precursor to human value systems.

Introduction

In *The Descent of Man*, Charles Darwin identified a moral sense as the cornerstone of what it is to be human. He suggested that, to a social species, the evolution of an ethical brain was essential. This is because no interdependent tribe could succeed if immoral acts, like murder, were to become commonplace. Although the author goes on to claim differences in morality between humans and other animals are a matter of degrees, crucially he suggests only the former can regret actions. Moreover, only they can categorize them in terms of rightness or wrongness. Methods of policing have been observed in other species such as chimpanzees. Yet it seems only humans are able to construct formal moral frameworks, like legal systems, to regulate aspects of life that are not directly linked to reproductive fitness (e.g., obscenity, blasphemy, and copyright legislation). This exception to the general rule within nature suggests that moral systems are a recent adaptation.

Another trait considered distinctly human is the capacity for language. Other animals harbor elaborate communication systems enabling them to alert peers to environmental features, and in some instances, there is even evidence of them attaching distinct sounds to particular targets. For example, vervet monkeys have been found to

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produce acoustically different alarm calls in the presence of different predators (Seafarth et al. 1980). Yet human language is thought to be remarkable because of its flexibility, i.e., the ability to combine symbols into understandable structures (Hauser et al. 2002). Humans appear to have been uniquely endowed with the recursive computational mechanisms required to create infinite combinations from a finite set of elements. It is this combinatorial property that has led to the emergence of these two capacities to be considered in parallel.

A Link Between Language and Morality

In addition to the recursive properties of language allowing humans to generate speech creatively, Poulshock (2006) reasons that they let humans moralize creatively. In as much as speakers can use terms to name an unlimited amount of concepts, the author argues they can also discuss them or catalogue them in terms of being positive or negative. People can moralize about cultural etiquette, or important contemporary concerns (e.g., current affairs or the use/abuse of modern technology), along with relatively inconsequential topics (e.g., if a comedian has gone too far). They can also moralize about events that are temporally displaced (e.g., dwelling on past events or considering future ones) or the consequences of things we may never do (e.g., "if I were to do X then Y will be the result"). What's more, the author points out that speakers can use language to meta-moralize about the nature of morality itself and whether or not it is actually moral to moralize.

Even if nonspeaking animals were to have the cognitive apparatus required for constant reflection, Poulshock argues it is unlikely they would be able to share or enforce morality. For instance, he asks, how would a clan with limited linguistic resources share nuanced ethical judgments beyond approach and avoidance behavior (e.g., violence is acceptable when used for self-defense)? In the absence of language, or paralinguistic features, the author claims it would not be possible to share these values or communicate an intricate social etiquette for others to abide by. Instead, the direct transference of beliefs and values is most easily achieved through oral or written means. Principles can then be supported by consensually agreed ethical frameworks which are shared within a given population.

Thus Poulshock hypothesizes the uniquely human property of language paved the way for the equally unique property of morality to be shared culturally or intergenerationally. Furthermore, he writes, a recursive communication system that is open to temporally and spatially displaced events can enforce moral behavior with linguistically encoded promises, threats, rewards, or punishments. The result is a socially constructed code which is capable of informing conduct, as well as providing individuals a criterion to evaluate others. On a microlevel, the ease of transmission means communities can establish their own unique information-sharing networks. Members can then be selectively raised or socialized to a specific moral tradition. The author cites the example of religious organizations promoting moral precepts through the use of language. For instance, the Deuteronomic tradition teaches followers to memorize commandments, verbally impart them to others, proselytize with nonbelievers, and record their values in the form of memes (e.g., paintings or signs). Accordingly, our language instinct has historically been treated as critical to the internalization and spread of a moral code.

Ergo Poulshock claims the evolution of language has allowed for members of a community to widen their influence to guide or manipulate the behavior of peers. Linguistically enabled morality lets individuals or groups benefit from cohesion and cooperation through cost-efficient means. It also eases the rapid communal good or bad marking of members, depending on their compliance. The second reason prompts another possible factor in the relationship between language and morality: formal or informal legal structures.

Knight (2008) studies this proposition in detail, theorizing language and the rule of law developed in tandem. Each is thought to stem from adaptations designed to enforce cooperation between strangers. So instead of viewing morality as a product of language, he contends the two capacities are synonymous in evolutionary history. To illustrate this notion, he juxtaposes the lack of shared morality in primate societies, where conflict is physically resolved, with the self-governing egalitarianism of hunter-gatherer ones. The difference, he suggests, is that the latter are not bound to the same laws of signal operation that are limited to acts of the body. Instead, the advent of language has enabled them to signal in the abstract. This means that humans can utilize institutional beliefs (i.e., protocols) so that important behaviors such as sexual activity and competition can be regulated without conflict. To paraphrase Goodall (1982), this difference may be why human societies exhibit law with order, whereas chimp societies are run by order without law. The role of language in determining these conventions empowers human societies to transcend primate politics, where the key arbiter is dominance.

Boyd and Richardson (2009) also support the coevolution of language and moral structures. They argue that humanity's linguistically determined ability to set social norms is what led to cultural evolution, resulting in the beginning of distinct populations. Within each given group, the extent that conventional behaviors are adopted is influenced by the three Rs, i.e., the impact on reputation, the likelihood of reciprocation, and the possibility of retribution, if they go ignored. In a civilization comprised of small clans, the authors deduce that group-level cooperation would predict the attainment and survival of different communities. They argue that within social systems, where morality is reinforced by punishment or reward, reproductive success is determined by the extent an individual fits in with their community. Accordingly, mating competition would favor those with phenotypes supporting pro-social vs anti-social behavior. Zlatev (2014) adds that an adaptation allowing individuals to be continuously tracked is core to this paradigm. Hence a communication system based upon displacement is highly advantageous for enabling others to compare opinions on fellow group members. Reputations can be systematically enhanced or reduced based upon the extent to which they deviate from ideals. He concludes that this led to the evolution of traits promoting pro-social behavior, like shame. This hypothesis is akin to Dunbar's (2004) gossip model, which claims one functional role of language is being able to identify individuals who reap benefits from a society they do not contribute to. In his framework, the benefits of being able to identify and share information about "free riders" are twofold. Firstly, speakers may be able to protect other group members from exploitation. Secondly, the spread of such a trait could deter those that wish to exploit them.

As well as the contrivance and enforcement of morals, Poulshock (2006) asserts another adaptive role for language is offering people a means to assess their behavior from its impact on others. Moreover, it lets group members make others privy to their own mental states. This bi-directional mind reading grants greater insight of moral relations than any closed communication system ever could. Similarly, Zlatev (2014) asserts that language and morality both emerged in tandem with capacities to share affective, perceptual, or reflective experiences, including joint attention and empathy. In addition, when met with convoluted circumstances, which can be expected in even the smallest communities, Poulshock (2006) argues oral and written exchanges support discussion and reconciliation. This is particularly common when an understanding of tensions requires third-level intentionality, i.e., knowing someone else who knows someone else who knows something (Dunbar 2004). Language can also be useful in instances where a trusted mediator can communicate an apology or peace offering on another's behalf: another scenario a closed system could not assist.

Moral Development

One way of testing a functional link between language and morality is to study the ontogeny of each trait. If a utility of language is the circulation of communal values, then both ought to emerge and develop simultaneously. However, research into this potential relationship is constrained by the need for subjects to possess the language skills necessary to describe their moral considerations. Consequently, young children are generally not suitable for testing. This means the modest literature has tended towards theoretical discussions.

Poulshock (2006) suggests parallels in acquisition are commonsense, since an appreciation of complex issues requires the vocabulary to comprehend and discuss them. He cites examples of stumped language and moral development in neglected children, implying a critical period for both that may be owed to crossover modules. Further indirect evidence of a functional link is shown by linguistic analysis from Snow (1990), who analyzed transcripts of 51 h of caregiver and child interactions. At two and a half, she found children spent the most time talking about bad behavior, whereas at three and a half, it was more good behavior. Finally, at four and a half, she noticed they became focused on "should" or "should not" discussions. Based on this, Snow describes the ontogeny of moral sense as following a negotiation of the meaning of words and acts, predicting a correlation between linguistic and moral sophistication. Likewise, Hanfling (2003) advocates a connection, asserting that as children are introduced to concepts (e.g., bullying), they are also taught about their community's judgments of them (i.e., it is wrong).

Combined, these studies present a consistent picture of language providing a means for parents or guardian to introduce and contextualize specific actions. Children's moral maturation is then, by necessity, led by language. Its recursive and displaced properties are therefore critical to reinforcing schemas in instances when a behavior does not proceed the lesson. This does not necessarily suggest infants do not possess an innate sense of wrongness. Though it does suggest that the way concepts are first introduced to them goes some way toward determining how they are later regarded.

Conclusion

Humans are not only special in their capacity for language but also in their ability to design, internalize, and enforce communal values. The key suggestion from the work cited above is that societies based on the collective codification of morality can only be achieved with a flexible means of expression. Unlike their closest evolutionary cousins, people can reason and reconcile with each other in the abstract and establish institutions plus communal values. In particular, the recursive aspect of language and the option to converse over events and people displaced in time and space are thought to play essential parts in regulating morality. A related avenue not touched here is the assumption that, in addition to facilitating moral codes, language facilitates immorality by making deception easier with the option to lie. Although as Poulshock (2006) deliberates, this does not contradict its role in enabling, extending, and maintaining morality.

Cross-References

▶ Evolution of Morality
▶ Function of Morality (Haidt)
▶ Morality

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Moral Philosophy

▶ Psychology
▶ Trolley Problem, or Would You Throw the Fat Guy Off the Bridge?

Moral Realism

▶ Objectivity

Moral Reasoning

▶ Would You Kill the Fat Man?

Moral Relationships

▶ Social Relations Theory (Fiske)

Moral Senses

▶ Moral Instincts

Moral Sentiments

▶ Moral Instincts

Moral Subjectivism

▶ Modern Moral Relativism

Moral Tribes

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Synonyms

Altruism; Cooperation; Tribalism

Definition

Moral tribes refers to a specifically value-based dimension of tribalism, which is a set of behaviors and cognitive mechanisms that lead individuals to favor themselves and the groups to which they identify. Groups that identify with a set of interests and values distinct from another set of interests and values constitute differing moral tribes.

Introduction

Evolution by Darwinian natural selection appears to be an extremely selfish process by which individuals compete against other individuals in a contest for survival and reproduction. However, Hamilton (1964) showed, and Dawkins (1976) articulated, that "selfish genes" can lead to altruistic and cooperative behaviors at the level of individuals, especially if those individuals are kin and therefore share genes with each other. When individuals are not kin, it can still ultimately favor the helper to aid strangers if it is done under the assumption that the stranger will repay the favor, or reciprocate (Trivers 1971).

Cooperating in groups was likely favored by evolution for a variety of reasons including having greater access to resources, as well as protection from and ability to fight against threats (Greene 2013; Tooby and Cosmides 2010). From an evolutionary point of view, kin selection paired with processes related to and as a result of reciprocal altruism explains a large amount of ingroup activity, including social exchange, helping behavior, and parental care. These features of human social life likely form a large part of the human moral architecture (Curry 2016). The notion that ingroup activity forms a large part of human life is also supported by the fact that humans evolved in small-scale conditions in which they interacted with a relatively small number of individuals many of whom were kin and close acquaintances (Dunbar 1992).

While reciprocal altruism is one biological mechanism that explains altruism and cooperative behaviors with people besides our kin, ingroup favoritism and cooperation between kin often limits the scope of altruism and cooperation. Greene (2013) argues that while morality evolved for cooperation, it primarily favors cooperation *within* groups and not necessarily between groups. In other words, people's morality is geared towards favoring themselves first along with loved ones and personal acquaintances. However, it becomes a stretch to expand that sphere of morality to strangers. Greene (2013) views this conundrum of human social life as analogous to the Tragedy of the Commons, where the pursuit of individual self-interest leads to a result that is collectively worse off for everyone involved. He refers to this dilemma as the Tragedy of Commonsense Morality.

The tendency for people to favor the groups to which they belong and act with antipathy and hostility towards the groups to which they do not belong is part of a larger psychological phenomenon than morality known as tribalism (Clark et al. 2019). Tribal phenomena include a number of cognitive and behavioral manifestations of ingroup–outgroup dynamics, which can be seen at every level of human social life.

Some of the tribal behaviors and cognitions people exhibit include ingroup favoritism (Balliet et al. 2014; Clark et al. 2019), discrimination and hostility towards outgroups (Balliet et al. 2014; Tajfel and Turner 1979), biased moral reasoning (Ditto et al. 2009; Clark et al. 2019), evaluating identical information more favorably when it supports one's political beliefs (Ditto et al. 2018), avoiding and being more critical of information that goes against one's beliefs (Kahan et al. 2017), ideologically motivated reasoning (Kahan 2013), and political polarization (Stroud 2010).

The Origins of Human Morality

Morality may be a feature that is unique to humans. Indeed, morality is fundamental to human life. Evidence shows that infants are capable of distinguishing good and bad behaviors within the first year of life. Six- and ten-month-old babies not only recognize helpful and hurtful actions in an experimental environment, but also have preferences for a helpful over a hurtful agent (Hamlin et al. 2007). By the second year of life, babies begin to develop the capacity to distinguish fairness (Sloane et al. 2012), prosocial sharing (Brownell et al. 2013), and empathy (Gubler and Bischof 1991).

Sharing behavior in primates exhibits rules and regularities similar to the moral concern in human behavior, to preserve a hierarchy, protect resources, and prevent cheating (Flack and De Waal 2000). After a fight occurs between two chimpanzees, they will sometimes show what looks like forgiveness and reconciliation, for example, "kissing" to reconcile and embracing for consolation (Greene 2013). However, significant differences in moral-like behavior in chimpanzees, such as the absence of anything like a moral conscience that involves internalizing group norms, and the absence of remorse for violent behavior, have prompted Boehm (2017) to make the analogy of chimpanzees to human psychopaths.

Coalitional Psychology

Similar patterns of human and chimpanzee intergroup conflict have led many researchers to study evolutionary parallels between these two species (Wrangham and Glowacki 2012). Intraspecific violence in chimpanzees and humans is often coalitionary, where members of the species band together and express intergroup hostility (Wrangham and Glowacki 2012). The capacity to coalesce into coalitions is a necessary condition for intergroup conflict and warlike behavior (Tooby and Cosmides 2010).

Coalitional ties can result in negative judgments towards outsiders, such as representing healthy outgroup members as being infectious

(Petersen 2017). Affiliation helps to explain why people of different group-orientations are differently affected by the same information. For example, differences in moral values between political affiliations affect how disturbed one is to moral violations like kicking a dog, burning a country's flag, and slapping one's father (Graham et al. 2009).

The fact that individuals have a need to belong to a group (Baumeister and Leary 1995) and affiliations are often tied to location (Balish et al. 2013) provides a plausible path for moral tribes to be constituted in sports. Team sports may provide individuals a representation of an adapted form of conflict (Sell et al. 2017). Teams seem to have a form of morality that is enhanced by different regional divides (Balish et al. 2013). The zero-sum nature of sports (i.e., winners imply losers) gives individuals the thrills of winning, but also the pains associated with defeat. Stressful matches being played have been shown to negatively affect one's health, increasing risk of strokes and cardiovascular problems (Wilbert-Lampen et al. 2008). The fact that people are physiologically connected to the outcomes of their team shows how salient an identification with a sports team can be for an individual.

Moral and Political Tribes

Curry (2016) posited a multifaceted nature of morality that corresponds to its many-solutions based nature. These multiple parts of morality evolved as game-theoretic solutions to distinct evolutionary problems of cooperation. The central problems of cooperation outlined by Curry (2016) are as follows: (1) the allocation of resources to kin; (2) coordination to mutual advantage; (3) exchange; and (4) conflict resolution. Curry et al. (2018) describe how humans have culturally and biologically evolved solutions to these moral problems.

For example, the problem of allocating resources to kin is largely solved by kin selection, which explains many instances of altruism and favoring behavior for kin (Hamilton 1964). The problem of coordinating to mutual advantage occurs in a variety of contexts that involve two or more agents with insufficient knowledge coordinating their behaviors for mutual benefit. Solutions include a host of behaviors that involve signaling, public communicative calls, focal points, among other such behaviors (Curry et al. 2018). In support of his morality-as-cooperation theory, Curry et al. (2018) found that among 60 societies seven cooperative behaviors that encompassed the cooperative domains of the theory were found to be uniformly positive.

Additionally, when it comes to politics, evidence indicates that conservatives and liberals emphasize different moral values, which constitute separate moral tribes (Graham et al. 2009). For example, liberals tend to emphasize harm and fairness concerns more than conservatives (Graham et al. 2009). Harm concerns include actions that make use of violent force against others which results in pain and suffering, while fairness concerns include actions that result in the unequal allocation and access to resources for individuals falls (Haidt and Graham 2007). On the other hand, morality concerns for conservatives are more equally distributed across the five foundations. Concern for the groups to which one belongs, tradition and authority, sacred values, and purity are more salient values for conservatives compared to liberals (Graham et al. 2009).

In addition to fundamental differences in moral values, evidence indicates that conservatives and liberals are equally politically biased and thus susceptible to ideologically skewed beliefs and tribalistic tendencies (Ditto et al. 2018). Together the findings on moral differences and political biases between liberals and conservatives help to explain why they are so often at cross-purposes; they are often not even speaking the same language (Haidt and Graham 2007).

Conclusion

For many decades, research from a variety of disciplines has converged on the finding that humans often engage in ingroup favoritism and express outgroup hostility, which constitutes the essence of tribalism. When this tribal tendency becomes moral, as it often does in intergroup conflict, politics, and when faced with cooperative problems, people are more likely to be tribal and the tribal behaviors are more likely to be severe. This is because problems that are moral are ones that humans value deeply and strongly identify with and thus are willing to sacrifice for. The last few decades have presented new and unprecedented challenges for the expression of our evolved tribal tendencies with the rise of the internet and social media. Only with a better understanding of our nature, can we better know how to meet these new challenges and successfully solve them.

Cross-References

- ▶ Cooperation
- ▶ Moral Foundations Theory
- ▶ Morality
- ▶ Tribalism

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Moral Values

► Harris' Moral Landscape
► Honor Code

Moralistic Aggression

► Robert Trivers

Moralistic Punishment

► Altruistic Punishment
► Altruistic Punishment and Strong Reciprocity

Morality

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Synonyms

Evolution; Evolutionary psychology; Moral behavior; Natural selection

Definition

Human morality constitutes the by-product of evolution as it emerges from a continuum of shared behavioral characteristics between human and nonhuman species. Natural selection has rendered moral behavior to seemingly constitute a highly developed trait expressed by human primates.

Introduction

Contrary to the belief that wants morality to encompass a trait unique to humans, it is argued that other animal species may possess moral behaviors including community concern, reciprocity, and empathy. Although it is not supported that a fully developed moral system is present in other animals, it is indicated that human morality is embedded in phylogenetic relationships with ancestral animal species. Evolutionary scientists depict the phenomenon of homology by rendering the presence of shared behaviors as the result of a common ancestor between humans and other animal species. They also support that two distantly related species can share a behavior while they inhabit similar environmental niches as it results from convergence due to shared environmental constraints. Intriguingly, through examining behavioral patterns of other species might explain mechanisms of human behavior and depict that a multitude of phenomena concerning morality are the result of continual evolution.

The Empirical and Normative Senses of Morality

Morality from a scientific perspective refers to a set of empirical phenomena, such as the observed capacity of human beings to make normative judgments and the tendency to have certain sentiments including sympathy, guilt, or blame as well as intuitions about fairness and violence. Scientists speak of the Evolution of Morality in empirical sense while utilizing tools as comparative genomics and primate studies to analyze the subject of interest. While morality in the normative sense is not an empirical phenomenon to be explained, there are vital points that require attention regarding evolutionary theory and principles (Rosenberg 2006).

The Normative Sense of Morality questions one's evolutionary background for moral guidance; through the normative approach, scientists might attempt to gain insight into the content of morality as claimed by proponents of prescriptive evolutionary ethics. Evolutionary theory can potentially explain metaethics and allow to resolve inquiries about the existence and nature of morality (Boniolo and de Anna 2006). Once topics of interest are selected, scientists begin to consider scientific explanatory projects concerning morality in the empirical sense.

Moral Behaviors in Nonhuman Species

Moral emotions have been identified as the principle mechanisms that uphold the rules of behavior underlying social norms and conventions. They constitute vital components in interactions as they normalize the behaviors of group members and foster the tendency to be in line with the norms (Haidt 2003). Equity and equality are the two primary elements that synthesize justice. Adam Smith refers to justice as "the main pillar that upholds the whole edifice of human society." This perspective lies within the rationale that wants animal species to react in the face of unfair treatment (Tinklepaugh 1928). This notion argues that based on instincts of moral behavior, primates tend to respond negatively when others are treated less well (Fletcher 2008), fact that contributes to the maintenance of harmonious coexistence among group members (Brosnan 2006).

Evolution of Justice

The sense of justice is seen in nonhuman primates as they are likely to respond negatively to earning less preferred rewards than a social partner (Frank 1988). In respect to natural selection, researchers argue that it might have been a critical environmental factor that led to selection for the behavior (Darwin 1964). It is important to underline that evolution of shared behaviors does not indicate identical approaches from species as well as it does not eliminate the novelty of behaviors that are unique to specific primates, but not others, due to their complex nature (Flack 2000). Nonetheless, inequity can be explained as the by-product of another behavior that led individuals to selectively focus on conspecifics, such as social learning, which is present in many species and can be beneficial (Tangney et al., 2007). On the other end of the spectrum, inequity superficially might not be perceived as rewarding from species due to unjust circumstances; however, reacting negatively to unfair treatment can cease harmful interactions with opponents and encourage them to seek out equitable partners (Bosnan 2006). There is also evidence that human and nonhuman species actively take steps toward rectifying inequity when the moral sense of justice is violated and they receive less than they deserve (Brosnan, 2008). In this stage, primates react by punishing others who do not play fairly (Fehr and Gachter 2002). It can be argued that reaction to disadvantageous inequity is more salient as there is immediate benefit to the partner. On the other hand, advantageous inequity leads individuals to abandon their own good and act against their short-term self-interest in order to give up rewards to benefit others. The definition in itself might seemingly appear disastrous for the individual; however, it derives long-term benefits for the community and the conservation of healthy social norms.

Studies on Moral Behavior in Primates

Naturalistic studies have shown that long-term behavior of the partner is more important than the outcome of the individual. In an experimental cooperation task, subjects were more likely to cooperate if they and their partner evenly shared the rewards. In one study, monkeys had to work together to pull a heavy tray of food. The foods differed every time, and the monkeys had to decide whether they were willing to work for the food reward they were offered. Pairs in which the partners shared the good food about half of the time were successful in the cooperative task, while pairs in which one partner eliminated the other rarely obtained cooperation (Burnham and Kurzban, 2005). Aside from the fact that non-sharing dominants received higher value rewards than their partners, their success rate was lower; thus, they tended to receive fewer rewards. The subjects seemed to expect that working together on a task leads to equity if reward outcome and dominant partners who failed to do this received a form of punishment (Brosnan et al. 2006). The findings indicate that a critical factor in a cooperative task is the partner's behavior to be equitable, rather than the actual value of reward. In experiments where participants are humans (Boesch 2007), responses to inequity differ depending on the source (Silk et al. 2005). Humans change their responses if there is a reason for inequity (Hagan and Hammerstein, 2006). Individuals offer less, and partners accept inequity more readily if the subject is perceived to have earned the right to it. Interestingly, humans tend to respond more negatively to inequity that is caused by another human directly than to inequity that is the result of chance.

Darwinism and Moral Emotions

Charles Darwin proposed that the basis of morality is innate and that humans and animals share an evolutionary background. He mentions in his book *The Descent of Man and Selection in Relation to Sex* that nonhuman primates possess social instincts, such as parental affection, and they are consequently characterized by a sense of conscience that would be otherwise intellectual power if it was well developed as it is in humans. It is argued that other animals experience some level of sympathy for others in their environment that also includes a strong need for familial interactions (Darwin 1964).

Darwin does not disregard the importance of advanced intellectual ability for meaningful judgment, but he underlines that well-marked social instincts are present in animals indicating moral constraints and conscience in species other than humans (Darwin, 1981). He supports that this notion functions as an evidence for evolutionary continuity between humans and other animals as he relied on comparative data regarding facial expressions across human cultures and other species. He addressed the reasons why universal behaviors are expressed in the form that they do (Darwin 1998).

Adam Smith and Morality

Smith appears to support Darwinian theory. In his book *The Theory of Moral Sentiments* discusses that both positive and negative passions are seen in animals and that they share many traits that are considered human. To support his views, Smith mentions that all animals, including children and dogs, are likely to become angry at inanimate objects that hurt. He highlights that human instincts were in place to promote survival and reproduction. He further believed that instincts were initially experienced as emotion, rather than rational thought, and were often lied outside conscious awareness (Smith 1817). The idea of evolutionary continuity of behavior and the recognition of moral emotions has its roots embedded in the ancient past (Ekman, 1998).

Conclusion

A better understanding of the evolution of moral behaviors can be achieved through examining outcome behaviors that allow for insight in the past. According to the notion of natural selection, salient behaviors were practiced more than others, rendering certain moral tendencies as vital for survival. Values of justice and equity appear to be among some of the most crucial components to maintain social norms and order intact in communities of human and nonhuman animals. Undoubtedly, intellectual ability and moral judgment of humans are not comparable with those of other species; nevertheless, the presence of social instincts functions as an evidence of moral conscience in other species. With respect to moral behaviors and emotions, through approaching the precursors, such as the inequity response, scientists may gain a better insight into the outcomes of evolution of moral behavior. Since clear continuity has been determined between the species, it can function as the foundation of future research on evolutionary approaches to human morality.

Cross-References

- ▶ Charitable Giving (Peter Singer)
- ▶ Evolutionary Biology and Feminism
- ▶ Evolution of Free Will
- ▶ Harris' Moral Landscape
- ▶ Human Enculturation
- ▶ Moral Instincts
- ▶ Moral Instincts and Morality
- ▶ Philosophical Foundations for a Modern Morality
- ▶ Psychology
- ▶ Science and Morality
- ▶ Wild Justice
- ▶ Would You Kill the Fat Man?

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Morality Is Cooperation

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Synonyms

Interdependence; Joint intentionality; Mutualism

Definition

Morality evolved as a mechanism to enhance cooperative behaviors and discourage cheating.

Introduction

Modern humans live in highly collaborative social groups. Although it has been theorized that human sociality only begins to develop as infants internalize symbols of the cultural world around them (Vygotsky 1978), recent research instead suggests that sociality may begin to form before infants are even able to take part in this cultural environment. Research by Michael Tomasello and Amrisha Vaish (2013) has shown that very young infants can already perform cognitive functions that other apes cannot, including cooperative communication. That infants who have yet to embrace any culture or language have performed with higher levels of cognition than adult apes supports theories that human sociality is not embedded entirely in the cultural world, but instead has its roots in the unique human evolutionary history. Additionally, as humans evolved into the ultra-cooperative beings seen today, the competition to be chosen as a partner may have contributed to an environment that became inundated by moral rules and expectations. Ultimately, there is much evidence to suggest that morality exists in humans as a mechanism to heighten cooperative actions and control cheating within groups.

The Interdependence Hypothesis

It is likely that there was a period of progression between the relatively low levels of cooperation of our distant evolutionary ancestors to the high levels that exist in today's human societies. Tomasello et al. (2012) present a theory to describe how humans came to be highly moral and cooperative individuals. They hypothesize that during human evolutionary history, humans began collaborating with others in order to survive and reproduce. Once individuals became interdependent, they also became altruistic, as their success was now contingent upon the success of their partner. This is called the *Interdependence Hypothesis*, and it involves two crucial steps. In the first step, there was a change in the ecology that required humans to search for their food collaboratively. This forced individuals to (a) make the decision to forage together even when it meant giving up the smaller food they already had, (b) solve the problem of dividing the food after the collaboration was over, and (c) control any potential cheating from individuals wanting the food without participating in the search. Individuals had to communicate their goals during their mutualistic collaboration, which enabled them to create new, joint goals. The joint intentionality present between collaborators forced each member to have an interest in the welfare and success of their partner, as their partner's success was now directly linked to their own. Tomasello and Vaish (2013) further argue that it was this interdependence that originally encouraged helping behaviors. Moreover, the competitive partner choice market would have stimulated cooperation and discouraged cheating, as reputation was undoubtedly an important aspect for being chosen as a partner.

The second step of the Interdependence Hypothesis occurred once group populations grew even larger and they began to compete with other groups (Tomasello et al. 2012). These pressures presented new problems for humans that required a much more dynamic social climate. The increasing group size led to difficulties learning about an individual's reputation, which then led to a decrease in many people's motivation to cooperate. This degradation of cooperation encouraged new group-wide goals and social norms to once again boost collaboration, which in turn created highly enforced mutual expectations for behavior that remained consistent throughout a group. Everyone within the group now had their own role and again depended on other members for their survival.

Morality Improves Cooperation

According to Tomasello and Vaish (2013), children will become moral beings once they can effectively collaborate with other individuals and fully join in the social norms and practices of their culture. Not only do individuals follow the norms of their culture; they also enforce them on others – and themselves – if rules happen to be broken. This leads to a collective morality within the group that individuals will work in accordance

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with. In order for a behavior to be considered moral, it requires an individual to either suppress their own well-being for the sake of someone else's or adopt another person's self-interest for their own as well.

According to Baumard et al. (2013), morality is an outcome of the cooperative actions within groups and exists to fairly allocate the benefits of cooperation. They propose a two-step hypothesis to suggest how it has evolved. In step one of the evolution of morality, partner choice would always favor those who were fair cooperators during collaborative activities, even if their motivation for cooperating was selfish. Second, competition among the ultra-cooperative individuals to be chosen as a partner created a further requirement: an individual motivation for fairness. This social selection for fairness contributed to the expectation of highly moral human behavior seen today. They suggest moral behavior exists to benefit one's reputation as a good cooperator, as this can only be done by an individual with a genuine moral sense. The researchers argue morality was adaptive when competition for cooperative partners was extreme, and the most effective way to be chosen was to be fair and equitable with the profits of cooperation.

Conclusion

The studies discussed here have suggested that morality evolved alongside cooperation to encourage cooperators and discourage cheaters. Many other researchers have also suggested the root of morality lies in the evolution of cooperation, with, for example, Joshua Greene (2015) arguing that the function of morality is to support cooperation. Similarly, Tage Shakti Rai and Alan Page Fiske (2011) describe morality by its ability to create and maintain long-term cooperative relationships. Frans de Waal (2006) has emphasized that just because natural selection is a ruthless progression through time does not mean it cannot also proliferate pleasant behaviors like fairness, cooperation, and morality. Natural selection simply supports organisms that successfully reproduce and ignores the mechanisms that make them successful. Additionally, although the process of natural selection did not classify what our particular morals and norms would be, it did set the groundwork for human kindness and allowed morality to flourish.

Cross-References

▶ Cooperation
▶ Evolution of Cooperation
▶ Evolution of Morality
▶ Michael Tomasello
▶ Reputation

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Morality of Creating New Life

▶ Responsibility for Child Suffering

Moralizing Gods and the Rise of Civilization

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Synonyms

Religion and development of civilization

Definition

Gods and their moralizing values, especially the powers regarding punishment, are related with how civilizations rose and developed into large societies.

The emergence of human civilization is related to the power of large-scale cooperation. Although cooperative behavior is seen in most mammals, large-scale cooperation extends beyond genetically related relatives. This type of cooperation, sometimes called ultrasociality, can be seen even in anonymous and one-time interactions but, without the reputational concern of any kind of large-scale cooperation, cannot be explained by standard evolutionary mechanisms. One of the hypotheses to solve this problem is the supernatural punishment hypothesis (Johnson and Krüger 2004). According to this hypothesis, the human surveillance mechanism does not observe and monitor every situation. However, the possibility of punishment by a supernatural agent eliminates the limitations of punishment for a person who violates moral rules. In other words, the mechanism of fear of supernatural punishment as an evolutionarily stable strategy explains why religions, and especially of religions with a moralizing high gods, spread (Johnson 2016). It is therefore thought that religious belief and especially the belief in fear of supernatural punishment is an evolutionary adaptation to eliminate the problem of cooperation between large groups (Johnson and Krüger 2004; Johnson 2016). In this perspective, the existence of punishment by an omniscient and omnipotent supernatural agent eliminates the possibility of being not able to observe a moral violator among the communities of human beings with certainty in punishing a person who will make a moral violation. It is thought that the emergence of the idea of supernatural punishment is what underlies the ultrasociality commonly seen in human societies (Purzycki et al. 2016; see also Norenzayan 2013).

In other words, it is thought that there is a relationship between the emergence of moralizing religions and cooperation. For example, the emergence of big moral religions is believed to force people to move together around the temple and thus lead to the agricultural revolution (Norenzayan 2013). The fact that the temples in Göbekli Tepe are dated to a time before the agricultural revolution supports this view (Schmidt 2000).

In addition, different findings using different methods seem to support the supernatural punishment hypothesis. For example, people who believe that God has a punitive character do less cheating behavior on a subsequent task, while participants who believe in God's forgiving aspects show significantly more cheating behavior in the task (Shariff and Norenzayan 2011). In a cross-cultural analysis of national crime rates, Shariff and Rhemtulla (2012) found that there is a negative correlation between the degree of believing in hell and the national crime rates, but there is a positive correlation with the degree of belief in heaven. In a study of 186 different cultures, Johnson (2009) found a positive relationship between the belief in a punitive god and cooperation. Moreover, in a study conducted in eight different cultures, it was observed that a factor that increased the level of helping toward distant coreligionists is the belief that God has a punitive character (Purzycki et al. 2016). Yilmaz and Bahçekapili (2016) also experimentally

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showed that activating the punitive aspects of religion caused a significant increase in prosocial intentions. More interestingly, in a way similar to the punitive effect of religion, activating the punitive aspects of secular institutions has similarly led to an increase in prosocial intentions in the same study. This shows that, in addition to religious authorities, punitive secular authorities have similar power in cooperation in contemporary civilization. Likewise, in the study of Shariff and Norenzayan (2007), activating religious or secular institutions had a very similar effect on prosocial behavior.

As a result, these findings support the argument that societies that believe in punitive and omnipotent supernatural powers are more cooperative, and less selfish, which in turn expand and create our large-scale civilization (see Norenzayan 2013).

Cross-References

- ▶ Altruistic Punishment and Strong Reciprocity
- ▶ Evolution of Morality
- ▶ Religious Beliefs

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Morals

▶ Evolution of Morality

Morbid Jealousy

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Synonyms

Pathological jealousy

Definition

It is an intense form of jealousy, mostly involving the intimate interpersonal relationship, and it produces significant distress and impairment.

Introduction

Jealousy is a complex human emotion. The Cambridge dictionary defines "jealousy" as "a feeling of unhappiness and anger because someone has something or someone that you want" (Dictionary 2015). Jealousy is often described as a negative emotional state. Jealousy as an emotion has a close link with anger and sadness (unhappiness). Jealousy often results in unhappiness or anger. Jealous feelings are often discussed in the context of ownership (power, possession), performance, sexual activity, and love (Bhugra 1993; Zandbergen and Brown 2015). Jealousy is also strongly linked to the emotion "love." Love gets colored with jealousy, when it demands the "exclusive, unconditional, continued and total possession of the romantic partner" (Maggini et al. 2006).

In the history of literature, the word "jealousy" was frequently used and denoted to a negative feeling state. Interestingly, it had caught the attention of poets and artists to be depicted in popular literary creations (Shepherd 1961). In literature, the words "jealousy" and "envy" are interchangeably used. A finer difference exists between these two words. Envy refers to the feeling that an individual experiences when the individual wants something that he/she does not possess. Jealousy refers to the feeling state that an individual experiences in response to a threat of impending loss of something that the individual possesses (Bhugra 1993).

Sometimes, jealous feelings become very intense, which lead to significant interpersonal impairment and produces clinically obvious distress. It is often referred as "pathological jealousy," "delusional jealousy," "erotic jealousy," or "morbid jealousy" (Miller et al. 2010). Morbid jealousy is often discussed in the context of spousal relationship. Morbid jealousy often results in irrational thoughts and behavior, which are extreme and unacceptable (Kingham and Gordon 2004). It is important to differentiate the feeling of jealousy experienced by normal people from the individuals who experience morbid jealousy. People often experience jealousy after a firm or robust evidence of unfaithfulness, and often they have the readiness to accept alternative thoughts; however, those with morbid jealousy often harbor their belief without robust evidence and offer strong resistance to accept alternative possibilities (Kingham and Gordon 2004). Certain predisposing factors have been identified for morbid jealousy in spousal relationships, which are attractiveness of the partner, partner working at a distant place from home, and partner having interaction with opposite gender (Singh et al. 2017). Morbid jealousy is broadly classified into two subtypes (Mullen et al. 1990):

1. *Symptomatic jealousy*: Here, jealousy evolves in the background of a psychiatric disorder. As the underlying psychiatric disorder progresses, jealousy evolves concurrently. For example, delusion of infidelity in paranoid schizophrenia or persistent delusional disorder.
2. *Reactive jealousy*: Here, jealousy develops as a reaction to underlying mental state (low self-esteem), psychosocial circumstances (e.g., relational difficulties with partner), or mental illness (e.g., depression). Reactive jealousy is considered as an independent psychopathology, different from the comorbid psychiatric illness or condition that provoked its development.

The Othello Syndrome

The Othello syndrome refers to "delusional jealousy" (Leong et al. 1994). Patients with Othello syndrome often display outbursts of anger and hostility. Their anger may result in verbal threats and physical fights to homicidal behavior in extreme conditions (Leong et al. 1994). Othello syndrome has been reported in the context of schizophrenia and related psychotic disorders, organic mental disorders, personality disorder, substance use disorder, and even some neurotic conditions (Alarcon 1980). The prevalence is almost equal in both genders, but higher prevalence is reported among old people (Seeman 2016).

The etiopathogenesis of Othello syndrome can be linked to certain biological (e.g., neurological conditions like dementia, organic brain syndrome, drug induced), psychological (e.g., paranoid personality, borderline personality, low self-esteem), as well as social (e.g., excessive dependency on the partner, rejection, or altercation with the partner) factors (Seeman 2016).

The core symptom of Othello syndrome is "delusion of jealousy" or "delusion of infidelity"

(Alarcon 1980; Cipriani et al. 2012). When it is developing in the context of dementia, cognitive deficits, hallucinations, and other delusions may accompany (Cipriani et al. 2012). Violent behavior in dementia can be secondary to the delusion of jealousy. Similarly, patients with Parkinson's disease on antiparkinsonian treatment may report about symptoms of Othello syndrome (frequently paranoid and sexual delusions), which often improves with reduction of dopamine agonist agents (Kataoka and Sugie 2018). In the background of obsessive-compulsive disorder, pathological jealousy related to the fidelity of spouse or partner has an obsessional quality (recurrent, repetitive, intrusive, unpleasurable, and irrational). In the context of schizophrenia and related psychotic disorders, morbid jealousy-related spousal relationship is often firmly held and cannot be convinced by contrary evidences. It is often accompanied with keeping a close watch on the partner and partners belongings (Singh et al. 2017).

As the patient may pose threat to self, partner as well as to the rival during the course of the disorder, there is a urgent need of treatment for prevention of untoward consequences (Seeman 2016). Treatment for Othello syndrome can be done pharmacologically (by using antipsychotic medications) or through psychological interventions (e.g., cognitive therapy, couple therapy) (Seeman 2016).

Morbid Jealousy and Alcoholism

Morbid jealousy is often discussed in the context of alcoholism. The prevalence of morbid jealousy among individuals with alcohol use disorder is found to be as common as 34% (Michael et al. 1995). Morbid jealousy in alcoholics can be reported during the state of intoxication, or it can be seen as a comorbid delusion (Michael et al. 1995). Morbid jealousy related to sexual relationships and intimacy is more commonly seen in individuals with severe form of alcohol use disorder (Shrestha et al. 1985). Violence particularly involving the domestic setting and spouse is commonly associated with alcoholism. Morbid jealousy involving the sexual relationship can be one of the major contributors to alcohol-related violence (Coid 1982).

Morbid Jealousy: Social and Evolutionary Significances

Jealousy has many evolutionary advantages. In the process evolution, there occurs competition in the environment. Those who are able to compete better survive, and those who fail to compete become extinct. Jealousy is one of the determinants of competition. Jealousy related to achievement, possession, power, and procreation (love and sex) acts as a triggering factor for competition. Jealousy as a "mate-guarding" phenomenon is also seen in other primates. Jealousy is capable of producing social pain in a partner, when the other partner is found to be engaged with a new partner or stranger as evident from the study on monkeys (Maninger et al. 2017).

As per the Darwinian hypothesis, jealousy has fitness advantages for both man and woman from evolutionary point of view. Jealousy may be a determining factor for mating (Harris 2004). Jealousy carries different evolutionary significance for males and females. Jealousy in males is mostly related to the risk of sexual contact of their partner with other males (rivals), which reduces their paternity probability, whereas in females, jealousy is not due to reduced maternity probability as females retain their maternity probability, irrespective of their partner contact. The jealousy in females is due to the risk/fear of digression of their committed sexual partner to other female partners for sexual contact (Buss et al. 1992).

Jealousy in sexual relationships may be advantageous as it facilitates the propagation of own genes (Singh et al. 2017). Jealousy may also stabilize the romantic relationship; however morbid jealousy destabilizes the romantic relationship (Sun et al. 2016).

Morbid jealousy may have evolutionary significance. It has been seen that greater percentage of males with morbid jealousy often doubt the sexual fidelity of their partner, whereas greater percentage of females with morbid jealousy doubt about the emotional fidelity of their partner (Easton et al. 2007). Males and females with morbid jealousy are often preoccupied with the status and attractiveness of their rivals, respectively (Easton et al. 2007).

Evidence supports that morbid jealousy or pathological jealousy falls in the same spectrum of pathological love; however, there are subtle differences in their love style and attachment style. Avoidant attachment is characteristic feature of morbid jealousy, whereas secure attachment is characteristic of pathological love (Stravogiannis et al. 2018).

When compared with healthy individuals, individuals with morbid jealousy are often high in impulsivity, trait anxiety, novelty seeking, and harm avoidance. They have low cooperativeness and self-directedness and poor social adjustment (Costa et al. 2015).

Morbid Jealousy: Neurobiological Implications

Morbid jealousy is seen commonly in the context of organic mental disorders. Evidence suggests that in approximately 30% cases, a neurobiological attribute can be traced for the Othello syndrome (Cipriani et al. 2012). Frontal lobe dysfunction is possibly implicated for morbid jealousy. Dopamine excess has been implicated in the development of morbid jealousy of Othello syndrome. Patients with Parkinson's disease with Othello syndrome reported reduction to resolution of their delusions of jealousy after decreasing the dose of dopaminergic agonists (Kataoka and Sugie 2018). Evidence supports that morbid jealousy (delusion of jealousy) develops after cerebrovascular accidents (stroke), head injury, and brain tumors (meningioma) (Kuruppuarachchi and Seneviratne 2011). Neuroimaging studies found that when people are exposed to scenarios of romantic jealousy, there occurs activation of the basal ganglia. Similarly, the intensity of morbid jealousy related to romantic relationship is correlated with the activation of ventromedial prefrontal cortex (VMPFC) (Sun et al. 2016). Studies among monkeys revealed that feeling of jealousy produces alteration of glucose metabolism in specific brain areas. During jealous state, there occurred increased uptake of fluorodeoxyglucose (FDG) in the left anterior and posterior cingulate cortex, whereas FDG uptake is lowered in the right medial amygdala (Maninger et al. 2017).

Conclusion

Jealousy is an essential emotion and does have a role in the survival of an individual as well as a race. Morbid jealousy should be always differentiated from non-pathological jealousy. Morbid jealousy can be a core feature of various psychiatric disorders and is treatable. One needs to be familiar with the antecedents and consequences of morbid jealousy for its holistic management.

Cross-References

- ▶ Delusional Jealousy
- ▶ Envy
- ▶ Jealousy: Psychiatric Diagnosis
- ▶ Othello Syndrome

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Morning Sickness

▶ Pregnancy Sickness

Mortality

▶ Death
▶ Medical Definitions of Death
▶ Survival and Death

Mortality Rates

▶ Calculating Starvation Risk

Mortality Risk

▶ Extrinsic Mortality

Most Perpetrators Are Men

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Synonyms

Male intrasexual competition; Male intrasexual homicide

Definition

Homicides that occur are overwhelmingly committed by males.

Introduction

Across cultures, homicides are overwhelmingly perpetrated by males. The UN Office on Drugs and Crime (2013) reports that 80–97% of homicide perpetrators are men. Men kill other unrelated men in cases of conflict over social status, face, pride, and reputation, as well as conflict over material resources. Men also often kill women as the result of sexual jealousy. It is very rare for women to kill other individuals. This, however, does not mean that women's level of competition with other individuals is trivial, rather women often use indirect tactics of competition, such as verbal assault or manipulation of other individuals.

Men as the Perpetrators of Homicide

From an evolutionary perspective, men are more likely to be the perpetrators of homicide because there are greater incentives for them to do so. Men's reproductive success is limited by their ability to gain access to mating opportunities. As a result, sexual selection theory predicts that, to avoid the possibility of reproductive oblivion, men must compete with other men for access to women (Trivers 1972). Competition over direct access to women, as well as competition over resources that will indirectly increase men's mating prospects, may turn fatal.

The homicide rate for male perpetrators increases sharply during the late-teens, peaks during the early 20s, and then rapidly decreases afterwards. A similar peak is present for women, but the magnitude is much lower than that of men. Moreover, across life span, the homicide rate for men exceeds that for women (Daly and Wilson 1988). The evolutionary explanation for this age-homicide curve is that this peak age of homicide corresponds with the life-stage where mating goals loom large, and therefore, competitiveness and the reputation for a readiness to fight is most crucial. Men of this age-category are much more willing to engage in a variety of risky behaviors, such as dangerous driving, sports participation, and excessive drinking, than men in other age-categories. This willingness to accept a high degree of risk likely contributes to the tendency for conflicts between men to escalate quickly to violence and, sometimes, homicide.

Independent of the age effect, male homicide perpetrators tend to be unemployed, poor, and of low education as compared to the general population (Daly and Wilson 1988; Hiraiwa-Hasegawa 2004). These individuals are likely to have difficulty attracting mates using other means and may therefore resort to physical violence to deter competitors. Along similar lines, having a family appears to have an inhibitory effect on men's violence (Daly and Wilson 1988). Although this effect has not yet been demonstrated clearly, it is consistent with the evolutionary perspective – that men who have access to mating opportunities already have less incentive to engage in risky, violent behavior in order to gain additional opportunities.

When men kill women, the most frequent motive is sexual jealousy. Men facing a situation in which their romantic partner leaves them, even when there is no new partner to compete with, may result in a violent, and even deadly, response. Sexual jealousy like this is thought to be an adaptive psychological mechanism for mate-guarding and paternity assurance (Daly et al. 1982). Women usually do not kill their male partners in the same situation, because female-female competition for acquiring mates is weaker among females than among males, and the costs of violence are greater in females than in males.

Women are generally unlikely to be the perpetrators of homicide, given their large investment in offspring. Indeed, women are essential to the survival of their offspring (Sear and Mace 2008). Thus, engaging in high-risk violent competition poses a serious threat to their reproductive success. Consequently, women adopt more subtle, indirect tactics such as gossip and social exclusion, thereby attacking the competitors without the danger of physical violence.

Conclusion

Across cultures, most perpetrators of homicide are men. Evolutionary psychology can explain this phenomenon from the theory of sexual selection. Men's reproductive success is limited by their ability to gain access to mating opportunities. As a result, men must compete with other men for access to women. Competition over direct access to women, as well as competition over resources that will indirectly increase men's mating prospects, may turn fatal. Men also more likely to kill women for the purpose of mate-guarding than women in the same situation. Women are unlikely to be the perpetrators of homicide, given their large investment in offspring. This does not mean that women to not compete each other, but women adopt more subtle, indirect tactics, thereby attacking the competitors without the danger of physical violence. The homicide rate for male perpetrators increases sharply during the late-teens, peaks during the early 20s, and then rapidly decreases afterwards. This is also explained by the sexual selection theory: that this peak age of homicide corresponds with the life-stage where mating goals loom large, and therefore, competitiveness and the reputation for a readiness to fight is most crucial.

Cross-References

- ▶ Contexts for Men's Aggression Against Men
- ▶ Contexts for Women's Aggression Against Women
- ▶ Meta-analysis of Sex Differences in Aggression
- ▶ Sex Differences in Same-Sex Aggression

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Most Victims Are Men

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Synonyms

Male homicide victims; Male intrasexual competition; Male intrasexual homicide

Definitions

Homicide that occur overwhelmingly victimize men.

Introduction

Across cultures, when homicides are committed, they overwhelmingly victimize men. The UN Office on Drugs and Crime (2013) reports that

95% of homicide victims are men. Indeed, most homicides are both perpetrated by, and targeted at, men.

Men as the Victims of Homicide

Statics in modern, developed countries show that the homicide victims, as well as the perpetrators, are predominantly men. For instance, the US Department of Justice (2010) announces that murder victims in the United States in 2010 consists of 10,058 men, 2918 women, and 20 cases where the sex was unknown: almost 78% are men. The Office for National Statistics in the United Kingdom (2015) reveals that, in 2015, the homicide victim rate for men was 11.7 per million whereas it was 6.4 per million among women. The same pattern can be found in the WHO's mortality statistics (2016), showing a rate of male deaths by assault that exceeds those of females for nearly all countries reported. The US data is broken down by each age category and demonstrate that across all age categories, with the exception of infant under 1 year of age, males are more likely than females to be the victim of homicide.

An evolutionary explanation of this pattern of findings is linked to the tendency for males to be the primary perpetrator of homicides. Men are more likely to be the perpetrators of homicide because there are greater incentives for them to do so. Men's reproductive success is limited by their ability to gain access to mating opportunities. As a result, to avoid the possibility of reproductive oblivion, men must compete with other men for access to women (Trivers 1972). Competition over direct access to women, as well as competition over resources that will indirectly increase men's mating prospects, can turn fatal. This violence is directed against men, not women, because other men are the obstacle to men's attempts to gain mating opportunities. The violence is not typically directed against women, as they are the desired resource to whom men are competing.

On the other hand, men's attempts to prevent their mating partners from leaving the relationship, or suspected infidelity, may turn deadly. However, this type of homicide is relatively rare in comparison to the proportions of homicides that are male on male.

Conclusion

Across cultures, when homicides are committed, they overwhelmingly victimize men. Statistics in modern, developed countries clearly demonstrate this pattern; the proportion of men in all the homicide victims usually amounts to more than two thirds. This pattern is linked to the tendency for males to be the primary perpetrator of homicides. Men's reproductive success is limited by their ability to gain access to mating opportunities. Competition over direct access to women, as well as competition over resources that will indirectly increase men's mating prospects, can turn fatal.

Cross-References

▶ Contexts for Men's Aggression Against Men
▶ Contexts for Men's Aggression Against Women
▶ Contexts for Women's Aggression Against Women
▶ Meta-Analysis of Sex Differences in Aggression
▶ Sex Differences in Same-Sex Aggression
▶ Violence to Control Women's Sexuality

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Mother Hypothesis

▶ Grandmother Hypothesis, The

M

Mother Tongue Hypothesis

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Synonyms

Proto-language; Proto-Sapiens; Proto-World

Definition

Either a person's native language or a hypothesis about the original human language.

Introduction

The phrase "mother tongue" is often used to indicate someone's native language, as in "what is your mother tongue?" But in linguistics and anthropology "mother tongue" is a hypothesis about the origins of human language. In this context "mother tongue" refers either to the original or first human language or to the most recent common ancestor of all extant or currently existing human languages.

Monogenesis of Language

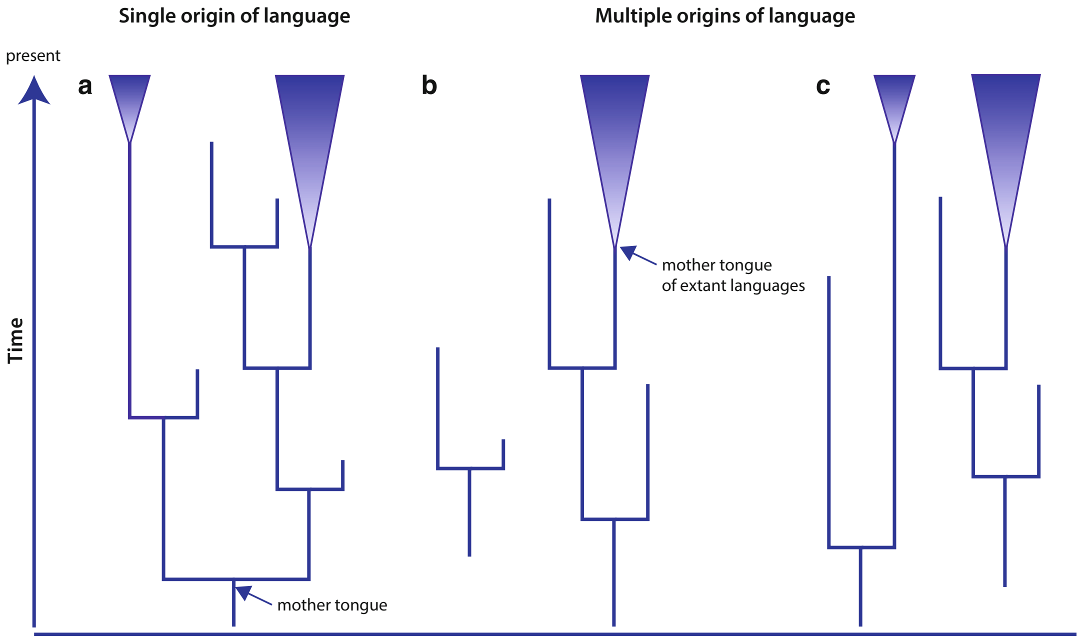
Many scenarios for the origin of human language are possible (Fig. 1). Here "origin of language" can be understood to mean not just people starting to talk, but to the origins of the apparatus – the cognitive, neurological, and anatomical specializations – that make language possible and the genetic architecture that gives rise to them. Some theoreticians believe that this language faculty arose as a specific adaptation for language (Pinker and Jackendoff 2005), while others argue that it evolved for more general cognitive reasons and was only later co-opted specifically for language (Hauser et al. 2002; Christiansen and Chater 2008).

Figure 1a depicts a scenario known as "monogenesis" in which the capacity for language evolved only once, and all modern languages descend from a single original or protolanguage (e.g., Bickerton 1990). Spoken language itself might descend from earlier "gestural" forms of communication (Corballis 2003). Early on in the evolution of spoken language, many private "protolanguages" might have existed within family units (Fitch 2004), but the evolutionary advantages of language meant that it eventually became a public and shared "technology."

In contrast to monogenesis, there might have been more than one origin of the capacity for language (Fig. 1b, c), and these separate capacities need not have evolved at the same time. In the case of more than one origin, all modern languages might descend from one of these founding events (Fig. 1b) or from more than one (Fig. 1c). Even if the modern languages descend from a single origin, they might trace their ancestry to a more recent common ancestor (Fig. 1b).

Proponents of "polygenesis" believe that the capacity for language arose early in the evolution of our species but that languages only emerged later and independently in various groups around the world (Coupé and Hombert 2005). "Polygenesis" should not be confused with the scenarios expressed in Fig. 1b and c which show separate origins of the capacity for language. Instead, here there is a single origin of the capacity for language, but specific social, environmental, or demographic factors are thought to have encouraged the formation and use of language and at different times in different places. This view of polygenesis has much in common with the argument that the language capacity emerged out of more general-purpose cognitive mechanisms.

Is it reasonable to think there was a single origin of the capacity for language (Fig. 1a)? All human groups that exist today have language, and they share a key gene known as *FOXP2* (Enard et al. 2002). This gene is thought to be involved in language and speech production and influences the fine motor control of facial muscles that are required for the production of speech (Enard et al. 2002). Genetic studies further show that all



Mother Tongue Hypothesis, Fig. 1 The diagrams show two scenarios for the origin and evolution of human languages. Panel (**a**) A single origin of language occurred in the remote past and all human languages throughout our species' 160,000–200,000 year history descended from that original or founding "mother tongue." Branches that end in the present represent extant or currently existing human languages. Blue triangles represent collections of languages with a common ancestor. Branches that end before the present represent extinct languages (such as Latin). Panel (**b**) The human language capacity arose more than once in separate populations, and not necessarily at the same time, but today all human languages trace their ancestry to just one of these origins. Panel (**c**) More than one origin of language and currently extant human languages trace their ancestry to these different origins. Other scenarios are possible

human groups have shared ancestry (e.g., Rosenberg et al. 2002).

In addition, all human languages rely on combining sounds to make words; many of those sounds are common across languages; different languages seem to structure the world semantically in similar ways (Youn et al. 2016); all human languages recognize the past, present, and future; and all human languages structure words into sentences with subjects (S), verbs (V), and objects (O) as in "I (S) kicked (V) the ball (O)" (Greenberg 1963). Finally, all humans are capable of learning and speaking each other's languages.

The similarities among languages and our apparently shared genetic influences on language in the form of *FOXP2* are consistent with the capacity for language having evolved just once, or if this capacity evolved more than once, all modern languages trace their ancestry back to just one of these events. Genetic evidence points to all modern humans descending from a common ancestor that existed between 125,000–150,000 years ago (Poznick et al. 2013), suggesting that the capacity for language is at least that old.

On the other hand, our species is probably older than this common ancestor (analogous to the scenario for the mother tongue depicted in the right side of Fig. 1b), with fossil evidence suggesting modern humans evolved 160,000–200,000 years ago (Fleagle et al. 2008). Language transmits information using symbols, so it is reasonable to expect that any species that possessed language would show evidence of symbolic behavior. The appearance of symbolic behavior in the form of the use of red ochre pigments (Henshilwood et al. 2002; Henshilwood and Dubreuil 2009) might, then, indicate that the capacity for language was present by 160,000–200,000 years ago.

Is Language Older than Our Species?

Could language be even older than our species? Might, for example, our genetic cousins the Neanderthals (*Homo sapiens neanderthalensis*) have had language? No one can know because this species went extinct by around 30,000–40,000 years ago. But if Neanderthals could speak, then the capacity for language might already have been present at the time of our common ancestor with them, now thought to be 550,000–765,000 years ago (Prüfer et al. 2014; Meyer et al. 2016).

The *FOXP2* gene in humans and Neanderthals is identical in its coding regions while differing at two key positions from the chimpanzee version of this same gene (Krause et al. 2007). The identity between humans and Neanderthals might suggest Neanderthals could speak. On the other hand, *FOXP2* is not "the language gene" as many genes will be involved in building a capacity for language, and very recent genetic evidence shows that the human and Neanderthal versions of *FOXP2* are expressed or used differently in our two brains (Maricic et al. 2013).

There is also scant evidence that Neanderthals had art or other symbolic behavior (Klein 1999); suggestions that they might have buried their dead or put flowers in the graves divide opinion and do not alone constitute evidence for symbolic thinking anyway. By comparison to Neanderthals, *Homo sapiens* who lived in Western Europe at the same time as Neanderthals had abundant and sophisticated art, they made musical instruments and had a far wider variety of tools and weapons.

We cannot be sure that Neanderthals lacked language, but it would be curious for a species that did have language to show so little evidence of putting it to use. By comparison, modern humans' cultural and symbolic variety and the evidence of innovation and cooperation are just what we expect of a species that possessed a powerful medium for transmitting information.

A More Recent Origin of Language

Some authors suggest that the human language capacity might have arisen a mere 40,000–50,000 years ago in the modern human populations which had occupied Western Europe by that time (e.g., Klein 1999). The archeological record shows that this was a time of a great flowering of art, symbolism, and technology. Klein (1999) suggests this outpouring of culture indicates that an important genetic development occurred which gave rise to the cognitive abilities that language requires.

The suggestion of a late genetic change making language possible confronts several difficulties. One is that at least some modern human groups had probably left Africa by 120,000 years ago, making it all the way to modern-day China (Wu et al. 2015). Migrating this far would have required adapting to new environments, climates, and sources of food while outcompeting the species already there. Our ability to do this by 120,000 years ago may attest to an already fully developed system of transmitting culturally acquired knowledge, and the most likely medium of this transmission would have been language (Pagel and Mace 2004; Pagel 2012).

The suggestion of a recent emergence of language also struggles to explain the presence of similarly structured languages in human groups all over the world. Thus, for example, the San Bushmen of Southern Africa have the same linguistic capabilities as any other modern human group, but their lineage might have separated from most of the rest of humanity perhaps 100,000 years ago (Knight et al. 2003). For the language faculty to have evolved just 50,000 years ago would require that, somehow, the early Europeans of that time made their way back to Africa, and perhaps interbred with other groups, thereby giving them the capacity for language. There is no genetic or archeological evidence in support of this scenario.

A variant of Klein's (1999) suggestion avoids the difficulties of proposing a late genetic change, by adopting the perspective of those who support polygenesis, as described above (e.g., Coupé and Hombert 2005). But even polygenesis runs into difficulties when trying to explain humans' early symbolic behaviors (Henshilwood et al. 2002; Henshilwood and Dubreuil 2009) and apparent technological abilities. These capabilities, expressed early in our history as a species, evidently allowed us to transmit information culturally and adapt to new environments well enough to have occupied East Asia by 120,000 years ago (Wu et al. 2015).

What Was the Mother Tongue Like?

Writing is only around 5000 years old so archeology, beyond identifying symbolic behaviors, is not of much use in identifying traces of a mother tongue or suggesting what it might have been like.

Linguists, on the other hand, can look for words that are shared among enough human languages to suggest that they may be echoes of a shared mother tongue. Statistical investigations of the rate over long periods of time at which words get replaced by new unrelated words lend credibility to this pursuit (Pagel et al. 2007). In particular, some numeral words and pronouns (I, you, who, two, three, five) and the "wh" words or *who*, *what*, *where*, *why*, and *when* tend to evolve slowly enough that it might be possible to trace their ancestry back 10,000 years or more (Pagel et al. 2007).

Evidence that words can retain traces of their ancestry even beyond 10,000 years comes from studies of shared words among language families of Eurasia (Pagel et al. 2013) that probably shared a common ancestor around 15,000 years (Pagel et al. 2013). Two languages that have been evolving separately since a common ancestor 15,000 years before represent about 30,000 years of linguistic evolution over which some traces of similarity in a small number of words have been found (Pagel et al. 2013). Among these words are thou (you), I, we, who, and what.

Still, even 30,000 years of independent evolution doesn't take us back to a common ancestor that might have existed 100,000–200,000 years ago. Historical linguists adopting an approach called mass comparison attempt to identify shared combinations of sounds in widely geographically separated languages, in hopes that these sounds might be echoes from this more distant past. The work is contentious and controversial among linguists (e.g., Campbell and Poser 2008) but one of its chief exponents, Merritt Ruhlen, working with another linguist, John Bengtson, has proposed 27 "global etymologies" (Bengtson and Ruhlen 1994).

The list of global etymologies includes words for *who*, *what*, *two*, *water*, and *finger* or *one*. For instance, Ruhlen points out that the sounds tok, tik, dik, or tak surface repeatedly in the world's languages as a word for the number one or toe or a digit. Other global etymologies include "?ak'wa" (water), "pal" for "two," "half" for "portion," "mi (n)" for "what," and "ku(n)" for "who." Might these be relics of a mother tongue? Intriguingly, Ruhlen's proposed global etymologies tend to be words that are frequently used in everyday speech, and it is these that tend to change the slowest (Pagel et al. 2007).

Conclusions

Genetics, the shared structure and the universal learnability of human languages, make it reasonable to believe that extant humans have a common language capacity that evolved just once and probably early in our history, perhaps 160,000–200,000 years ago. But the rate at which words evolve means that traces of our common language or "mother tongue" have proven difficult to find. Further clues for evidence linking the world's languages may yet come from studying vocabulary, but other sources such as patterns of tones and styles of prefixes and suffixes (Diamond 2011; Vajda 2010; Hruschka et al. 2015) may also prove useful.

Cross-References

▶ Communication and Developmental Milestones
▶ Darwin on the Origin of Language
▶ Early Theories of Language
▶ Human Storytelling

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- ▶ Language Acquisition
- ▶ Language Acquisition in Infants and Toddlers
- ▶ Language and Handedness
- ▶ Language Development
- ▶ Language Dysfunction
- ▶ Language Modularity
- ▶ Linguistic Evolution
- ▶ Metaphor and Analogy
- ▶ Modeling Language Transmission
- ▶ Modern Theories of Language
- ▶ Moral Language Regulation
- ▶ Müller's Language Hypotheses
- ▶ Neurobiology of Language
- ▶ Noam Chomsky and Linguistics
- ▶ Pinker's (1994) The Language Instinct
- ▶ Universal Grammar
- ▶ Verbing and Nouning
- ▶ Vocabulary
- ▶ Words and Rules

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Mother's Brain Size Decrease During Pregnancy

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Synonyms

Baby brain; Momnesia; Pregnancy brain

Definition

During pregnancy, mothers' brains gradually decrease in size until parturition.

Introduction

During pregnancy, a mother's body goes through numerous physiological changes in order to become adequately prepared with the tasks of motherhood. One change that is not readily apparent to the mother or others is the shrinking of the overall volume of her brain. It has been found in both rats and humans that the size of a mother's brain decreases gradually throughout the gestation period, beginning at implantation of the fertilized egg (Chan et al. 2015; Oatridge et al. 2002). The amount of size decrease can vary between individuals, ranging from a 2% to 7% size decrease (Oatridge et al. 2002). Women with preeclampsia, a hypertension disorder acquired during pregnancy, can experience an even greater decrease in brain size (Oatridge et al. 2002). Although the total size of the brain decreases, there are actually increases in the size of the ventricles and the pituitary gland (Oatridge et al. 2002; Gonzalez et al. 1988). After parturition, the brain gradually returns to its pre-pregnancy size, taking up to 6 months (Brizendine 2006; Oatridge et al. 2002).

Causes of Size Decrease

While there is no consensus about the cause of brain shrinkage during pregnancy, there are a number of possibilities. Holdcroft et al. (2005) note that changes in brain size during pregnancy seem to be related to a change in cell metabolism. Six weeks after delivery, while the brain is returning to its original size, there is a significant increase in intracellular pH. While it is interesting to note that this change is only evident after the pregnancy has terminated, it may be a small part of the mechanism that causes the size change. In research with rats, Chan et al. (2015) postulated that a decrease in taurine leads to the change. Taurine helps prevent cell shrinkage in the brain, and its levels diminish during pregnancy.

Effects of Size Decrease

While the effects of a mother dealing with a shrinking brain are not life-threatening or large enough to have significant effects on daily life, there are some reported changes that come with brain size fluctuations. When returning to its original size during the postpartum period, there is evidence showing that there are changes in the neuronal connections of certain areas of the brain (Chan et al. 2015; Kim et al. 2010). In rats, there is a reported increase in hippocampal bilateral connectivity during pregnancy (Chan et al. 2015). In humans, there is evidence of gray matter increase in certain areas of the prefrontal cortex, gyrus, parietal lobe, insula, and thalamus (Kim et al. 2010).

There are conflicting results in studies investigating changes in the cognitive abilities of pregnant women. Some find no diminished abilities, while others have found significant impairments in memory (Crawley et al. 2003; Buckwalter et al. 1999). Interestingly, in a study by Crawley et al. (2003), there were no significant differences in the abilities of verbal memory, focused attention, or divided attention between pregnant and non-pregnant women; however, pregnant women rated themselves as less skilled in those areas during their third trimester. This general view of cognitive impairment held by pregnant women is apparent in common media as well; the reported pregnancy-induced dimness is colloquially referred to as "baby brain" or "momnesia" (Knapton 2014).

Evolutionary Implications

Researchers have speculated about the possible evolutionary significance of a shrinking brain during pregnancy. According to the results of the study by Kim et al. (2010), the shrinking and then subsequent regrowing could help facilitate the refocus of neuronal circuitry in areas of the brain that facilitate motherhood and motherly behaviors. Because of metabolic changes, expectant mothers may limit their physical activity keeping them close to home and safe from predators and other dangers.

Conclusion

During pregnancy, there is a significant decrease in the size of the mother's brain up until parturition, after which the decrease is reversed (Oatridge et al. 2002). The effects of this phenomenon are not dramatic, but they are noticed by pregnant mothers (Crawley et al. 2003). While there is some intriguing evidence regarding causes and evolutionary implications of the size decrease, further studies are needed to help provide a more detailed understanding of how and why this aspect of pregnancy occurs.

Cross-References

▶ Pregnancy

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Mother's Function

▶ Maternal Role

Mother-Child dyad

▶ Maternal Role

Motherese

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Synonyms

Baby talk (BT); Infant-directed speech (IDS)

Definition

Motherese refers to the ways humans (typically mothers or caregivers) alter their speech when addressing young infants or children compared to speech addressed to another adult.

Introduction

"Motherese," also commonly referred to as infant-directed speech (IDS) or baby talk (BT), refers to the ways in which adults alter various aspects of their speech when addressing children and infants. This can be done in a variety of ways such as using simplified speech, repeating words, shortening sentences, as well as altering the acoustic properties of speech such as using higher pitch, increased pitch range, more pitch variability, and slowing down their speech rate.

The ways in which an individual changes their speech style depends on many factors, including the age and sex of the child as well as the culture they are living in and the relationship to the child (Cooper and Aslin 1994). IDS can be measured using structured or natural observations between an adult and a child (typically a caregiver and their infant) and comparing the speech to that between the adult and another adult in the same setting. Comparisons are then made between the speech samples to determine whether and how the adults modify their speech in the presence of the child. Furthermore, studies have shown that infants as young as 7 weeks prefer infant-directed speech to adult-directed speech.

There is evidence to suggest that this behavior is common across human societies and highly recognizable (Broesch and Bryant 2015; Bryant et al. 2012; Bryant and Barrett 2007; Fernald 1992), although there has been debate about the extent to which some societies address young children and alter their speech style. Research shows that vocal changes happen similarly across industrialized populations (Fernald 1992) as well as in small-scale, traditional societies (Broesch and Bryant 2015).

Several possible functions of IDS have been proposed. For example, IDS might aid in language learning specifically or learning in general (Thiessen et al. 2005; Csibra and Gergely 2009). There is evidence that IDS contains acoustic features that help children learn vowel categories of their language (Kuhl et al. 1997). IDS also likely helps communicate emotional information,

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including effectively grabbing and maintaining the attention of infant listeners. In general, IDS also can facilitate an affective relationship between the adult and the child (Fernald 1989; Trainor et al. 2000). Infants demonstrate a preference for IDS over adult-directed speech (ADS) even when the IDS is in a language that is unfamiliar to the infant (Werker and McLeod 1989) suggesting that it is an excellent source of information for infants in early stages of perceptual and conceptual development.

Conclusion

Adults modify various aspects of their speech including acoustic properties, and speech style, often simplifying the nature of their speech and drawing attention to particular aspects of the language or the interactive context. There is evidence indicating that humans across the globe produce motherese. Infants also prefer IDS over ADS and learn more easily from it. It is thought that this behavior facilitates an affective relationship between a caregiver and his/her infant as well as enhances or enables the learning process.

Cross-References

► Language Acquisition in Infants and Toddlers
► Modern Theories of Language

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Mother-Fetus Conflict

► Conflict Between Parents and Offspring

Mother-Infant Attraction

► Maternal Bonding

Mother-Infant Co-sleeping

► Co-sleeping

Mother-Infant Proximity

► Co-sleeping

Mothers and Maternal Grandmothers and Childhood Survival

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Synonyms

Allomothering; Alloparenting

Definition

After mothers, maternal grandmothers have the largest effect on child survival.

Introduction

Arguably the greatest threat to reproduction faced by humans is infant and child mortality. Once an individual reaches adulthood, mortality rates are relatively low, with average lifespan ranging between 68 and 78 years among hunter-gatherers (Gurven and Kaplan 2007). Adults are also likely to find a mate, with virtually 100% of female and at least 90% of male hunter-gatherers doing so (see Volk and Atkinson (2008) for a brief review). The largest fitness challenges humans face, therefore, are first surviving their childhood and becoming reproductively successful adults and then themselves raising children to adulthood. Prior to modern history, infant mortality rates ranged from 25% to 40% and child mortality rates from 36% to 50%. These rates are very similar to modern hunter-gatherer groups (23% and 46%, respectively) (Volk and Atkinson 2008). Indeed, Hrdy (2009) notes that in many hunter-gatherer groups, a woman might give birth multiple times and never raise a child to adulthood.

Given the helplessness of infants, it is not surprising (though only recently acknowledged by scholars) that human mothers require the help of others in raising offspring. Human mothers are remarkable in their propensity to allow others to interact with infants, even immediately after birth. This is not the case with other great apes, who will not allow any adult contact with infants until 3–6 months following birth (Hrdy 2009). Hewlett (1991) found that among many foraging societies, for the 20–50% of the time an infant is held, it is held by someone other than the mother, usually a close female relative or trusted caretaker.

There are conflicting thoughts on the importance of paternal investment, but collectively, they show that in contrast to mothers, fathers have less impact on child survival (Sear and Mace 2008). However, Gray and Anderson (2015) reviewed the influence fathers have on their children and concluded that paternal investment is typically resource-based (e.g., financial) and may impact on children's survival, health, socio-emotional outcomes, social competence, and educational attainment. Thus, while paternal investment does not seem to affect child mortality rates, Sear and Mace (2008) reported in roughly one-third of the studies where they examined paternal death that father's absence increases the likelihood of child death, but only for a limited time following the father's death.

The question of who keeps children alive then becomes a vital question, and Sear and Mace (2008) after examining 45 studies of kin-child-rearing patterns among people living in nonindustrialized groups, found that after mothers, maternal grandmothers have the largest effect on child survival. Hawkes et al. (1997) found that among Hadza hunter-gatherers, seasonal fluctuations in foraging returns among women predicted variations in child weight for non-nursing mothers. For nursing mothers, the foraging returns of the maternal grandmothers had a positive effect on the child's weight gain. Researchers in the USA have found that among kin, the majority of caretakers are single, African-American grandmothers (Cuddeback 2004), while in the UK, it is married white grandmothers (Aldgate and McIntosh 2006; Farmer and Moyers 2008; Hunt 2003).

Following mothers and maternal grandmothers, it is females in the group who invest

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the most in childcare. Barry and Paxson (1971), using a controlled sample of 186 societies, found that after mothers, the principle companions and caretakers of children were adult female relatives, then female children, and then other females. Crittenden and Marlowe (2008) documented that among the Hadza, while female kin are most likely to engage in allomothering behavior such as holding an infant, female nonkin also assist more than males. In their review, Weisner et al. (1977) repeatedly stated that when care is needed, it is a woman more than a man, even if she is more distantly related, who is preferred. Likewise, as mentioned, in foraging societies, when a woman's gathering production decreases after the birth of a new child, the slack is most often picked up by female relatives and then friends (Hill and Hurtado 1996).

Female assistance in childcare and food provisioning begins at a young age. Older children in Bwa Mawego in Dominica take on adult domestic tasks, and in some households, girls around 12 years of age spend as much time in domestic activities as adult women (Quinlan et al. 2003). This activity does not only include childcare and overall relieves mothers' domestic burden which allows them to spend more time caring for their children. Similarly, among the Abuluyia females were the caretaker of small children more than twice as often as males (Weisner 1987). The same trend exists among the Kikuyu of Kenya (Leiderman and Leiderman 1974); Kikuyu child caretakers are usually 7–12 years of age, female, and may be siblings, cousins, or neighbors. They may care for the child for more than half the day, and while they may accompany the mother on various tasks, their primary responsibility is to take care of the child. This care occurs even if the mother is still nursing; the caretakers will take the child to the mother when it is hungry.

Based on the existing evidence from these societies, it seems probable that among humans, there has been selection for food sharing, work sharing, and by extension cooperation, particularly among mothers and grandmothers of young children (Hawkes et al. 1997), and other female group members (Kelly 1995). For example, Silk et al. (2009) found females who formed strong bonds with their mothers and adult daughters experienced the highest rates of offspring survival (Sokol-Chang et al. 2017).

Conclusion

Overwhelmingly, childcare is prioritized and performed by women, even during childhood and until they are grandmothers. Due to this preference and the prioritization of the care of the child, women are much more reliant on other women, particularly female relatives like grandmothers. This reliance on a cooperative system has a crucial purpose: the care of one's offspring so as to safeguard their investment and, hence, the continuation of their genetic line. This is not only an important relationship but a labor-intensive one as well. When examined in its entirety, cooperation among women has been a vital factor in the survival of the species.

Cross-References

- ▶ Alloparenting and Grandparenting
- ▶ Cooperative Breeding
- ▶ Cooperation
- ▶ Grandparental Investment

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Motion Parallax

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Definition

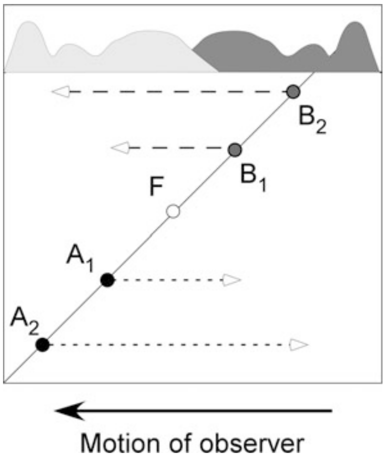
Motion parallax is a monocular depth cue produced by a difference in apparent direction and speed of displacement of the objects located at different distances from an observer, as that observer moves through the environment.

Introduction

Unlike binocular disparity, motion parallax is a monocular depth cue that does not require integrating information from two retinas. However, it is conceptually similar to binocular disparity in that the visual system must still detect the differences between two retinal images obtained successively rather than simultaneously (Rogers and Graham 1982). Motion parallax requires side-to-side (or translational) movement of the body or the head of an observer as depth information cannot be obtained from rotational movement alone. It has comparable precision to binocular disparity (Rogers and Graham 1982), and its use for depth perception is well documented for both humans and nonhuman animals.

Use of Motion Parallax for Depth Perception in Humans and Nonhuman Animals

As an observer moves through the environment, he or she will typically focus on some point in that environment (Fig. 1); in vertebrates, vestibuloocular reflex will trigger eye movements ensuring that the object will remain fixated on the fovea during movement. As a result, the features of the environment will form a bidirectional optic flow: The objects behind the fixation point will be



Motion Parallax, Fig. 1 Pattern of retinal motion created by the translational movement of observer to the left and fixation at the intermediate distance (*F*). The points located farther than fixation point (B_1 and B_2) will move in the same direction as the observer, whereas the points located nearer than fixation point (A_1 and A_2) will move in the opposite direction. Moreover, the speed with which each point moves across the retina will be proportionate to that point's depth from fixation; for example, the points A_2 and B_2 will be perceived as moving faster than the points A_1 and B_1

viewed as moving in the same direction as the observer, and the objects in front of the fixation point will be viewed as moving in the opposite direction. The speed of the perceived movement will be proportional to the distance from fixation point affording the relative ordering of the objects in depth. Sensitivity to motion parallax is already present in infants as young as 6 months (Condry and Yonas 2013).

Behavioral demonstrations of motion parallax have been carried out in several vertebrate species (barn owl, van der Willigen et al. 2002; gerbils, Ellard et al. 1984; Goodale et al. 1990; rat, Legg and Lambert 1990; rhesus monkeys, Cao and Schiller 2002). In addition, many birds (e.g., pigeons, chickens, gulls, and many shorebirds) also exhibit repetitive backward and forward head movements during walking as well as during landing ("head bobbing") that are thought to provide depth information via motion parallax. Supporting this hypothesis, Xiao and Frost (2013) have recently described neurons sensitive to motion parallax in pretectum of pigeons.

Similarly to vertebrates, many insects show behaviors that appear to produce motion parallax and result in better depth perception. For example, locusts and mantids periodically move their head sidewise prior to striking at a target; if the target is moved in the same direction as the head, then they tend to overestimate the distance to it suggesting that these movements are used as a source of motion parallax information (Kral 2003). Relatedly, bees and wasps circle around objects of interest, presumably using motion parallax to access their three-dimensional structure (Werner et al. 2016). Finally, fruit flies appear to use motion parallax generated by walking toward different objects to judge the distance to them (Collett 2002).

Conclusion

Depth information provided by motion parallax cues is as powerful and precise as that derived from binocular disparity (Rogers and Graham 1982). Unlike binocular disparity, motion parallax does not require binocular integration of information from overlapping visual fields and is therefore likely to be used by the animals with laterally located eyes, even when their visual fields have some degree of binocular overlap (Kral 2003).

Cross-References

▶ Binocular Disparity
▶ Depth Perception

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Motivates Punishment

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Synonyms

Functional benefits of punishment; Reasons to punish wrongdoers

Definition

Elicitors of the desire to punish a wrongdoer.

Introduction

People are motivated to punish cheaters, but substantial disagreement exists among philosophers and social scientists regarding why. Evolutionary psychologists have been drawn to human punishment because sometimes it appears to be *altruistic punishment*, costly but non-beneficial to the punisher, making it an anomaly for evolutionary, functional explanations – that is, explanations of how the fitness consequences of a psychological phenomenon enable and shape its adaptive evolution. Functional explanations provide answers to the question of why humans punish – that is, they articulate aspects of human experience that make people want to punish wrongdoers. Evolutionary psychologists have proposed two main classes of functional explanations for human punishment. *Adaptationist* explanations assume that selection is driven exclusively by the punisher's *inclusive fitness* – the fitness effects of punishment on the punisher and the punisher's relatives (Hamilton 1964). By contrast, *multilevel selection* (or group selection) explanations assert that important selection occurs both on the punisher's inclusive fitness (individual level) and at the level of groups by enabling groups to compete against one another based on their complement of punishers. Adaptationists deny the importance of group-level selection, arguing it is too weak to influence the evolution of adaptations. A brief overview of the evolutionary psychology of human punishment provides substantial insight into what motivates humans to punish.

Adaptationist Approaches

Evolutionary theorists frequently used group-level or species-level explanations for biological adaptations until George Williams (1966) showed how individual-level selection could explain adaptive designs better than the prevailing group-level or species-level explanations (i.e., "good for the group" or "good for the species" explanations). Williams persuasively argued that group-level selection is weak compared to individual-level selection, so individual-level

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selection should drive the evolution of biological adaptations. Williams not only undermined the credibility of group-level explanations, he also placed a foundational emphasis on the importance of articulating design features of adaptations. This emphasis developed into the form of evolutionary psychology known as *adaptationism*, an approach to evolutionary psychology that relies exclusively on individual-level selection in an empirical search for theoretically derived, a priori characterized design features of adaptations that comprise human psychology. Adaptationism has produced many complementary explanations and predictions regarding what factors motivate people to punish wrongdoers.

Seeking Better Treatment

According to *social exchange theory*, people use punishment as "a bargaining strategy" to enhance subsequent outcomes from their reciprocal interactions with specific individuals with whom they have potentially long-standing relationships (Krasnow et al. 2012, p. 3). That is, punishment modulates later behaviors of relationship partners to provide increased, direct benefits to the punisher. From this perspective, people forage for social exchange partners, using reputation information about how potential partners treat others as a cue to how they might themselves be treated, but preferring to use information about actual treatment by the person toward themselves if available. People invest in relationships hoping for long-term payouts. Such investment includes punishing defection when detected and directing punishment only toward those with whom a long-standing relationship can be expected so that punishment costs can be recaptured through subsequent benefits from that person. This perspective makes the surprising prediction that people will prefer to interact with cheaters more after they have punished compared to before they are punished. This is because punishment is theorized as an attempt to negotiate for better treatment. If however the actor perceives that prior losses are unlikely to be recouped by better outcomes later, the theory predicts the actor will abandon the relationship rather than paying costs to punish. By this approach, people are more likely to punish someone who defects against them but is expected to continue as a long-term relationship partner but unlikely to punish someone who wrongs another person (a third party) or whose relationship the person is abandoning.

Social Relationships

Lieberman and Linke (2007) argued that an actor's punitive motivation should be moderated by the degree to which an actor's biological fitness is tied to others. For example, *inclusive fitness theory* shows how an actor's fitness is tied to genetic relatives (Hamilton 1964). A brother's children increase inclusive fitness and more so per child as compared to a cousin's children. This is because we share more genes with a sibling than a cousin. Inclusive fitness is important for punitive motivation both because punishing a relative reduces one's own fitness and because one's fitness is reduced more when the victim is a relative compared to a nonrelative. Lieberman and Linke (2007) also argued that social exchange happens more with members of coalitions and in-groups and less with out-group members, so people should be more concerned with the welfare of coalition and in-group members. In summary, punitive motivation is moderated by the degree of genetic and social interdependence of the punisher with both the perpetrator and the victim: punitive motivation is stronger when victims are more interdependent and when perpetrators are less interdependent.

Reputation

One context in which apparently anomalous altruistic punishment occurs is when people pay costs to punish unknown third parties with whom the punisher expects no future interactions through which costs paid to punish the wrongdoer could be recouped. Kurzban et al. (2007) identified a means by which other sources of benefits and cost avoidance could be reaped by punishers of third parties even in such circumstances. They argue that punishment is costly due to dangers such as retaliation, making punishment itself a *costly* and therefore an *honest signal* – reliable source of information for observers that the punisher is willing and able to punish transgressions.

By punishing wrongdoers a punisher can develop a reputation as being someone who would be costly to mistreat. Kurzban et al. (2007) proposed that reputation benefits of punishment resulted in evolutionary selection for punishment-related adaptation. In particular, they predicted and found that punitive motivation is greater in the presence of an audience.

Offsetting Fitness Differentials

The exceptional cooperativeness observed in human groups is a theoretical problem because human groups produce benefits for members, but members who do not contribute to group efforts while still taking benefits – free riders – would have greater fitness outcomes than those who do contribute. Price, Cosmides, and Tooby (2002) argued that compared to contributors, free riders enjoy a *fitness differential* over contributors. The fitness differential is equal to (a) the costs paid by contributors, added to the (b) benefits free riders take greater than those received by contributors. This fitness advantage must be eliminated for adaptations enabling collective actions to evolve. Price et al. (2002) argued that *punitive sentiment* is an adaptation that evolved to offset these fitness differentials in the context of collective actions. Punitive sentiment is activated by free riders, detected as those "who (1) benefit from a collective action, (2) could have contributed . . . without incurring a cost disproportionately greater than their share of the expected benefit, and (3) did not expend adequate effort towards the collective action" (p. 206). Punitive sentiment's output is "desire that the target of the sentiment is harmed [and perhaps] that (1) the target be aware of being harmed, (2) the target know why, and (3) others be aware of the reason for punishment" (p. 206). Indeed, participant punitive motivations tracked these and not other variables in response to free riders failing to contribute to a group-defense collective action (Price et al. 2002).

Recruiting Labor

Another perspective that argues punishment is motivated by free riding was provided by Krasnow, Delton, Cosmides, and Tooby (2015). They proposed that punishment can be considered a labor recruitment method: it can transform would-be free riders into contributors. Krasnow et al. (2015) demonstrated how this could work using individual-level selection in agent-based modeling by enhancing the biological plausibility of model features, such as allowing agents to punish only some defections to avoid excessive costs of overpunishing as occurs when all punisher agents punish every defection, by allowing free riders to become cooperators rather than merely endure fitness reduction while continuing to free ride and by allowing agents to interact with members of other groups to enhance the strength of natural selection. Finally, agents remembered who punished them in the past and selectively cooperated when in the presence of prior punishers.

Multilevel Selection Approaches

As mentioned above, in the 1960s, George Williams largely undermined the scientific legitimacy of group selection theory. Over the last few decades, however, evolutionary scholars have reconsidered the issue. Sober and Wilson (1998) convincingly demonstrated how, mathematically, natural selection could in fact happen at multiple levels, including the group level, within a narrow set of circumstances. They further argued that human morality, including punishment, might be comprised of adaptations formed in part by group-level selection.

The problem for explaining group-level selection of human punishment is articulating how selection could favor individuals who pay costs to punish wrongdoers when the punisher himself or herself would not receive extra benefits as a result of the punishment compared to group members who did not punish. The group-level selection solution, in brief, is that (a) groups with more norm-compliance-enforcing punishers are more cohesive and functional than groups with fewer punishers and so (b) groups with more punishers outcompete and displace groups with fewer punishers, resulting in (c) a relative increase in the number of groups with more punishers, and therefore (d) punishers increase in number overall *despite* a reduction in the number of punishers within groups (because within groups such punishers are at a disadvantage relative to non-punishers). That is, punishers proliferate because punisher-rich groups become more common and spread faster than punisher-deficient groups, despite punishers become less common than non-punishers within groups.

Punishing Norm Violators

One prominent multilevel selection theory of human cooperation and punishment is *cultural group selection* (Richerson et al. 2016). According to cultural group selection theory, group-level selection is enabled by diverse, culturally inherited social norms and institutions that produce substantial between-group differences, which in turn impact competition among groups. The phenotypic differences across groups can have strong impacts when groups compete with one another, enabling genetic selection on even small amounts of genetic differences between groups, a form of gene-culture coevolution. Punishment plays multiple important roles in this theory. Punishment of deviant behaviors increases phenotypic homogeneity within groups by enhancing conformity to group-specific norms, which also increases phenotypic heterogeneity between groups. Punishment, sometimes as prescribed by cultural justice institutions such as religion, also enhances within-group cooperation and helps solve public goods dilemmas. Finally, within groups punishment decreases fitness for genes related to antisocial behavior and increases fitness for genes related to prosocial orientation, another case of gene-culture coevolution. Core claims of cultural group selection theory were supported by a study looking at anonymous costly punishment and cooperation across 15 diverse societies. It found that people tended to punish unfair offers in all societies, but critically cooperation was higher in societies whose members punished more (Henrich, et al. 2006).

Conclusion

Evolutionary psychologists have developed many explanations regarding why humans are motivated to punish wrongdoers based on functional considerations of punishment. Explanations relying on individual-level selection focus exclusively on effects of punishment on the punisher's inclusive fitness. From this perspective, people are motivated to punish others in order to (a) negotiate for better treatment; (b) protect the fitness interests of themselves, their relatives, and their in-group members; (c) develop a reputation that deters defection; (d) offset fitness differentials that free riders reap over contributors in collective action; and (e) recruit labor in collective actions. These explanations are complementary – all could be true – as could the more controversial group-selection explanations, which argue that people are motivated to punish norm violators to enhance their group's ability to compete with other groups.

Cross-References

▶ Ability and Willingness of Victim to Retaliate
▶ Altruistic Punishment
▶ Altruistic Punishment Enhances Reputation
▶ Altruistic Punishment and Strong Reciprocity
▶ Benefit Group Relative to Other Groups
▶ Coalitional Relationships
▶ Competition Between Groups
▶ Costs of Punishing
▶ Cross-Cultural Punishment
▶ Cultural Evolution
▶ Evolution of Culture
▶ Evolution of Punishment
▶ George C. Williams
▶ Inclusive Fitness
▶ Mechanisms to Detect and Punish Cheaters
▶ Norms
▶ Punitive Sentiment

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Motivational States Promoting Survival and Reproduction

► Relevance of Evolved Interests to Modern Morality

Motor Aphasia

► Broca's and Wernicke's Aphasia

Motor Control

► Inhibition

Motor Development

► Mobility: Crawling and Walking

Mouth-to-Mouth Kissing

► Kissing

Movement

► Selectionist Models: Implications for Behavior

mtDNA

► Mitochondrial DNA

Müller's Language Hypotheses

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M

Synonyms

Origin of language; Theories of language evolution

Definition

Early speculative hypotheses concerning the origin of spoken language proposed by Friederich Max Müller.

Introduction

Müller was a German philologist who proposed some early theoretical formulations concerning the origins of language. After the publication of the *On the Origin of Species* (Darwin 1859), Müller was fascinated of the idea of human evolution. The tradition of giving disparage names to the theories regarding how language evolved commenced by him and this was an important factor in his campaign against Darwin's evolutionary theory. Müller (1877) established associations between Darwin's evolution on humans and many other ideas about development, and he concluded that languages are subjected to historical radical change. However, he later realized that he underestimated Darwin's arguments and his suggestions concerning the history of human evolution. In 1864, Müller presented his *Lectures on the Science of Language*. In this period of time, he started to reevaluate Darwin's theory. In his book *The Descent of Man*, Darwin had a section about language, in which he proposed his own theories about the evolution of language. Müller had different ideas about how language emerged, and therefore, he talked in public about "Mr. Darwin's Philosophy of Language" in 1873 (Knoll 1986).

In these talks, Müller suggested that the language faculties of humans are distinguished from other creations; therefore, it was impossible that the human development evolved from a lower form without language acquisition. His point of view was supported by his opinion that the linguistic roots are considered the original building blocks of language. In his *Lectures on the Science of Language*, he proposes that Indo-European verbal communication had developed from hundreds of verbal "cells." Muller hypothesized these linguistic roots as fundamental of the theory of mind. He assumed that they establish the principles for the Kantian intellectual classifications used by the mind to establish cognizance out of raw sense of perception. This capacity of the individual's mental structure separated man from animals since it formed the basis of man's exclusive capacity to form a concept and think rationally. Without these capacities, neither language nor thought was possible. Muller concluded that because animals do not have this intellectual structure, their evolution into humans was impossible (Müller 1877). In 1861, Max Müller published a list of theoretical concepts concerning the origins of language. He later abandoned most of his theories which include the bow-wow theory, the pooh-pooh theory, the ding-dong theory, the yo-he-yo theory, and the ta-ta theory.

Bow-wow Theory

The overall idea of the bow-wow theory is that the beginning of language can be linked to the era when the progenitors begun to mimic the sounds in their natural environment, a process called "onomatopoeia." The natural world was of paramount importance in the origination of language since the sounds of animals played a significant role in the invention of the vocabulary used in verbal communication. On the one hand, Thorndike (1943) was doubtful about the credibility of this theory as the beginning of language. On the other hand, recent studies support that being able to mimic sounds in the natural environment was of paramount importance in the evolution of language (Malle 2002).

A major drawback of this theory is the fact that people use different words to describe the sounds of animals according to their country and more specifically due to their language. For instance, the sound of a pig can be translated differently across the countries. In English a pig makes the sound "oink-oink"; in Russian it makes the sound "hyru-hyru," while in Chinese it makes the sound "oh-ee-oh-ee." Therefore, the sound and pronunciation of the words are not similar among the different languages. Another challenge is that there are few onomatopoeic words in the languages worldwide. In case the theory was credible, then it should have plenty of words out from the basic rule of onomatopoeia. The hypothesis that language is determined by how a person interprets a sound was rejected, since there are many languages and nobody can suggest that human vocabulary arose from these sounds (Thorndike 1943).

Ding-Dong Theory

The idea behind the ding-dong theory is sound symbolism which is a non-arbitrary connection between phonetic features of linguistic items and their meanings. Max Müller proposed the ding-dong theory by stating that meaning comes from sounds. According to this hypothesis, the origin of language emerged evolutionary with the names our ancestors gave to objects, activities, and natural occurrences after an identifiable sound associated with it in real life (Mandavilli 2016). An important characteristic of this notion is the fact that words having high front vowels were used to name small or jagged objects in comparison to words having round vowel which were used to name large or round-shaped objects. It is suggested that people used sound symbolism to give meaning to words through onomatopoeia, which is the formation of a word by the imitation of a sound made by or associated with its referent in real space and time. "Boom," for example, was the word given for explosion (Mitchell et al. 2013).

A major drawback of this theory is that onomatopoeic words are not the same in every country because the sound and pronunciation of the words are not similar among the different languages. People in different nations used their onomatopoeic words which are formed in regard to how they interpret different sounds. These words are a very limited part of their vocabulary since many ideas such as love, hate, harmony, and the sea do not make any noise. Thus, the relationship between meaning and sound is not established by any convincing evidence. For this reason, this theory was later abandoned by Max Müller (1877).

Pooh-Pooh Theory

The early speculative language theory of pooh-pooh refers to the notion that language emerged from unintentional vocal responses to agony, fright, astonishment, enthusiasm, satisfaction, or other emotions (Barber 1965). Although animals produce sounds as well, language is unique only to humans. "Ouch!," "Aaah!," and 'Wow!' are some examples of the unintentional vocal responses emerged from this hypothesis. This theory can be found in the literature as "the expressive theory," "the interjectionist theory," or "the expressions of emotions theory" (Barber 1965). Max Müller supported the pooh-pooh theory for a while, but later he abandoned it (Müller 1877).

The notion of the pooh-pooh theory that language has many emotional exclamations has been widely criticized by many reviewers. It has been argued that the explanations provided by this theory constitute a very limited part of the vocabulary of different languages. Further, the various languages worldwide have different interjections. For instance, in English pain is "ouch" while in Greek it is "aou." Therefore, it is indicated that sounds could have been different if their origin was from interjections. To conclude, although early human beings could use words based on interjections, these may have been ultimately overruled by more sophisticated and abstract terms (Mandavilli 2016). Finally, it is important to emphasize that although animals react to the sudden feelings they experience using sounds, they didn't develop language as human beings did.

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Ta-Ta Theory

According to Richard Paget, who was specialized in speech science and the origin of speech, the ta-ta hypothesis supported that speech and the development of sound were produced to endorse the hand and body movements of the individual. So as to better demonstrate the meaning behind the gestures, these sounds developed different words or combinations of sounds unavoidably leading to speech patterns (Paget 1930). For instance, when a person would say ta-ta, he was basically saying good-bye with his tongue. At some point, this theory was even supported by Charles Darwin (Mandavilli 2016). In the literature, there are several theories such as the oral gesture theory and the chew-chew theory that support the idea that the origin of language begun from early human gestures made from their mouth. It is of paramount importance, though, to understand that it is not feasible to explain the origin of some words using the ta-ta theory.

A major drawback of this early theory on the origins of language is the fact that in order for people to be able to communicate between them, they must see each other in person. Also, there must be daylight, or firelight, and with nothing blocking one's view. In the case of hand gestures, it is important that hands are free from doing something else (Meir et al. 2010). Last but not least, in order for this hypothesis to be the origination of communication, it required a number of gestures be in place already for humans to imitate with their mouth gestures, but it didn't explain how these gestures arose in the first place.

Yo-He-Yo Theory

Verbal communication commenced as growls, moans, and rhythmic chants which occur from the overwrought efforts of ancient menfolk to move a bole or heave a rock, as it is suggested by the yo-he-yo theory (Mandavilli 2016). Every time these people were working hard to move something heavy, they produced a sound which slowly developed into words such as "heave" and "lift." Therefore, if they didn't want to memorize their favorite growl-sound and use it over and over again, this sound would seem to be random. However, these growls would not be accounted as words in a lingual meaning since they would seem unlikely to fall into consonant-vowel patterns (Rice 1976). The yo-he-yo theory does, however, have a benefit since it is responsible for the inducement, the cause, and the aim of the language. It describes language as emerging from a need to communicate and for making the engagement among men possible in organized activity, rather than seeing language resulting from a cave dweller imitating an animal's howl to assist him while he was away for his dinner (Barber 1965). The yo-he-yo theory can be found in the literature as "Yo-heave-ho theory" and "Social interaction source," all of them proposed that the origin of speech begun as growls and moans (Mandavilli 2016).

Notwithstanding the above, evolution plays an important role since it claims that human society developed gradually because men began thinking of new things to do and began doing them collectively. What makes this theory interesting, however, is the query of what motivates the speechless cavemen move a bole in the first place since language emerged from the growls and moans of their teamwork. Unfortunately, the origin of grammatical structure is not explained or how a social system got started void of language (Rice 1976).

Conclusion

It is a matter of fact that language is a human trait. All efforts to explain how language emerged have failed because of the absence of knowledge concerning the origin of any language. An important factor for the inability to acknowledge the origin of language is because none of the animals possesses any transitional form of communication. As a consequence, Muller could not differentiate between humans' linguistic faculties and the growls or sounds of animals. Also, the cultural roots of various hand gestures did not suggest verbal communication. Last but not least, most of his theories did not explain the origination of different ideas or things that do not produce any sound, and this resulted to a limited vocabulary of the languages. As a result, there could not be an explanation of the origin of names of such concepts, if his theories were chosen as the origin of language.

Cross-References

► Darwin on the Origin of Language
► Early Theories of Language

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Muller's Theories on Language Evolution

► Early Theories of Language

Müllerian Association

► Müllerian Mimicry

Müllerian Mimicry

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Synonyms

Arithmetic mimicry; Müllerian association; Mutualistic mimicry

Definitions

When two or more harmful or unpalatable animals develop similar appearances as a shared protective device.

Introduction

Many prey possess chemical compounds or other noxious substances that they advertise to potential predators via conspicuous warning signals (Sherratt 2008). Mimicry describes the resemblance of one animal (a mimic) to chemically defended species (the model), such that a third animal (usually a predator) is deceived by the similarity. Mimicry is conventionally divided into two separate categories depending on the phenotype of the mimic: Müllerian and Batesian mimicry.

In Batesian mimicry, named for its discoverer, Henry Walter Bates (1862), an undefended harmless species evolves a warning signal (e.g., visual, auditory, or even behavior) that imitates a harmful model species. Batesian mimics gain protection against predators due to their being perceived as harmful. In contrast, Müllerian mimicry describes the phenomenon where two or more defended species, that may or may not be closely related and share one or more predators, mimic each other's warning signals. It is named after the German naturalist Fritz Müller who first proposed the concept in 1878.

M

A History of the Idea

In his travels in the Amazon, Henry Walter Bates (1862) observed that as well as palatable species evolving to resemble defended species, there were cases where a number of defended species evolved to resemble one another. He proposed that this was due to species sharing similar environmental conditions. However, Fritz Müller suggested an alternative reason. In one of the first clear mathematical treatments of natural selection, Müller hypothesized that to learn to avoid prey with a particular visual signal, predators must kill a fixed (i.e., density-independent) number of that prey. If true, resemblance between a number of prey species would result in a divided cost of educating the predator species. Consequently, an increase in the number of visually similar co-mimics spreads out the costs of predator education between the co-mimics, and Müllerian mimics enjoy strength in numbers: the relative risk of predation on each individual species is reduced. Hence, two defended species that have separate signals pay a higher mortality cost during predator learning than two species that, through mimicry, resemble each other. Although this idea initially met with opposition from Müller's contemporaries (see Ruxton et al. 2004), the benefits of shared resemblance of unpalatable species are now commonly accepted.

Müllerian mimicry has been explored comparatively, theoretically, and empirically (Sherratt 2008). Selection for color pattern similarity among mimetic butterflies in the wild is strong, and research in controlled laboratory settings has uncovered the selective pressures leading to signal similarity.

Examples of Müllerian Mimicry

Müllerian mimicry is common in nature, with perhaps the most obvious example being the black and yellow striping of bees and wasps (hymenopterans). Müllerian mimicry has also been documented in frogs, larval and adult lepidoptera, snakes, coleoptera, fish, and birds (Sherratt 2008). The most celebrated example of Müllerian mimicry is observed in the heliconid butterflies of South America, which involves a large collection of similar-looking species – a "mimicry ring."

There is a long and distinguished history of research on mimicry in heliconids. The Heliconiinae is comprised of 46 described species that all contain cyanogenic glycosides and have conspicuous brightly colored wing patterns – aposematism. Mimicry in *Heliconius* often involves convergence between species within the same genus, and frequently between pairs of species. For example, *Heliconius cydno* generally mimics *H. sara*/*H. sapho*, while *H. melpomene* always mimics *H. erato*. These latter two species are distributed throughout South and Central America, and mimic one another wherever they co-occur.

The Evolution of Müllerian Mimicry

There have been two schools of thought on how mimicry evolves. One proposes that shared resemblance evolves through gradual change, the other suggests that mimicry evolves by large mutational leaps toward similarity, followed by gradual evolutionary fine-tuning of the mimic's appearance toward the model. The second proposal, called "the two-step hypothesis," is supported by recent comparative genomics research showing that large-effect chromosomal inversions give rise to different morphs, but that the evolution of accurate mimicry likely requires a more gradual accumulation of multiple genetic changes (Nadeau et al. 2016). The two-step hypothesis relies on the assumption that predators generalize their avoidance between models and mimics with a level of shared resemblance (Leimar et al. 2012).

Müller's Idea About Predators Attacking a Fixed Number of Prey During Learning

One of the specific assumptions of Müller's theory is that predators must attack and consume a fixed number of each distinct defended species before learning to reject them. To date this hypothesis has only been tested in laboratory experiments with captive predators. There has been no support for the "fixed n" idea. In fact, several experiments have reported increases in the numbers of unpalatable prey attacked as the abundance of that prey type increases (Sherratt 2008). This is perhaps unsurprising, because animal cognition is rarely expected to be as simple as learning by a fixed amount of experience. An outstanding question of fundamental importance is how predator learning proceeds in a wild setting – how often does a predator meet a model or a mimic in a season, or during its lifetime, and does it ever learn to completely avoid aposematic prey? Questions like these are challenging to address, but could reveal more of the underlying dynamics of mimicry.

Evidence That There Is Selection for Signal Similarity

Several field studies have found evidence for the benefit of sharing a common warning signal (reviewed by Sherratt 2008). In mark-recapture experiments on *Heliconius* butterflies, wild avian predators select against rare and unfamiliar warning signals, and locally common signals have better survival. For example, Mallet and Barton (1989) transferred *H. erato* across the hybrid zone and monitored the survival of the transferred individuals compared to control species which were marked and released. The authors found that more of the rare artificial migrants were attacked and less recaptured than controls. Selection for color pattern similarity among mimetic butterflies in the wild was strong.

However, evidence that predators select for increasing signal similarity is less clear when research is conducted in the laboratory. By observing the foraging behavior of wild-caught captive great tits (*Parus major*) on artificially defended prey, many researchers have found that co-mimics with visually dissimilar appearance can still benefit from strength in numbers, with predators generalizing their avoidance to dissimilar signals (discussed in Rowland et al. 2010). However, mimicry *can* provide substantial protection during predator learning in the lab, and in the manner proposed originally by Müller, especially if prey are visually distinct and not easily generalized (Rowland et al. 2010), of if the system is more similar to the real world. In a study by Ihalainen et al. (2012), great tits foraged in simple or complex prey communities. Ihalainen et al. found that predator learning was slower and more costly to nonmimetic prey in visually complex prey environments. Therefore, mimicry would be beneficial in complex prey communities, because of the costs of predator education, just as Müller predicted. Interestingly, many studies have found evidence that pattern similarity might also benefit prey in the later stages of the learning process (note that Müller's ideas are about naïve predators). These findings in conjunction with the field experiments discussed previously suggest that once birds are familiar with a particular warning color or pattern, experienced predators can also select for pattern convergence (Rowland et al. 2010).

Species Differences in Defensive Chemical Concentrations, and Its Consequences for the Evolution of Müllerian Mimicry

Several authors have proposed the existence of a "mimicry spectrum," of which Batesian and Müllerian mimicry could be thought of as the extremes. This spectrum comes from the continuous range of chemical defense levels of mimics, and the role of predator psychology in selecting for signal similarity (Sherratt 2008). It is well documented that different predators vary in their ability to overcome the chemical defenses of aposematic prey, and it has been suggested by some authors that this can produce a hybrid between Batesian mimicry and Müllerian mimicry. In cases where predators profit from attacking prey with lower levels of chemical defense, model species may not benefit from coexistence with a visually indistinguishable but less-defended mimic, and lose protection when the mimic form is abundant. Therefore, mimicry between two species can be parasitic and Batesian-like (or quasi-

Batesian) and not necessarily mutualistic and Müllerian. Quasi-Batesian mimics could face much stronger selection for signal accuracy than might Müllerian co-mimics (Sherratt 2008).

Müllerian Mimicry and Speciation

Differences in color patterns between co-mimics can lead to reproductive isolation via pre- and post-zygotic mechanisms. This is because recombinant wing patterns are usually nonmimetic and are selected against by predators – a form of hybrid incompatibility. Furthermore, model species are also less attracted, or not attracted at all, to hybrids as potential mates – a form of post-zygotic isolation. Furthermore, *Heliconius* species' mate preference for similar wing patterns is also genetically linked to the genes that underlie the wing color that distinguishes them from other species. This reinforces post-zygotic isolation.

The Molecules Behind Mimicry

Research on heliconius butterflies has unraveled the genomic bases of how mimicry evolves and varies. Advances in sequencing technology and comparative genomics, coupled with recombination mapping have enabled scientists to accurately pinpoint candidate genomic regions responsible for mimicry.

The major loci involved in wing pattern development are collectively referred to as the "wing patterning toolkit" (reviewed in Kronforst and Papa 2015), and comprise: *optix* – a transcription factor that places red patches onto the wings, as well as the black regions on the central part of the wing; *WntA* which is associated with melanization of the fore and hind wing; *poik* – a cell-cycle regulator that is involved in patterning of the central areas of wings, and a gene, *cortex*, that probably regulates pigmentation patterning (Nadeau et al. 2016).

Conclusion

The study of mimicry has been a test bed for the theory of natural selection. Since Müller's hypothesis, the molecules responsible for mimetic resemblance have been uncovered. We now also have evidence that mimicry evolution is driven by predation pressure: species exposed to the same predators benefit from sharing warning signals. However, there is still a need to investigate how avoidance learning and predators' response to mimetic symbols occurs in the wild. This is an important question because if predators learn to avoid unpalatable prey differently in simple and complex communities in the lab, then how does learning proceed in natural conditions where we don't know how often they meet aposematic prey, if they meet them sequentially or simultaneously, and if they ever fully learn about aposematic prey. There is still much to discover.

Cross-References

► Aposematism
► Butterfly Mimicry
► Communication, Cues, and Signals
► Frequency-Dependent Selection

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Multi-cultural

► Cross-Cultural Variation

Multiculturalism

► Cross-Cultural Similarities

Multilevel Selection Theory

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Synonyms

Group selection; Inclusive fitness theory; Individual selection; Kin selection

Definition

Theoretical framework that presumes that selection can occur at an array of organizational levels from the gene upwards.

Multilevel Selection Theory

Multilevel Selection (MLS) Theory is the synthesis of individual selection as popularized by researchers including Williams (1966) and group selection as championed by researchers including DS Wilson (1975). More precisely, MLS theory offers a pluralist and flexible analytical framework that recognizes that selection occurs at different levels of organization as a function of context. As Wilson and Sober (1994) illustrated, when there is both phenotypic uniformity and shared fate at a given level of organization, then selection can be expected to occur. Consequently, if there is no phenotypic uniformity or shared fate among individuals in a given set of individuals, then one would expect selection to occur at the level of individuals. On the contrary, if individuals in a given set do share a common fate with one another and demonstrate phenotypic uniformity, then one would expect selection to occur at the level of sets wherein sets are being selected across evolutionary time relative to other sets in a given population of sets.

In the case of humans, there is obviously not genetic uniformity within any groups (outside of twin-pairs); however, there can certainly exist phenotypic uniformity as well as shared fate within groups since cultural transmission is dynamic and not isomorphic with underlying genetic variation (Wilson and Kniffin 1999). Indeed, Wilson and Sober (1994) review the fact that it is undisputed that genes function as “replicators” as a fundamental level of evolutionary accounting; however, conditions can and do emerge whereby units of organization above genes and above individuals can emerge and function as “vehicles” on which selective retention can occur. For example, Kniffin and Wilson (2010) review three case studies where group-level incentive structures within a population allocated rewards so that individuals would benefit if they served group-level interests.

Among non-human species, there exist numerous laboratory demonstrations whereby selection at group or community levels has been cultivated – often for beneficial outcomes. For example, Swenson et al. (2000) successfully selected for soil ecosystems that more effectively degraded an environmental toxin. More recently, with the benefit of genetic sequencing technology, researchers have demonstrated that selection at the level of soil microbiomes can be effected in ways that yield valuable traits in economically important plants

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(Garcia and Kao-Kniffin 2018; Panke-Buisse et al. 2015). Criticisms (e.g., Williams 1966) of group- or multilevel selection have traditionally acknowledged that it can be artificially effected but have been skeptical that it occurs in nature except for species where there is uniquely high genetic relatedness among individuals within groups (i.e., social insects [Seeley 2009]). Reviews of naturalistic evidence from species such as African buffalo indicate, though, that such dynamics are visible in naturalistic conditions (Prins 1996; Wilson 1997).

MLS theory has been the subject of contentious debate at various points over the past five decades partly because it has been central to debates on the evolution of altruism and selfishness where a variety of additional theories (e.g., kin selection and inclusive fitness theories) have been presented as competing frameworks. Consistent with the intention of MLS theory to be integrative and holistic of more specific theories, Wilson and Wilson (2007: 345) summarize the MLS perspective with the pithy set of statements that "Selfishness beats altruism within groups. Altruistic groups beat selfish groups. Everything else is commentary." In other words, selection occurs at the level of individuals when individuals are selfish while selection occurs at the level of groups when individuals are altruistic within their own groups. Debate continues on these perspectives; however, there has, at least, been important refinements over the past five decades in how such positions are argued whereby (1) MLS theory offers an integrative framework that is inclusive of both individual- and group-level selection and (2) relatively new genetic sequencing technologies are increasingly facilitating a better understanding of the mechanisms through which selection at multiple levels (from the gene "upwards") can be examined.

Cross-References

- ► E. O. Wilson
- ► Examples of Group Selection
- ► Group Selection
- ► Kin Selection Hypothesis
- ► Problems with Group Selection
- ► Selfish Gene, The

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Multilingualism

- ► Bilingualism
- ► Critical Period of Acquisition

Multiple Mating

- ► Polygamy (Behavioral Ecology)

Multiple Matings

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Synonyms

Extra-pair copulations (EPCs); Promiscuity

Definition

Copulate with more than one sexual partner.

Introduction

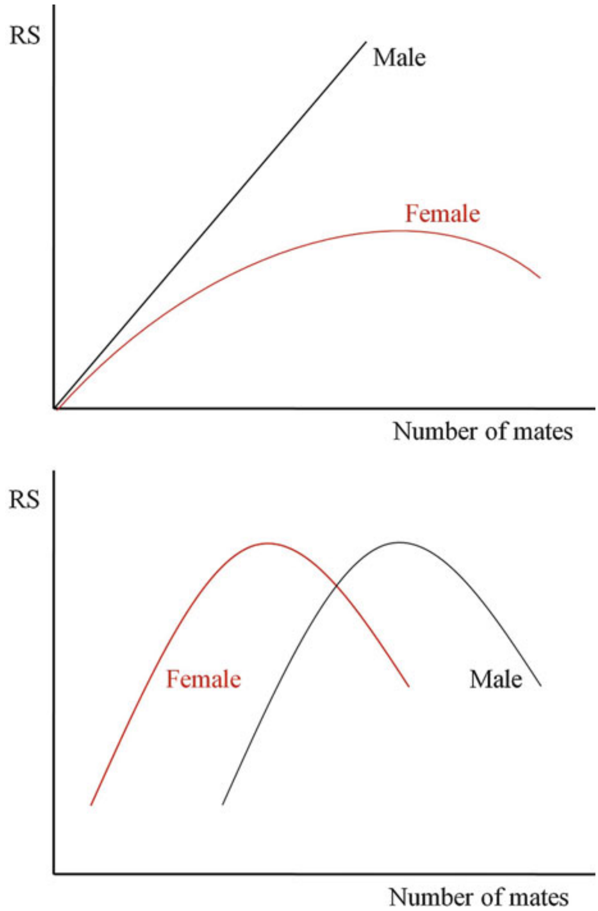
Empirical studies and theoretical models have used the number of sexual partners (i.e., the level of *multiple mating*) as a proxy for lifetime reproductive success (RS) or fitness. This is not surprising considering the close connection between mating and reproduction. In the case of males, the relationship between these variables is considered positive, whereas for females an intermediate optimum is predicted (Fig. 1a). However, when males invest heavily in each mate an intermediate optimum is also expected (Fig. 1b). On the other hand, it is recognized that sperm competition, postcopulatory cryptic choice, and postcopulatory sexual conflict could strongly modify the relationship between reproductive success and number of mates (Shackelford and Hansen 2015). These processes could partially explain why in different human societies the high proportions of cuckolded males, resulting from female multiple mating, are not reflected in high rates of extra-pair paternity (Larmuseau et al. 2016) (Fig. 1).

The relevance of the number of sexual partners for evolutionary interpretations is also evident in definitions of animal mating systems, which include the number of mates as an essential component. Mating systems are usually considered species-specific, although alternative strategies are included in recent discussions. Mating system definitions have been extremely helpful for organizing interspecific variation in, and identifying selective pressures acting on, reproductive behavior. Nevertheless, even when alternative strategies have been identified, each strategy encompasses individual variation in the number of mates due to strategic decisions, constraints, and chance. Hence, the study of individual variation in number of mates could provide interesting perspectives on the proximate causes and selective pressures acting on individual decisions, especially in species with a long reproductive lifetime, such as humans.

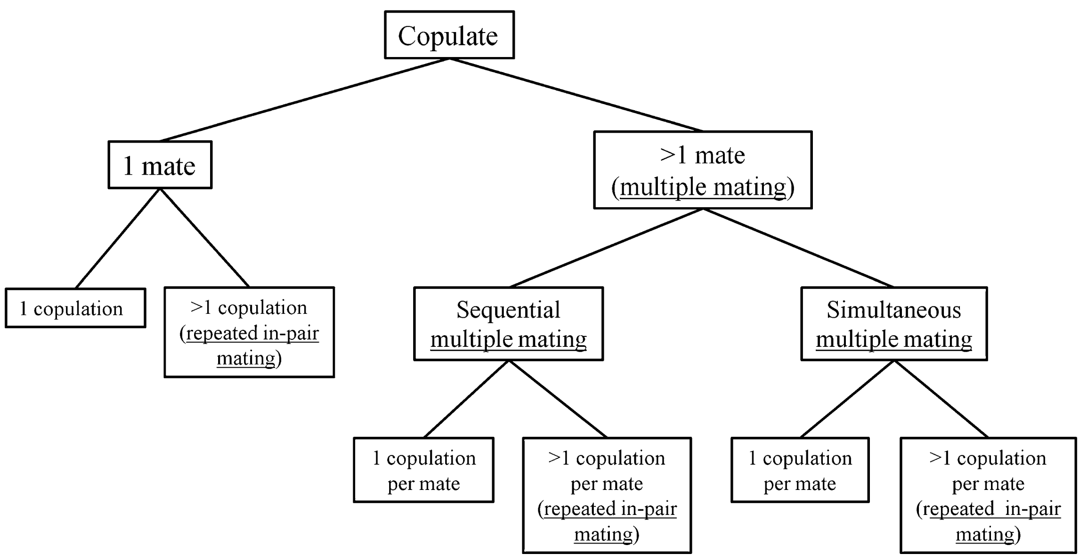
Identifying the Alternatives

When an individual reaches reproductive age she/he will face a number of lifetime options regarding the number of reproductive partners. In Fig. 2 these options are illustrated. An individual could decide or be able to copulate with just one mate or with multiple mates (the alternative of remaining virgin for life is also a possibility). The route taken could depend, for example, on the sex appeal of the individual or the amount of resources she/he controls. For example, an unattractive male or a male without resources could be constrained to mate with only one female, probably of low quality, or try an alternative strategy such as become a rapist. In addition, if he is able to get one female, he could just mate once (this would be the case if he is a rapist or if he is involved in a short-term interaction with the female) or establish a long-term relation in which he performs multiple *in-pair copulations* with the female (which could reduce, for example, the negative effects of female extra-pair copulations [EPCs] or increase the strength of the pair bond) (Fig. 2).

In contrast, if the individual is able, and decides, to copulate with multiple mates, she/he could opt for a series of short-term relationships in which she/he mates once with each partner



Multiple Matings, Fig. 1 Relationship between individual reproductive success (RS) and lifetime number of sexual partners (number of mates). (**a**) Male RS is limited by number of mates, while female RS has an intermediate optimum determined by the costs and benefits of multiple mating. (**b**) Both male and female RS have an intermediate optimum determined by the sex-specific costs and benefits of multiple mating (in the example, the female optimum is smaller than that of males)



Multiple Matings, Fig. 2 Diagram showing the alternatives open to an individual female or male regarding lifetime number of sexual partners (number of mates) (sequential multiple mating) or she/he could perform repeated in-pair copulations with each of a number of mates; a mixture of these alternatives is also possible, as when a female exhibits simultaneous multiple mating by establishing a long-term relationship with a male and, at the same time, having EPCs with other males (this could benefit the female if she improves the quality of her offspring by mating with a high-quality "lover" who fertilizes her eggs, while being supported by a male that provides better parental care than the attractive male).

Analyzing individual mating frequencies improves the ability to identify ecological and phenotypic constraints on multiple mating and thus to understand the selective pressures and adaptive traits affecting variation in the number of mates and of copulations. For example, several studies have focused either on multiple mating or in repeated in-pair copulations, but trade-offs between these variables (due, for example, to constraints in time or ejaculate production) have been rarely considered, as well as the influence of factors such as health or age on these trade-offs. However, as shown in the following sections, most evolutionary perspectives on human mating frequencies have used the traditional mating systems approach.

Multiple Mating in Humans: Patterns

Although most contemporary human societies practice socially imposed monogamy, this system often enables simultaneous multiple mating. An example of simultaneous multiple mating is EPCs in which sexual couples involved in long-term romantic relationships simultaneously seek EPCs with additional partners. EPCs are very common in humans, appearing approximately in 84 % of societies (Simmons et al. 2004; Anderson 2006). Nevertheless, its frequency and intensity of occurrence is highly variable. For instance, in the "Tsimane," a forager–horticulturists society living in the lowland rainforests in the Beni region of Bolivia, 31 % of married males reported EPCs during the first 5 years of marriage (Scelza 2013). Similar rates were reported by the Himba society wives, a seminomadic pastoral population living in the northwest corner of Namibia (Scelza 2013). Female EPCs are also common in several societies from Northwestern Europe and North America, with approximately 20 % of women reporting cases of sexual infidelities, slightly lower than those reported by men (Simmons et al. 2004). Nevertheless, in the "iKung," a hunter-gatherer society from the Kalahari Desert of Namibia (Anderson 2006), EPC rates of 2–5 % have been reported, whereas in Hong Kong EPC rates of 8 % in men and 1 % in women have been reported (Anderson 2006).

Multiple Mating in Humans: Selective Pressures and Adaptations

The close relationship between mating and reproductive success, as well as the convergence and divergence of male and female reproductive interests, frequently will result in intense selection on male *and* female traits related to mating frequency and, thus, in adaptations to achieve optimal mating rates (Fig. 1). The RS of men is constrained by the number of fertile women they can mate with, but other factors, such as high paternity uncertainty, also could favor multiple mating in males as a way of increasing the probability of fathering at least some children. On the other hand, selection also can favor male traits that reduce multiple mating in females, such as sexual jealousy, mate guarding, and frequent in-pair copulation (Shackelford et al. 2006).

In the case of women, multiple mating could be selected to obtain genetic benefits for their offspring (Scelza 2013). For instance, female EPCs may allow producing offspring with higher heterozygosis, thus increasing their probability of survival, or offspring whose fathers are of better genetic quality. Moreover, female multiple mating could result in material benefits. For example, a woman could receive resources from each of their sexual partners, she could decrease the time to establish a new marriage when the previous relationship dissolves through death or divorce, or she can reduce infanticide risk (Scelza 2013). However, adaptive EPCs by women select for male adaptations against female EPCs, such as different forms of punishment, including reductions in paternal investment or even homicide.

An intriguing aspect of human sociosexual behavior is why socially imposed monogamy is exhibited by most contemporary societies, instead of another mating system, such as polygamy, which could allow males and females to gain some of the potential benefits previously mentioned. In fact, most current societies have norms that sanction multiple mating (i.e., EPC), even though such sanctions also affect those who impose the norms (Hauert et al. 2007); for instance, most contemporary societies who practice socially imposed monogamy have a patriarchal origin and punish men (although usually not as hard as women) that incur in EPCs (Hauert et al. 2007). However, besides reducing cuckolding risks, other selective pressures could favor the development of norms against multiple mating. Human pathogens are a fascinating possibility. Exposure to sexually transmitted bacterial infections (STIs), such as syphilis, gonorrhea, or Chlamydia, may impose an enormous selective cost (Bauch and McElreath 2016), particularly in populations living in conditions similar to those experienced by humans during most of their evolutionary time, such as limited access to health services and contraceptive methods (Apari et al. 2014). In these populations multiple mating increases morbidity of STIs and thus the costs for individuals and their offspring, such as high rates of infertility and of mortality (Apari et al. 2014). In addition, there is evidence that seminal fluid proteins have a negative effect on the immune status of females, probably resulting from changes in the expression of immune genes in the cells of the female cervix following reception of a male ejaculate (Sharkey et al. 2012). Furthermore, persistent postmating immune changes produced by multiple mating can result in environments favorable for the invasion of other sexually transmitted pathogens, such as herpes simplex, human papilloma, and hepatitis B or C viruses (STVs; Sharkey et al. 2012). It is interesting to note that the negative effect of seminal proteins on female immune status could be increased not only via multiple mating but also as a result of repeated in-pair mating, leading to the hypothesis that those seminal proteins evolved as a sexually antagonistic adaptation against female cuckoldry. Thus, norms against multiple mating could be favored not only as adaptations for increasing paternity certainty but also as mechanisms for decreasing the risk of STIs and STVs contagion (Bauch and McElreath 2016).

Conclusion

Multiple mating is observed in most human populations, although at different frequencies, and it seems likely that it has been present during most of the history of the species as attested by the psychological, physiological, and cultural adaptations observed in men and women. Sexual differences in benefits and costs derived from multiple mating indicate that sexual conflict over multiple mating is common. Although sexual selection probably had a strong effect in the evolution of multiple mating in humans, other selective pressures, such as those derived from diseases directly or indirectly related to sexual promiscuity, need to be carefully studied for a full understanding of the causes and consequences of multiple mating.

Cross-References

- ▶ Aggression
- ▶ Copulation
- ▶ Human Copulation
- ▶ Infidelity
- ▶ In-Pair Copulation
- ▶ Marriage
- ▶ Mate Retention
- ▶ Mating Strategies
- ▶ Mating Systems
- ▶ Sexual Behavior
- ▶ Sexual Conflict
- ▶ Sexual Conflict Theory
- ▶ Sexual Reproduction
- ▶ Sperm Competition Theory

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Multiple Memory Systems

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Synonyms

Declarative memory; Episodic memory; Implicit memory; Localization of memory substrates

Definition

"Multiple Memory Systems" is a theory that is based on scientific and evolutionary evidence, which lead us to conceptualize memory, not as unitary system, but multiple systems that work independently, in synchrony, and/or in competition with each other.

Introduction

There are several memory systems, which have been identified (Poldrack and Packard 2003). Depending on the memory function(s) that are needed for the organism to encode, learn, maintain, or recall information, these systems are thought to be able to work independently, in unison/cooperation, or even compete with each other (Sherry and Schacter 1987).

They work independently when, for example, you are driving with memorized directions, given to you for a place in the city, while at the same time you are not forgetting the specific time that you have to attend an important meeting and you are concurrently contemplating the details of your presentation at the meeting.

The cooperation of memory systems is apparent, for example, when positive emotional memories come to you when you walk into a bookstore and the smell of pencils and school supplies hit your nose, and remember suddenly small details from your school years start coming back to you.

Their competition is evident in our everyday life as well. For example, a memory system responsible for encoding factual information could be "interrupted" by a competing memory system responsible for emotional information. For example, the memorized information for a neuroanatomy test that you are taking in class right now is not very clear in your mind during the test, as you suddenly remember last day's upsetting phone call from a friend.

We theorized that our memory and its systems have been evolving along with their associated structures in our brain. Our memory systems, which are defined as "…an interaction among acquisition, retention, and retrieval mechanisms

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that is characterized by certain rules of operation..." (Sherry and Schacter 1987) are therefore evolved traits that are in place for improving our survival chances and reproductive success as a species. There are a few different approaches in categorizing and defining memory systems, but these approaches are not in contrast with each other, but coexist and represent different types of scientific approaches. Therefore, the neuroscientific approach will naturally approach the matter from a structural neuroanatomical worldview, while the evolutionary perspective will capitalize on the changes observed through the evolution of our species.

Neuroscientific and Cognitive Approach

Murray et al. (2017) postulated that neuroscience and its subspecialties, such as neuropsychology, neuroimaging (e.g., functional Magnetic Resonance Imaging-fMRI, spectroscopy, Computerized Axial Tomography-CAT, just to name a couple well-known neuroimaging methods), neurodevelopment, pushed towards learning massive information about our brains following centuries of knowing minimal information about our brain and its function. This vast amount of knowledge generated more research and thoughts about our memory and its functioning, and therefore, we started developing evidence supported hypotheses and theories about our brain's memory systems.

In addition, the associations between brain damage in certain locations and the followed behavioral disorders (as in the case of H.M., the famous patient with global amnesia) allowed making inferences about memory dysfunctions and brain areas associated with certain memory systems. The first and second world wars (followed my numerous other regional wars) produced a large number of brain damaged individuals, who needed to be cognitively assessed and, in more recent years, scanned for deciphering the presence and extent of brain damage. These individuals, via their acquired brain injuries, taught us a lot about memory, brain areas that control memories, and in return directed us to theories about memory systems. After all, memory is one of the first cognitive functions, which are affected by brain injuries (Rabinowitz and Levin 2014).

In neuroscience, it is widely accepted that the following memory systems exist (Murray et al. 2017):

1. A memory system that involves our personal learned experiences and habitual information is associated with our basal ganglia. For example, our memories about the name of a location that means a lot to us or a movie that we really liked.
2. Our explicit memories are mostly connected with functioning in medial structures of our temporal lobes (medial structures). For example, factual information about statistical formulas that you learned for a test.
3. Emotional information and emotional memories are mostly correlated with our amygdala. For example, negative emotions that are brought up after visiting a hospital that reminds you of health issues that you had in the past.
4. Memories about movement and motor processes are connected with the function of our cerebellums.
5. The priming of behavior is connected with our sensory cortices.

A cognitive science and cognitive psychology take on the memory systems follows a similar world view with the neuroscientific approach with mostly semantic differences between the two. Here three memory systems are recognized. Each memory system receives sensory and emotional information and in return, each memory system produces its output. Here are three memory systems:

1. The Procedural Memory System, which is highly associated with the cerebellum and caudate nucleus, processes action-related memories and memories related to motor processes, such as cycling (Hsiao and Chai 2003).
2. The Declarative/Explicit Memory System is mainly associated with the hippocampus and is responsible of mindful encoding of information and memories of facts and events. Our semantic (e.g., news pieces) and episodic

memories (private living experiences) are processed here.
3. Amygdala is associated with our Emotional Memory System, and therefore it is related to our emotional memories.

Evolutionary Science Approach

According to evolutionary cognitive scientists (Murray et al. 2017), there are seven memory systems that were developed through human evolution. These are:

1. Our Reinforcement Memory System, which describes memories that are made stronger or weaker via reinforcement schedules. For example, memories that are strengthened because they are associated with rewards (e.g., remembering your partner's birthday strengthens your relationship).
2. Our Navigation Memory, which we developed in order to store locations and directions to help us find our way in our environment. For example, remembering how to get around and short cuts in a large Business Building.
3. Our Biased Competition Memory, which compares new information we come across to older stored information in order to identify the need for new learning or expanding on older learned information. For example, in a child, learning that a dog is not a larger cat may create a new memory category for dogs.
4. Our Manual-Foraging Memory, which developed after we started walking on two legs and our arms and hands became free for action and collaborate better with our vision on visual-spatial integration tasks. This is the memory that allowed us to learn how to farm, hunt, and work manually in an environment that requires visual motor integration.
5. Our Feature Memory, which stores newly learned information according to its individual features, such as color, size, weight, surface. For example, learning that bears with black and white large spots are pandas.
6. Our Goal Memory, that helps us use stored and new information with our goal-directed behavior. For example, recalling what history taught us may lead to better politics, in order to avoid wars.
7. Our Social Subjective Memory that encodes information about how we view our selves and how we view others in order to maximize our self-awareness and social behavior. For example, our stored self-image and awareness of our social self is compared with other more social individuals, who seem better than us, and this leads us to improve our social behavior.

Conclusion

There are multiple memory systems and in theory, at least, organize our memories according to their nature, type, and brain location(s) that they are stored or processed. Our Memory systems could work in full cooperation with each other, individually, or compete with each other in our everyday lives. All memory systems exist in order to help us survive and succeed as human species. In order to fully comprehend and keep learning about our memory systems and their evolution and value, we need to work in a multidisciplinary environment that combines knowledge from numerous sciences such as (just to name a few) cognitive psychology, neuroscience, neuropsychology, evolutionary science, and neuroanatomy.

Different approaches in defining and identifying memory systems are not in competition with each other but coexist and are completing each other. For example, a neuroscientific approach to memory systems would certainly focus on brain loci, while an evolutionary approach to memory systems would concentrate on how our memory systems changed through our evolution to better serve our survival and reproduction needs.

Cross-References

▶ Evolution of Memory, The
▶ Different Functions of Memory
▶ Inceptive and Derived Memory
▶ Personal Memory

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Murder

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Synonyms

Homicide, to kill

Definition

Taking the life of another living being unlawfully.

Introduction

The act of murder has existed as long as humanity itself has existed. It is very unlikely, if not impossible, that murder will cease to be an issue during the course of humanity's future.

An Evolutionary Overview of Murder

Murder, or the act of killing another, has been around since the beginning of time and most likely well before recorded history. The act of killing another human carries the greatest cost to the victim. After all, the recently deceased is no longer able to provide for themselves or their surviving families. From an evolutionary standpoint, this would be the greatest cost to one's fitness; premature death without reproduction is a significant issue. It is important to note, however, that humans are not the only beings that commit murder. The act of murder has been observed within chimpanzees and other primates as well. Researchers Watts and Mitani found that "Observations support the arguments that infanticide has been an important selective force in chimpanzee social evolution and that females with dependent infants can be at great risk near range boundaries, but why male chimpanzees kill infants is still uncertain" (Watts and Mitani 2000).

Historically, there has always been some form of punishment for the act of killing. Some of the earliest recorded history showed records of punishment, including the famous Hammurabi's Code, which was recorded to approximately 1754 BCE. Hammurabi's Code covers one law pertaining to murder which states, "If the wife of one man on account of another man has their mates (her husband and the other man's wife) murdered, both of them shall be impaled (Claire n.d.)." This example demonstrates that ancient civilizations had laws in place to discourage and punish people from committing murder. In ancestral times, it is quite possible that the act of murder carried consequences as well. Ancestral groups are anticipated to have travelled within groups of approximately 150; therefore consequences to murder would have been severe. For example, the murderer could potentially face ostracization from the group and could have disastrous results for the survival of the murderer.

In modern society, there is no country where murder can legally take place. For example, the very definition of murder is the *unlawful* killing of another. The term "lawful murder" can be considered an oxymoron; the very meaning contradicts itself. Further, all modern judicial systems consider crimes that involve murder to be very serious, and these crimes often carry serious punishments. When it comes to murders within the family unit, studies conducted within the realm of evolutionary psychology seem to help offer an explanation. A study conducted by

Daly and Wilson reported that "Most 'family' homicides are spousal homicides, fueled by male sexual proprietariness. In the case of parent-offspring conflict, an evolutionary model predicts variations in the risk of violence as a function of the ages, sexes, and other characteristics of protagonists, and these predictions are upheld in tests with data on infanticides, parricides, and filicides" (Daly and Wilson 1988). This study helps shed light into some potentially relevant motives regarding murder in ancestral times.

Conclusion

In conclusion, the act of murder is as timeless as life itself. Humans are not the only species that is known to commit murder, as intentional killings have been documented within certain ape populations. Further research into the motivations of murder will hopefully shed additional light into this topic.

Cross-References

▶ Evidence of Intentional Killing
▶ Homicide Adaptation Theory
▶ Humans: Within-Group Conflicts
▶ Nonhuman Primates: Within-Group Conflicts
▶ Victims of Violence

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Murder by Design

▶ Homicide by Design

Muscle

▶ Hollywood Actors

Muscle Conduction Velocity

▶ Cortical Neurons and Conducting Velocity

Music

▶ Music and Hormones

Music and Hormones

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Synonyms

Ancient music; Art; Artistic expression; Contralto; Dopamine; Early castration; Emotions; Estradiol; Hormones; Mezzosoprano; Music; Nitric Oxide; Octave; Oxytocine; Prepuberal gonadectomy; Soprano; Tenor; Testosterone

Definition

Music is an art form and cultural activity whose language is organized in time with logic and sensitivity the sounds and silences. Originally, the word derives from Greek *mousiké [téchnē]* that means the "technique of Muses." It combines harmony, melody, rhythm, pitch, meter, timbre, etc. to create psychoanimic states with hormones, steroids, and other metabolic endogen compounds contributing to those effects.

Introduction

Musical stimuli in humans have been proved to be related with emotional areas of the brain such as insula, cingulated cortex, hypothalamus, hippocampus, amygdala, and prefrontal cortex (Boso et al. 2006). Probably nobody would disagree in the asseveration music has an effect over our moods, reducing stress, increasing well-being, improving social boundaries, and even over our motivation. When people needs to have energy to work, to drive, to train, to have fun, etc. music is a recurrent and common resource for natural energizing. Music has an effect on our metabolism and bodies which might guide toward those desirable mental states; in a human study, it was found an increase in saliva of 26 healthy young men, for their secretions of oxytocin release (positive social ties hormone) and a decrease for their cortisol release (stress, anxiety, and depression hormone) after exposure to music (Further details in chapter ▶ "Anxiety" and ▶ "Depression" of this encyclopedia.). Moreover, those cortisol levels positively correlated with the arousal level after listening several piano music pieces composed by Chopin (Ooishi et al. 2017). Nitric oxide (NO), endorphins, endocannabinoids, and dopamine have been suggested to play a role being released to the bloodstream with the musical experience (Boso et al. 2006).

Music also informs about our personal stories or cultural backgrounds, since we use to like what we have most listened, whether chosen or not. Or because inside a certain cultural or historical context, a virtuous musicians pursuits to do better than the most of the other people inside a group (Sloboda 1994). For instance, the question "W*hat kind of music do you like?*" is a very typical question to know someone better, at the beginning of any friendship, next to the question "*Do you come here very often?*" Not only a bidirectional connection to our mood or mental states but also what sort of music we rather, might be giving also information about ourselves, which in turn, enlightens about our personalities, specific personal characteristics, or cognitive styles (Greenberg et al. 2015). Two different forms of well-defined personalities were given depending on our most pronounced musical preferences: type E (low rhythm, melodic songs, soft arousal) and type S (animated, intense, tension, or thrilling music). For this well-documented classification, they used NEO-PI-R tests results beside the participants physiological reactions to their preferential variety of musical stimuli (Greenberg et al. 2015).

Wherever they come from, most songs and music in general are written in a mathematical language, following cycles, loops, logic sequences (waves, mirror, dialogues) using several parameters well understood by anyone, rhythm, tonalities, accelerations, intensities, etc. The nature of learning to play music will depend on practicing, specifically how much and what is done during that practicing (Sloboda 1994), because playing music combines motor skills besides a high cognitive learning. And it magically happens, that even the simplest tune with the simplest notes in some order sequence might have a sense and good meaning for all of us in many different cultures, languages, latitudes (i.e., Fur Frère Jacques) (Sloboda 1994). Music gives comfort to our brains. However, beyond a doubt understanding music is not a matter of perceiving single stimuli into patters and relating those patterns one another, as Gestalt Psychologist has proved. The same musical stimuli may have many different meanings (Furtger detausk Meyer 1994) (Further details, see chapter ▶ "Evolutionary Musicology" in this encyclopedia.).

Although, on the other hand, there are some sort of music with dark legends or "diabolic attributes" associated. These types of music have been affiliated with delinquent manners, such as violence or aggressive actions, drugs abuse, or promiscuous sex. A recent study proved listening to this type of music might have an effect on risk-taking likelihood behavior, although those risks did not happen in areas of health or safety (Enström and Schmaltz 2017). It has been proved that heavy metal might cause different autonomic responses because it's loud and intensity heavy change heart rhythm rate, respiration rate, and finger temperature in a group of young women (Cheng and Tsai 2016). The soft and most relaxing musical passages induced stronger parasympathetic activity compared with the loud passages (Cheng and Tsai 2016). Specially for the high intensity or loudness, dancing heavy metal moving head with long hair might cause injuries to the neck or vertebrates. It has been calculated the injury is depending on the constant gravity, the acceleration of movement, and the velocity of the head movements relative to T1 vertebra and the center of balance of the head (Patton and McIntosh 2008). Heavy metal songs have an average tempo of 146 beats per minute and that tempo in head banging of a range of motion of the head over than 45 might predict to cause at least headaches and dizziness (Patton and McIntosh 2008). One clinical teenager case of vertebral artery aneurism after head banging by heavy metal young drummer was described (Egnor et al. 1991).

But not only heavy metal has been considered as diabolic, also pagan music such as folkloric songs or music full of intensity and energy, such as Rock and Roll or Jazz had negative moral values at some point or historical moment. In a study performed over professional women musicians (pianists), it was defined the positive concepts for music in terms of morality and logic are represented by regularity and prediction in musical structures, without much dissonances, roughness, and the narrower pitch range, while the opposite is true for the negativity for music (Mesz et al. 2015).

Castrati: male sopranos in opera

The ancient concept of castrato was given by Jacques Rousseau in 1778, as "*a musician (singer) who, in infancy, has been deprived of his genital organs in order to preserve his high voice*" (Sundberg et al. 2007). Castrati singers were able to produce a distinctive resonance as well as the high pitch, which cannot be matched even by great non mutilated male singers of today (King et al. 2001). They used to be taller than their noncastrated fellows because of the growth plates did not close and growth hormone continued its release. Their facial features used to be feminine (King et al. 2001), consequently they were playing and singing both roles male and female in opera for more than 200 years (1600–1850) (Melicow 1983; Zegers 2009), having their own specific compositions such as George Frederic Handel did (1685–1759) with opera *Rinaldo* fitting their particular voice range. Castrati had developed the chest of a man, but their vocal cords were like those of a woman having their voices an incredible power, resonance, and control of breathing (Melicow 1983). The ranges of their voices were sometimes of four octaves, and they were carefully educated in vocal art and music (Sundberg et al. 2007). Sometimes they presented obesity, gynecomastia, and steatopygia, but they preserved their heterosexual desires, despite of their sterility (Melicow 1983). The voice of the last castrato, Alessandro Moreschi, can be listened nowadays to gramophone recordings from the beginning of past century. Each castrati voice had their particular color and characteristics; however, they all used to be described as sweet, soft, and ingratiating, or hard, brilliant, and penetrating (Sundberg et al. 2007).

In a study performed over Farinelli's skeleton (Andria, 1705 – Bologna, 1782), who was an Italian singer castrato very famous, it was observed his bones had interesting characteristics probably related to the effects of castration before puberty. Farinelli's real name was Carlo Maria Michelangelo Nicola Broschi. Farinelli presented long limb-bones, periphyseal lines and osteoporosis, bone mineral density, together with skull frontal bone affected by severe hyperostosis frontalis interna (HFI) (Belcastro et al. 2011). It is known the production of interleukin-6 (IL-6) by osteoblast is inhibited by estrogen and androgen, being the bone loss increased by ten times after menopause in woman or following castration in men mainly because of overproduction of IL-6 (Zanatta et al. 2016). Actually, Farinelli's HFI was classified as the most severe grade (type D), because the bone thickness ranges was from 9 mm at the sagittal sulcus to 21 mm at the maximally thickened area. Farinelli was a very good companion for the Spanish King Philip V who suffered for melancholia. He used to listen to Farinelli and his voice helped the King to improve his mood (Melicow 1983).

In general population, a study on 1532 autopsies during 4 years searching for HFI found 13 cases with this disease, 12 women and 1 man (an incidence of 0.8%). It is characterized by thickening of the frontal skull bone associated with roughness of the endocranial surface in a graduation from type A (less severe) to type D (more severe) (Devriendt et al. 2005). Some behavioral disorders and hormonal diseases have been described with this pathology such as virilism, hirsutism, fatigue, somnolence, obesity, irritability, aggressive behavior, vertigo, polydipsia, polyuria, loss of sense of smell, Pick's disease, Alzheimer disease, cerebral atherosclerosis, dementia praecox, etc. (Devriendt et al. 2005). Many of them are related with the hypothalamus and its homeostatic metabolisms.

The preauricular sulcus was found to be different in castrati too. Farinelli had an absence of preauricular sulcus in his left hip bone, in the male common value range for this parameter (Belcastro et al. 2011). In nonmutilated people, 500 *ossa coxae* were analyzed and it was found a sex difference in this bone morphology, being in males more likely to display a short and narrow sulcus in bone, while in female, probably because of their different pelvic shapes and bells, their sulcus shape was wider and longer (Karsten 2018). Only 10.04% of females analyzed had nonpreauricular sulcus or very narrow.

Farinelli's voice could span over three octaves and his thoracic development allowed him to hold a note without interruptions for breathing during a whole minute. Unfortunately, for his honor and memory, his bones were not well preserved since they were located at the feet of his great-niece grave (Belcastro et al. 2011). He was described as having an inclination to melancholy that got worse with advancing age, perhaps because of T deficiency, having his androgen receptors functional. Perhaps some metabolic loop could have been modified and having some other indirect effect over corticosterone because of a different metabolic instructions for homeostasis balance. On the other hand, castrati were relieved from some testosterone effects over the metabolism, besides the lowering of voice, such as alopecia, thickening of the dermis, development of acne, and formation of male pelvis (Zegers 2009).

Another castrato singer gonadezcomized before 12 years old was Gaspare Pacchierotti (1740–1821) and his skeleton was also studied. He was posterior to Farineli and also presented in his skeleton some androgen deprivation effects. The posture for singing requires an elongated neck compared with the shoulders, to allow air and sounds passing, specially for acute sounds such as erosion of cervical vertebrae. In Pacchierotti skeleton was found an insertion of respiratory muscles and muscles of the arms, and osteoporosis (Zanatta et al. 2016). The cause of his death was dropsy, being related the liquid metabolism with vasopressin and other hormones in the body, including testosterone. His teeth were white and a dental health preserved. However, an advanced dental erosion, probably due to bruxism, was described that might have happened because of a persistent distress (Zanatta et al. 2016) or for the physical pain caused by muscular atrophies.

As mentioned previously referring to heavy metal, bodily musical performances might induce injuries to the body. Musical ecstasy itself might drive into impossible body postures for long time without perceiving pain, perhaps because of endorphins or other hormones released. Body movements of wrist, torso, and head made by musicians while playing use to be related to specific structural musical features (Sloboda 1994).

Another well-known castrato singer was Domenico Caffareli (1710–1783). He was a rival of Farinelli but with an unpleasant personality; he refused to sing for Royalty and he returned gifts from Royalty (Melicow 1983). Cafarelli's real name was Gaetano Majorano, but he adopted his artistic name the same of his "benefactor" who castrated him (Melicow 1983). For the surgical intervention of gonadectomy, narcotics such as opium were sometimes used for neutered kids, and the majority of victims died as result (Zegers 2009). A heavy psychological tax after this irrevocable intervention should still be considered for fully understanding of the betrayal happened to them who did not chose to lose their chances for procreation (Zegers 2009). However, a psychological defense mechanism to deal with this injustice might be the arrogance that is a common characteristic for some castrati singers; they used to believe their voices were more than divine, beyond god design. For instance, the castrati Giovanni Battista

Velluti (1781–1861) once quoted "*My throat is worth as much as a queen*!" (Melicow 1983).

However, on the other hand, testosterone might also have an effect over voice on adult singers. A case of a 22 years old man singer with hypogonadism (only one testicle and hided in the inguinal canal) who was treated with adult exogenous testosterone was described. While hormonal treatment was supplied, his voice control for singing was elusive, a decreased pitch and an alteration of quality for his voice control happened. After hormonal treatment was over, his final voice range changed to a different level from a tenor to a baritone (King et al. 2001).

Many studies have been done trying to understand how these steroids might affect art, quality of sounds, or even long lasting of singer expectancies. A correlational analysis were done just comparing ages of decease for singers of both genders. They compared mean ages of death for singer women and man in a particular population studied (from years 1900 to 1995, 292 female singers and 505 male singers) and they concluded that androgens might have a protective effect in woman lives (Nieschlag 2003).

Hormones in Musicians

A different gender expression of roles has been described for professional musicians with musical talents; male musicians show a higher rate of female-typical gender role behavior while the opposite (male-typical gender role behavior) is true for female musicians, compared to nonmusicians people (Hassler 1991). As a hypothesis to this fact is an excess of testosterone levels during development embrionary organizational period inside the womb, from weeks 20 of gestation on, has been discussed. Also a different brain hemispheres balance might happened for this particular early hormonal change which leads to a different distribution of tasks among brain hemispheres (Hassler 1991). Men, but not women musicians, have been described more often to be non-right-handed than their fellows nonmusicians (Hassler 1991).

Measuring the digit ratio and musical aptitude some interesting results were found, suggesting the testosterone levels are related to musical talent but only in female groups (Borniger et al. 2013). Spearman correlation coefficient was found in between those with greater musical ability and their lower digit ratio, supposedly related to higher prenatal androgen exposure. Their musical talented and professionalism in music were measured with the test Advanced Measures of Music Audition (AMMA). A lack of differences in AMMA total raw scores in relation to testosterone was found, but testosterone exposure might facilitate the acquisition of musical aptitude, in terms of competitive skills for music inside a such as an orchestra (Borniger et al. 2013) (Further details, see chapther ► "Resource Competition" and ► Status Competition in this encyclopedia). Concerning to this difference in testosterone levels, some autoimmune disorders, related to the immune system and the complex interaction of testosterone with other hormones and neurotransmitters, have been described more often in musicians than in nonmusicians (Hassler 1991).

It has been proved that in humans and rodents, steroid hormones confer some sort of neuronal plasticity but also they have effects over the immune system or mood states. Several empirical evidences would make plausible to assert that music confers neuronal plasticity by affecting steroid hormones and alleviating or relieving stress (Fukui and Toyoshima 2014). Moderately higher levels of testosterone in women (positive correlation) and lower levels of testosterone in men (negative correlation) compared to their basal nonmusicians groups was found, indicating an effect of T in better talents for music. Moreover, a tendency but not significant to a positive relationship between a polymorphism (repeat length) in the androgen receptor (AR) and T levels was described (Fukui and Toyoshima 2014).

Studies with Rodents

A study on rats proved that exposing animals to music during 21 consecutive days significantly enhanced neurotrophins brain-derived neurotropic factor (BDNF) and reducing nerve growth factor (NGF) in rats hypothalamus (Angelucci et al.

2007). Music presented were recordings with no voice added, such as a compilation of New Age songs (Enya).

In a different study, classical music (Mozart) was exposed to Wistar rats and after that, they found changes in dopamine levels (DA), prolactine (PRL), and corticosterone happened in different brain areas (striatal nucleus-SN, prefrontal cortex-PFC, and mesencephalon-MS) (Tasset et al. 2012). Haloperidol is known to block dopamine receptors in the adenohypophisis, having an effect over the PRL releasing inhibiting its secretion. Music exposure was able to prevent the drop in DA prompted by haloperidol administration, specifically marked at the PFC and MS, having a lesser effect over the SN (Tasset et al. 2012).

Conclusion

Music is an ephemeral creation of art that exists in all the cultures in the world. They do happen in our brain while playing and just some time after played, producing emotions and metabolism reactions (For further details see chapter ▶ "Emotions" of this encyclopedia). Hormones have very relevant aspects for music production such as voice qualities and range of singers. Past century, castrati were prepuberal gonadectomized and their entire metabolism suffered from activational and organizational changes for a forced to a rebalanced not programmed. Even the skeleton, bones, and skulls might have morphological differences after early castration. On the other hand, testosterone has also an effect over good musicians and excellent musical performances. Another molecules related to emotions are involved in musical art, such as NO, endorphins, endocannabinoids, and dopamine; they are released to our bloodstream while listening music as the artistic creation which remains in memory.

Cross-References

- ▶ Anxiety
- ▶ Depression (Paul Gilbert)
- ▶ Emotions
- ▶ Evolutionary Musicology
- ▶ Resource Competition
- ▶ Status Competition

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Music as Auditory Cheesecake

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Synonyms

Music as by-product; Non-adaptationist view of music

Definition

As per Pinker's hypothesis, music is not evolutionary adaptive, but rather is a by-product of other evolutionarily evolved traits, and one that serves no biological function. Pinker suggests that music is an "auditory cheesecake." Cheesecake fulfills our evolved desire for fat and sugar, but the desire for cheesecake did not evolve and is merely a by-product, pleasurable but biologically useless. Similarly, Pinker regards music as a highly pleasurable by-product of the evolution of other mental faculties, primarily language.

Introduction

In *How the Mind Works* (Pinker, 1997), Steven Pinker offers an account of how the major faculties of mind were designed by natural selection. However, not all mental faculties have evolved through natural selection. Music, for Pinker, represents a clear example of a nonadaptive "pure pleasure technology, a cocktail of recreational drugs that we ingest through the ear to stimulate a mass of pleasure circuits at once" (p. 528), an auditory cheesecake. Although it may contribute to the well-being of the individual, music is biologically useless: "It shows no signs of design for attaining a goal such as long life, grandchildren, or accurate perception and prediction of the world. Compared with language, vision, social reasoning, and physical know-how, music could vanish from our species and the rest of our lifestyle would be virtually unchanged" (p. 528).

Instead of being an adaptation that was the direct target of natural selection, Pinker sees music as capitalizing on preexisting adaptive mental faculties: (1) language, in particular prosody, which involves the analysis of intonation contours, metrical structure, and hierarchical grouping of phrases; (2) auditory scene analysis, which draws on attention to harmonic relations for the process of segregating components from different sources and integrating components originating from the same source; (3) emotional calls that use acoustic signals associated with specific emotions and thus serve a communicative function; (4) habitat selection which involves attention to evocative environmental sounds; (5) motor control, involved in the rhythmic movement to music. All of these are fitness enhancing and are rewarded by a sensation of pleasure emitted by the pleasure centers of the brain. Music, as a pleasure

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technology, pushes all the right buttons, without enhancing fitness itself.

The Pleasure of Music

Before turning to the issue of the biological function of music and its evolutionary roots, it is important not to overlook the emphasis, embedded in Pinker's cheesecake hypothesis, on music's unique ability to evoke pleasure. Whether or not the pleasure derived from music is an upshot of music's ability to stimulate the mental faculties suggested by pinker as providing the mental machinery for music, research provided support for the notion that music is experienced as a highly potent emotion-evoking pleasure-inducing stimulus. Furthermore, the rewarding effect of music appears to be mediated by the same reward brain circuits as the effect of biologically relevant rewarding stimuli; for example, research has implicated the mesolimbic dopaminergic reward system as involved in musical pleasure (see Chanda and Levitin 2013).

Music: Adaptation or Cultural Creation?

Pinker's position lies at one extreme of the debate regarding music's adaptive function. For Pinker, music is not an adaptation, produced through the process of natural selection for its current function, but rather a spandrel, a by-product of other adaptive mental functions. But moreover, even music's current use is biologically functionless (as suggested by the cheesecake analogy). A less extreme position is Patel's theory that sees music as a human invention, albeit a biologically powerful one (Patel 2010). Like Pinker, Patel argues that music is rooted in non-musical functions, and thus a parsimonious explanation would not assume it has been shaped by natural selection. However, Patel sees music as a "transformative technology of the mind" in the sense that musical engagement and training can have beneficial enduring effects on other brain functions, such as language, attention, and executive function, thereby transforming the biological functions underlying it. These effects span individual timescales, as music needs to be learned anew by each generation and does not involve any further modification of the human brain specifically for musicality.

At the other extreme of the debate lie positions that deem music, or musicality, an adaptation. Two kinds of arguments have been advanced in support of music being an adaptation. The first kind of arguments concerns its adaptive function; we will review these arguments in the next section. The second kind focuses on characteristics of music that make it a likely candidate for an evolutionary explanation. For example, music needs to have been around long enough to be shaped by natural selection. Although it is not known at what point in history music came into existence, archeological evidence points to the existence of flutes approximately 40,000 years ago (Conard et al. 2009), suggesting an even older musical tradition. Music is universal and observed across cultures. Furthermore, music needs to be innately constrained to be the target of natural selection. Evidence suggests that at least some aspects of music are indeed innately constrained; for example, even young infants show sensitivity to musical structure, such as melodic pitch contours (see McDermott and Hauser 2005 for a review). However, it is not clear whether music satisfies the criterion of domain-specificity. As suggested by Pinker, sensitivity to pitch can result from adaptations for language and auditory scene analysis (see McDermott and Hauser 2005). The lack of domain-specificity suggests that music may be an exaptation (Gould 1991), not produced by natural selection for its current function but rather coopting traits evolved for other functions. Unlike Pinker's cheesecake idea, the idea of music as an exaptation leaves open the possibility that it does serve a biological function, albeit a novel one that did not result through natural selection.

The Adaptive Value of Music

Different theories have suggested several ways in which music can be fitness-enhancing. One possibility, suggested already by Darwin in *The Descent of Man*, is that music evolved through sexual selection and functions to attract sexual mates. Music, and the dance that accompanies it, has features that may act as indicators of fitness, coordination, strength, and health (Miller 2000). On the other hand, in contrast to other species in which song is performed by males, mainly during mating season, music is performed by both males and females in a wide range of contexts. Currently, empirical evidence for the function of music in sexual selection is lacking.

Another possibility is that the function of music lies in its role in childcare. Cross-culturally, caregivers use lullabies and play songs to soothe infants and regulate their arousal; such expressive vocal communication also strengthens mother–infant affiliative bonding. Thus, songs and rhythmic vocalizations may have adaptive value for the infant. Infant Directed Speech ("motherese"), with its exaggerated intonation contours and stress patterns, may fill a similar function (Trehub, 2003).

Another potential function of music is in increasing group cohesion. Dunbar (2004) proposed that music functioned as vocal grooming, replacing physical grooming when group size increased too much to allow for physical grooming. According to Dunbar, vocal grooming mimics the endorphin-releasing effects of social grooming, thereby creating feelings of warmth and trust that contribute to social bonding. Music may also promote rhythmic synchronization and coordination at the group level, thereby contributing to group identity and cooperation (Brown 2000).

Conclusion

The idea of music as an auditory cheesecake was suggested by Pinker as a counterexample to his adaptationist account of the major faculties of mind. Pinker suggests that features we see as beneficial or adaptive, in the colloquial sense of the word of contributing to better adjustment, may not be adaptive in the biological sense, and were not the target of natural selection. Music is a by-product of the evolution of other metal functions, pleasure-inducing, but nonetheless biologically functionless.

Although Pinker's point regarding the pleasure-inducing effects of music is uncontested, his view regarding the evolutionary roots of music and its function has been disputed by theories that suggested various potential adaptive functions of music. In effect, Pinker's controversial suggestion has sparked a renewed interest in the evolution of music. However, much of the debate remains speculative and is in need of clear empirically testable hypotheses and empirical evidence.

Cross-References

▶ Adaptation
▶ Evolutionary By-Products
▶ Evolutionary Musicology
▶ Music
▶ Steven Pinker

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Music as By-Product

▶ Music as Auditory Cheesecake

Music Evolution

▶ Evolutionary Musicology

Musical Protolanguage

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Synonyms

Evolutionary precursor to language and music; Prosodic protolanguage

Definition

Musical protolanguage refers to a hypothetical intermediary stage in the evolution of language, intervening between a non-linguistic stage and the possession of true language. Musical protolanguage at that stage consisted of non-referential song and served as the common ancestor of both language and music.

Introduction

Theories of language evolution often invoke a hypothetical protolanguage stage, a stage that serves as a precursor and as a scaffold for the evolution of modern language. The idea of such an intervening stage, and the notion that this protolanguage was akin to song and is the common ancestor of both music and language, can be traced back to Darwin. In *The Descent of Man and Selection in Relation to Sex* (Darwin, 1875), Darwin considers the evolution of language and claims that "[. . .] primeval man, or rather some early progenitor of man, probably first used his voice in producing true musical cadences, that is in singing, as do some of the gibbon-apes at the present day; and we may conclude from a widely-spread analogy, that this power would have been especially exerted during the courtship of the sexes,—would have expressed various emotions, such as love, jealousy, triumph,—and would have served as a challenge to rivals" (p. 87). Drawing on an analogy to birdsong, which like language needs to be learned and is practiced by the young members of the species, Darwin argues that human vocal imitation was driven by sexual selection. He then traces the emergence of meaningful words to the vocal imitation of natural sounds, other animal vocalizations, or man's own instinctive vocalizations: "It is, therefore, probable that the imitation of musical cries by articulate sounds may have given rise to words expressive of various complex emotions" (ibid.).

Fitch (2010), in a modern framing of Darwin's theory, says "prosodic protolanguage was a system with phonological generativity, using a small set of elements to build hierarchical structures. These structures were vocally generated, voluntarily controlled, and learned, sharing core aspects of speech and song" (p. 476). Fitch argues that the musical protolanguage model can explain the design features shared by language and music, such as the use of a vocal/auditory channel, discreteness, productivity, and cultural transmission (for a complete list, see Fitch 2010, p. 469). He further argues that comparative data documenting the evolution of vocal learning in several non-bird species, as well as neuroscientific data showing an overlap in the mechanisms underlying the processing of music and language, provide support for Darwin's musical protolanguage hypothesis. For example, people with congenital amusia, who are impaired in their ability to process music, also exhibit reduced ability to process emotional speech prosody, suggesting a common acoustic code (Thompson et al. 2012). It should be noted though that overlap in processing mechanisms, while consistent with Darwin's hypothesis, can be explained by seeing music as dependent on processes that evolved for language or vice versa, and does not necessarily imply a common evolutionary ancestor.

In Fitch's terms, musical protolanguage can be described as bare phonology, that is a system with generative phonology such that units can be hierarchically combined but devoid of semantics, with some elements of syntax (units that can be combined and hierarchically arranged in phrases). The main problem for the musical protolanguage model is explaining how such "bare phonology" evolved into a meaningful system that combines phonology and semantics.

Modern Versions of the Musical Protolanguage Hypothesis

The idea of a musical protolanguage, a common precursor of language and music, has been revived by modern scholars. Brown's (2000) *musilanguage* model provides one such example. Brown traces the emergence of musilanguage to primate emotive referential vocalizations – emotive calls elicited in response to a specific class of objects, for example, the alarm calls of vervet monkeys. According to Brown (2000), in the first protolanguage stage, musilanguage involved discrete-level pitches and the use of pitch for conveying semantic meaning, similar to the way lexical tones can differentiate meanings in tonal languages. The second stage involved combinatorial formation of phrases and expressive phrasing. However, the idea of discrete-level pitches, found in music but not in speech, was later (Brown 2017) replaced with the idea of an imprecise relative pitch system (high vs. low). Another way tones can potentially convey meaning is by capitalizing on cross-modal mappings between pitch and non-auditory properties such as size, brightness, and position (Brown 2017). Indeed, research has shown that such cross-modal mappings underlie some sound symbolic words and can also be exploited for conveying meaning through prosody (see Perniss et al. 2010).

Another model is Mythen's *Hmmmm* model (Mithen, 2005), termed so because it sees protolanguage as **h**olistic, **m**anipulative (used for manipulation of emotion and behavior rather than for reference), **m**ulti-modal (using movement, such as gesture and dance, as well as sound), **m**usical, and **m**imetic. Unlike the musilanguage in Brown's model that consists of discrete-level tones, in Mithen's Hmmmm model, utterances are holistic, rather than combined out of elements that can be recombined. The evolution of modern language then involved the segmentation of Hmmmm.

Non-music Models of Protolanguage

Other theories accepted the idea of a protolanguage stage but rejected the notion that this protolanguage was musical in nature (for an extensive review of protolanguage models, see Fitch 2010). One such model is Bickerton's (1990) lexical protolanguage model, according to which protolanguage is a propositionally meaningful system, consisting of a lexicon of meaningful words, but no syntax. Bickerton sees language as primarily representational, rather than communicative. He thus rejects the continuity between language and animal communication systems, focusing instead on the conceptual system as the precursor underlying protolanguage.

Another model is the gestural protolanguage model (see Arbib 2005; Corballis 2002). Although speech can iconically represent meaning through speech sounds or prosody (Perniss et al. 2010), gesture allows far more possibilities for iconicity. Gestural protolanguage is propositionally meaningful, thereby avoiding the difficulty that musical protolanguage faces in explaining the emergence of semantics. The modern discovery of the mirror neuron system suggests a mechanism that enables parity in gesture understanding, which may serve as a precursor to the complex imitation observed in humans and to protolanguage (Arbib 2005). The problem facing gestural theories is explaining the transition from gesture and the manual modality to speech.

Conclusion

The idea of a musical protolanguage, that is a common precursor to both music and language, was originally suggested by Darwin in *The descent of men* and has since been revived and reformulated by contemporary theorists. Although all models share this idea of a common ancestor, the presumed characteristics of the musical protolanguage differ between theories (e.g., synthetic vs. holistic). New evidence for shared brain mechanisms underlying music and language has been interpreted as supporting theories that posit a common evolutionary origin and led to renewed interest in Darwin's musical protolanguage model. The transition from musical protolanguage to meaningful speech remains a challenge for the musical protolanguage hypothesis.

Cross-References

- ► Darwin on the Origin of Language
- ► Evolutionary Musicology
- ► Gestural Theory
- ► Music
- ► Music as Auditory Cheesecake

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Mutation

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Synonyms

Genetic alterations

Definition

Mutation is defined as a permanent change occurring in the DNA, i.e., nucleotide sequence of a gene or chromosome.

Introduction

Gene is the functional unit of DNA. The characteristic of an individual is determined by its genetic pool. The genes are transferred to the organism from the parental genetic pool through complex process of reproduction. Mutation is a process of alteration of nucleotide sequence within a gene. Altered gene sequence in DNA ultimately leads to alteration in all copies of proteins encoded by that DNA. The alterations in proteins can lead to change in development and functioning of an organism (Lodish et al. 2000).

The genetic configurations keep on changing due to mutational process (Moorjani et al. 2016). The gene-environmental interaction is involved to a larger extent in this complex process. By this evolutionary process, the organism attempts to meet the needs for survival, i.e., mutation keeps an organism evolving. However, mutation also attributes to the extinction of species.

Mutation rate is defined as “the probability that a change in genetic information is passed to the next generation” (Sanjuán and Domingo-Calap 2016). The rate of mutation reflects the rate of evolution (Peck and Lauring 2018). The rate (speed) at which the mutation occurs represents the speed at which genetic diversity occurs in a given population. In humans, the most frequently mutated genes are those genes that are associated with various cancers (oncogenes) (Gorlov et al. 2018). Mutability of a gene (i.e., ability of the gene to mutate) depends on several characteristics of the gene. The genes that have higher level of expression, slowly replicating in nature and higher chromatin accessibility, are more associated with somatic mutations (Gorlov et al. 2018; Park et al. 2012). Information is carried by the genes. Genes determine the characteristics inherited from the parents. Germline mutation arrests the passage of information or transmits erroneous information (characteristics) to the next generation (Scally 2016).

Types of Mutation and Their Health Implications

Mutation of various types has been implicated in different health conditions. Mutation that involves germ cells (sperm, ovum) is known as germline mutation. Mutations involving non-germline cells are known as somatic mutations (Loewe 2008). Germline mutations are transmitted to progeny and lead to inherited disorders. Somatic mutations, on the other hand, are not transmitted to the progeny. Somatic mutations are mostly manifested as cancers (Kumar et al. 2017). Mutations in specific genes produce alteration in myeloproliferation in an unregulated manner resulting in hematological malignancies (Ortmann et al. 2015). Genetic mutation too attributes to the development of prion diseases like Creutzfeldt-Jakob disease (Gao et al. 2019).

Alleles are different forms of a gene (e.g., normal and mutant). In diploid organisms, genes occur in pairs. If mutant phenotype appears only when both alleles are mutated (i.e., individual is homozygous for the mutant allele), it is termed as recessive mutation. If mutant phenotype can appear even with one mutated allele (i.e., individual is heterozygous for the mutant allele), it is termed as dominant mutation (Lodish et al. 2000). Recessive mutations mostly lead to a loss of function, whereas dominant mutations lead to gain of function. Genetic disorders thus can be dominant and recessive depending on their pattern of inheritance. Autosomal disorders (due to mutation in autosomes) can be dominant (e.g., Huntington’s disease, Marfan syndrome, type 1 neurofibromatosis) and recessive (e.g., sickle cell anemia, thalassemia, cystic fibrosis, phenylketonuria, glycogen storage diseases). X-linked disorders account for almost all sex-linked disorders (mutations occurring in X and Y chromosomes) and are mostly recessive (e.g., hemophilia, Duchenne muscular dystrophy).

Alteration in a single base pair is called a point mutation (Lodish et al. 2000; Loewe 2008). Such mutations lead to either the production of a different protein (missense mutation) or the premature termination of a protein (nonsense mutation). Sickle cell anemia is autosomal recessive disorder due to missense point mutation that causes GUG codon (coding for valine) to replace normal GAG codon (coding for glutamic acid). Alteration in β chain of hemoglobin (Hb) caused due to this mutation causes decrease in oxygen-carrying capacity of Hb and sickling. Nonsense mutations lead to degradation of RNA. Frameshift mutation is a type of mutation that occurs when there is insertion or deletion of one or two base pairs, leading to altered reading frame of the DNA strand. Trinucleotide repeat sequences in an unregulated fashion occur due to certain dynamic mutation processes (Pearson et al. 2005). Fragile X syndrome characterized by mental retardation is an example of trinucleotide repeat where there are tandem repeats of CGG codon in *FMR1* gene. Mutations that result in expanded repeat sequences of trinucleotides result in development of several neurological as well as neuropsychiatric disorders (Pearson et al. 2005).

M

Mutation in Nonhumans

The process of mutation occurs in all living organisms. Mutation in microorganisms results in their evolution (Peck and Lauring 2018). Mutations can be spontaneous or can be induced by mutagens like UV radiations and alkylating agents (Watford and Warrington 2019). Mutation can alter the pathogenic characteristics of a microbe (Hershberg 2015; Sanjuán and Domingo-Calap 2016). Mutation may accentuate or attenuate the virulence of pathogenic microorganisms. Oral polio (Sabin) vaccine, for example, is produced by attenuating polio virus. These vaccine strains of poliovirus were developed by serial culture in monkey kidney cells. Mutation and selection produced this strain of poliovirus which retains its immunogenicity but has lost its pathogenicity (Fleischmann 1996). Different microorganisms have different rates of mutation. The rate of mutation in RNA viruses is higher than that of DNA viruses. Human immunodeficiency virus (HIV), for example, is a highly mutable virus. Sequential isolates of HIV from the same person can show antigenic variation. Similarly, faster mutation is reported among viruses with single-stranded nucleic acid than viruses with double-stranded nucleic acid (Sanjuán and Domingo-Calap 2016). Viruses with segmented nucleic acid (e.g., influenza virus) also show higher rate of mutation than those with unsegmented nucleic acid. Mutations in HIV lead to development of drug resistance. Drug resistance in bacteria like *Mycobacterium tuberculosis* and *Helicobacter pylori* is mainly due to mutations. Mutation attributes to the continued evolution of acquired resistance genes too, like TEM family of beta-lactamases which are generally transmitted from one bacterium to another by horizontal gene transfer methods (e.g., conjugation) (Woodford and Ellington 2007).

Mutations: From the Perspective of Evolution

In the process of evolution, mutation is always a possibility as well as probability, as gene-environmental interaction is happening continuously in an intricate manner. The genetic variations seen among organisms during the developmental process are the result of mutation (Lynch et al. 2016). However, the probability of mutations varies according to their nature. Some forms of mutations are more likely to happen, whereas certain others are rare to happen (García and Traulsen 2012). Epigenetic mutations that result from gene-environmental interactions have beneficial effects. They help the organism in adapting with the changing environment. Successful adaptation is highly essential for survival. However, certain epigenetic mutations slow the adaptation process and may stand as a hindrance too (Kronholm and Collins 2016). In the process of evolution, nature selects organisms that are adaptable and fit by shaping their phenotype. The role of mutations in this process of natural selection is still debatable (Svensson and Berger 2019; Uller et al. 2018).

Mutation rate is important from the evolutionary point of view. Mutation rates vary among different species of organisms as well as with same species of organisms. It has been seen that the mutation rates humans have are lower than gorillas and chimpanzee. Similarly, among human race, mutation rate is found to be higher within certain DNA regions of Europeans than Africans and East Asians, as found in some research (Harris and Pritchard 2017).

Mutated genes code for specific proteins that subserve a particular function. The proteins coded by the mutated genes ensemble various permutations and combinations to a final configuration, which is unpredictable. So, as per the existing knowledge, prediction of evolution remains uncertain (Sailer and Harms 2017).

Conclusion

The phenotypic variations among living organisms trace back to mutation. Many illnesses and their risk factors can be understood through the knowledge of mutation. Extensive research for in-depth understanding about mutation and its aftereffects may enable us in understanding the evolutionary process and even may guide us in predicting evolution.

Cross-References

▶ Genetic Predispositions

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M

Mutation Accumulation Theory

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Definition

Mutation accumulation theory describes two interconnected bodies of theory: (i) *Germ-line mutation accumulation theory* attempts to understand both the consequences of and the processes by which a population's *load* or *burden* of spontaneously occurring germ-line mutations increases in the absence of purifying selection, which results from the fitness diminishing effects of these mutations, and (ii) *Somatic mutation accumulation theory* encompasses the related phenomenon of cell lineages within multicellular organisms accumulating mutations as a result of failures of DNA repair in the course of mitotic cell division. These phenomena have important implications for understanding the origin of variance and covariance among fitness salient traits and the maintenance of population viability in the case of germ-line mutation accumulation and senescence and oncogenesis (the mechanisms by which cancer-causing neoplasms form) in the case of somatic mutation accumulation.

Introduction

This entry will start by introducing relevant background concepts, such as the nature of purifying selection, the relationship between fitness, pleiotropy and trait variance, and the sources and rates of new germ-line and somatic mutations. As the target audience for this entry is evolutionary psychologists, whose principal organismal focus is *Homo sapiens*, the entry will then summarize the human mutation load paradox, its resolution, and potential manifestations of ongoing mutation accumulation in human populations. Experimental evidence of the fitness reducing consequences of accumulating germ-line mutations in relation to work on nonhuman eukaryotic taxa will also be discussed. Finally, the possibility that there exists a group-selection level manifestation of purifying selection will be considered.

Review

Purifying Selection

Purifying selection describes the processes by which mutations are eliminated from a population as a consequence of their deleterious effects on fitness. This can occur via *natural selection* under conditions where the carriers of deleterious mutations exhibit suboptimal levels of traits critical for survival, such as in the case of defenses against predation and disease, or where mutations that interfere with critical biochemical processes lead to prereproductive organismal death at some point in the lifecycle or sterility. A parallel process of *somatic selection* operates at the cellular level, whereby newly acquired deleterious mutations trigger immunological policing responses, such as *apoptosis* or preprogrammed cell-death, which eliminate cells that are at risk of undergoing neoplastic transformations.

Among sexually reproducing organisms such as humans, deleterious germ-line mutations can influence fitness outcomes in ways related to their pleiotropic effects on traits, meaning that they can influence the development of multiple traits simultaneously. This has led to the prediction that there exists a latent *fitness factor* among traits believed to signal the sensitivity of development to perturbation stemming from the presence of deleterious mutations (Penke et al. 2007). Thus, in sexually reproducing taxa, there should exist genetic correlations among seemingly quite distinct traits, encompassing aspects of physical condition, behavior, health, and reproductive viability, high-levels of which are *unconditionally favored* by *sexual selection* across different environments (Penke et al. 2007).

Under conditions where purifying selection occurs in proportion to the fitness costs of spontaneously occurring germ-line mutations, individual differences in and genetic correlations among fitness salient traits will be maintained within a population, as spontaneously arising deleterious mutations conferring few fitness disadvantages to their carriers (by virtue of recessiveness) can persist for many generations, with the addition of each new mutation cumulatively reducing the fitness of the lineage relative to others that are burdened with fewer such mutations. This mechanism by which variance is maintained with respect to fitness salient traits in a population is known as *mutation-selection balance* (Fisher 1930).

Sources and Rates of De Novo Mutations in Humans

Germ-Line Mutations

Paternal age is a major source of common germ-line de novo base substitution mutations (i.e., point mutations involving the substitution of single nucleotides) in humans and other eukaryotic taxa also. These new mutations arise via mitosis in germ-line precursor cells (spermatocytes) as a consequence of age-related declines in the efficiency of DNA repair mechanisms and are transmitted via meiotic cell division to the haploid genomes of spermatozoa (Kong et al. 2012). The generational base substitution rate varies widely between mammalian species. In humans, each year of paternal age adds approximately two of these mutations to the genomes of offspring, with fathers aged 35 transmitting around 70 base substitution mutations to their offspring. Mice (which have considerably shorter generation lengths) transmit about half as many mutations per offspring. The chimpanzee generational transmission rate is about equal to the human rate (Lynch 2016). It is important to note that these rates are based on the assumption of linear increases in the numbers of de novo mutations transmitted to offspring. In practice, DNA repair mechanisms undergo accelerated deterioration as a function of advancing paternal age, leading to an accelerating rate of increase in the numbers of de novo mutations (Kong et al. 2012). Maternal age is furthermore a source of additional de novo point mutations, but to a lesser degree than paternal age (the rate is around 0.5 de novo mutations per haploid genome per year of maternal age; Wong et al. 2016).

While paternal age is the principal source of common and mildly deleterious base substitution mutations in human germ cells, other less common classes of germ-line mutation also occur spontaneously, such as insertion-deletion events (i.e., where a single nucleotide is either added or removed), which occur at around 8 % of the base substitution mutation rate. Even less common mutations involving the addition or removal of large numbers of nucleotides such as copy number variation and chromosomal abnormalities also occur spontaneously. It has been estimated that once all sources of spontaneous germ-line mutations are accounted for, the numbers of such mutations transmitted to offspring may be as high as 100 per generation (Lynch 2016).

Human studies examining the influence of paternal age as a strong proxy for the numbers of spontaneous germ-line mutations on offspring characteristics not only yield information about which trait variances may be maintained by mutation-selection balance but also yield evidence for the existence of the fitness factor, with paternal age effects having been found on diverse traits, such as the risk of offspring acquiring neurodevelopmental disorders such as autism and schizophrenia (Kong et al. 2012) and attention deficit/hyperactivity disorder (D'Onofrio et al. 2014), diminished physical attractiveness (Huber and Fieder 2014), and reproductive failure (i.e., childlessness) (Fieder and Huber 2015).

Finally, a critical factor that moderates a population's load of mutations is endogamy. More inbred (endogamous) populations being more homozygous for mutant alleles.

Somatic Mutations

Among somatic cell-lines, the rates of de novo mutation are much higher than the rate found in germ cells. Each mitotic event is associated with a de novo base substitution mutation rate that is on average 50 times that found in germ cells, with brain cells experiencing the lowest rate (eight times the germ-line rate) and skin cells experiencing the highest rate (112 times the germ-line rate) (Lynch 2016). Somatic mutation accumulation is a major cause of neoplastic transformation in defective cells that fail to be eliminated via somatic selection and may furthermore be a major source of contingency in terms of the underlying risks associated with developing certain neoplasms (Lynch 2016). Somatic mutations are also fundamental to *mutation accumulation theories of aging* (i.e., Medawar 1952), which treat the accumulation of mutations as a major source of the general and progressive cellular dysfunction characteristic of *senescence*.

Critically, somatic mutation accumulation is the principal mechanism through which selection occurs on mutation rate itself, as within an organism, both somatic and germ-line cells share common DNA repair systems, hence differential rates of somatic mutation can serve as a phenotypic target for selection on germ-line mutation rates (Lynch 2016).

The Mutation Load Paradox and Estimates of Human Fitness Loss Due to Germ-Line Mutation Accumulation

The Mutation Load Paradox

Although the human spontaneous germ-line mutation rate is relatively high when compared with that of other species, it is not especially high when considered in the context of human genome and population size and generation length (Lynch 2016). Despite this, it has been estimated that the human germ-line mutation rate, specifically under the assumption that approximately two out of the 100 or so de novo mutations will be harmful, would necessitate the reproductive failure of around 88 % of all humans born. In other words human populations should have undergone *mutational meltdown* whereby the fitness loss causes populations to shrink, increasing the rate at which these new mutations drift to fixation, leading to population collapse. In order to offset this each female should produce in excess of 16 offspring simply to maintain population viability. A *mutation load paradox* arises when this prediction is contrasted with the observation that reproductive failure and excess on this scale is not observed in human (specifically industrialized) populations (Kondrashov and Crow 1993).

One solution to the mutation load paradox is that relative fitness differentials between individuals of different genetic loads might constitute a source of cryptic or soft purifying selection in human populations, operating via factors such as sexual selection, and that large reproductive failure rates and excesses only emerge when mutation load-free genomes are used as the baseline for fitness comparisons against those with relatively high load (Lesecque et al. 2012).

Another interpretation of the paradox concerns the observation that high reproductive failure rates, especially among males, coupled with reproductive excess among females, were in fact a feature of historical, preindustrial populations. For example, among European populations in the year 1600 AD the average individual had around a 25–40 % chance of dying in infancy, a 50 % chance of dying during childhood (Volk and Atkinson 2008), and only around a 40 % chance of fully participating in reproduction (Rühli and Henneberg 2013). The average family size was close to five in 1600s England (Arkell & Whiteman, 1998) – given the high rates of preterm, infant, and child mortality, the *numbers ever conceived* would likely have been considerably higher. These historical Western infant and child mortality statistics are similar to those observed in contemporary hunter-gatherer populations (Volk and Atkinson 2008). Furthermore, the historical pattern of reproductive failure does not appear to have been random with respect to indicators of social and economic success, with far higher rates of reproductive failure being found among those with low relative to high social status, indicating that survivorship and reproductive success in this historical context might have been strongly coupled with elevated levels of certain competitiveness and fitness enhancing physical and psychological traits (Clark 2007).

It has been suggested therefore that historically observed levels of infant and child mortality, coupled with low reproductive participation and large compensatory reproduction among females, might account for a substantial portion of the variance in the estimates of both reproductive failure rates and magnitude of reproductive excess required by theory to purify de novo germ-line mutations in human populations (Woodley of Menie 2015). The paradox is therefore *only* apparent when the reproductive patterns of modern industrialized populations are considered, where perinatal mortality is extremely low (~0.5%; Woods 2008), infant and child mortalities are extremely low (~1 %; Volk and Atkinson 2008) and where the opportunity for full reproductive participation is extremely high (90 %; Rühli and Henneberg 2013).

The Consequences of Relaxed Purifying Selection in Industrialized Populations

The modern situation results from the extraordinary degree to which industrialized populations in particular have been shielded from potential sources of purifying selection stemming from reproductive impediments (such as stillbirth and infertility), disease, climate, famine, and violence.

Furthermore, conscious control over reproduction, facilitated by the wide-scale utilization of contraception, appears to have attenuated the correlation between potential indicators of low underlying genetic load, such as physical attractiveness and fecundity in some modern populations (Pflüger et al. 2012). Similarly, while paternal age predicts the risk of childlessness in contemporary samples of the US population, it does not appear to predict fecundity among the fraction of the offspring that go on to reproduce – a situation different from that of historical North American populations in which inverse associations between paternal age and offspring fitness have been observed (Fieder and Huber 2015). Lynch (2016) also notes the potential for cosmetic interventions to mask the effects of mutations on condition in modern populations. Findings such as these have been interpreted as indicating a potentially attenuated role for relative fitness differentials among individuals of differing genetic loads as a source of soft purifying selection in industrialized populations (Lynch, 2016; Woodley of Menie and Fernandes 2016).

These factors have therefore permitted population growth in industrialized nations to occur largely in the absence of major sources of purifying selection, which has facilitated an increase in the generational burden of germ-line mutations. Such mutations may have had significant fitness costs in the face of historical ecological stresses but no longer incur those same costs under modern milder conditions and are thus free to accumulate, perhaps exerting more subtle and direct effects on reproductive fitness. Consistent with this, Lynch (2016) has estimated that under the assumption of strongly relaxed purifying selection, it would be reasonable to expect fitness losses of approximately 1 % per generation among the populations of industrialized countries. Lynch also suggests that this generational fitness loss may even underestimate the true impact of mutation accumulation on fitness salient traits in these populations, as it does not take into account potential interactions between germ-line and somatic mutation rates, stemming from the damaging impact of the former on genes responsible for regulating the latter. Deregulation of the rate at which somatic mutations accumulate might therefore lead to behavioral deterioration in human populations at a rate greater than the predicted loss in reproductive fitness, owing to the large genomic target size of the brain – with as many as 84 % of all genes in the human genome being involved in some aspect of brain development and maintenance (Hawrylycz et al. 2012).

This should not be taken to imply that there is absolutely no purifying selection operating on modern industrialized populations. Rarer forms of spontaneous mutation involving copy number variation or chromosomal abnormalities or relatively more common mutations interacting via *synergistic epistasis* (Crow 1997) can potentially cause significant functional and reproductive impairment, especially when such mutations lead to loss of viability early in development (such as in the case of spontaneous abortion, which terminates around 10 % of all pregnancies in the West). It is important to note also that some tiny subset of these mutations may even be beneficial for the fitness of certain individuals under certain conditions. Thus, purifying selection continues to operate, even under modern conditions, albeit likely on a much narrower range of the mutational spectrum than was the case historically.

Experimental Evidence for the Effects of Germ-Line Mutation Accumulation on the Fitness of Nonhuman Eukaryotes

There exists an extensive literature on the results of germ-line mutation accumulation experiments conducted on a variety of eukaryotic taxa with short generation spans, which has been taken as support for the approximately 1 % loss in fitness per generation estimate in humans (Lynch 2016). These experiments typically involve the maintenance of a line of descent in which, via experimental intervention, random individuals are selected for mating, thus eliminating the opportunity for purifying selection to operate on these populations via the effects of sexual selection. The impact of accumulated germ-line mutations on fitness is quantified via the measurement of offspring yields, viability (i.e., the proportion surviving to adulthood), and also via direct measurement of levels of fitness salient traits.

Insects

Much of the eukaryotic germ-line mutation accumulation research has focused on insects, with a number of studies having examined *Drosophila melanogaster* (the fruit fly) in particular. The earliest research into mutation accumulation involved fruit flies and found fitness losses (measured in terms of egg-to-adult viability) amounting to 1–2 % per generation in the case of populations that were heterogeneous for a specific chromosome that had been experimentally sheltered (via transmission through the male line) from the effects of purifying selection. Homozygosity for the target chromosome was associated with a larger fitness loss per generation (5–10 %) (Mukai 1964). Subsequent studies involving fruit flies replicated the finding of fitness loss via germ-line mutation accumulation; however, these studies also typically found smaller fitness losses and smaller rates of mutation accumulation than that observed by Mukai (e.g., Fernandez and López-Fanjul 1997). Importantly, it has also been found that in fruit flies the degree to which these mutations impact fitness is sensitive to ecological context, with substantially greater fitness losses occurring under conditions of competition relative to conditions where competition was relaxed (2 % per generation vs. 0.2 %) (Shabalina et al. 1997). This is precisely analogous to the situation as it pertains to the populations of industrialized countries, where aspects of behavior and physiology which have a potentially high sensitivity to spontaneously occurring deleterious mutations may have been far more critical for survival and reproduction historically under conditions of ecological stress than in relatively much more mild contemporary environments.

Indications of reproductive fitness loss have been observed in other insect species bred under mutation accumulation conditions such as in the case of houseflies (Bryant and Reed 1999) and flour beetles, which have been found to exhibit diminished starvation resistance relative to controls (Lomnicki and Jasieński 2000).

Yeasts

Mutation accumulation experiments involving yeasts (a unicellular eukaryote) have also yielded indications of declining fitness (when offspring growth rates are compared with those of the parental generation), but with considerable line-to-line variability, indicating that some small minority of beneficial mutations may actually raise fitness when allowed to accumulate (Dickinson 2008).

Nematodes

Mutation accumulation experiments involving the roundworm *Caenorhabditis elegans*, which exhibits a lower spontaneous mutation rate than *D. melanogaster* (Lynch et al. 2008), find no indications of declining fitness when viability is used as an indicator (Keightley and Caballero 1997); however, behavioral deterioration in terms of diminished levels of behavioral traits believed to be highly fitness salient, such as those relating to chemosensing and locomotion, has been noted (Ajie et al. 2005).

Mammals

Recent research involving *mutator* strain mice (i.e., mice that are specifically bred for very high rates of germ-line mutation) have found evidence of rapid declines in reproductive fitness when bred under mutation accumulation conditions, relative to *wild-type* mice bred under the same conditions (Uchimura et al. 2015).

Potential Historical Micro-Evolutionary Manifestations of Germ-Line Mutation Accumulation in Humans

It is not possible to run mutation accumulation experiments on humans; however, historically, the nineteenth and twentieth centuries saw the biggest changes in evolutionarily significant aspects of the reproductive ecology of Western populations, such as dramatically falling infant and child mortality rates and rising opportunity for reproductive participation (Rühli and Henneberg 2013; Volk and Atkinson 2008), coupled with the wide-scale utilization of contraception. On this basis, it has been suggested that very recent microevolutionary indications of the effects of germ-line mutation accumulation on fitness salient traits might already be present in secular trend data on population levels of various physiological and psychological indicators of developmental stability (Rühli and Henneberg 2013; Woodley of Menie and Fernandes 2016).

Consistent with this expectation, secular trends towards reduced cranio-facial symmetry have been noted among both male and female European Americans, starting in the early nineteenth century. These have been interpreted as stemming in part from a microevolutionary trend towards declining developmental stability due to germ-line mutation accumulation in this population. Declining developmental stability has also been proposed as an explanation for parallel secular trends in the incidence of neurodevelopmental disorders, such as autism and ADHD, and also for historical increases in the incidence of sinistrality or left-handedness among Western populations (Woodley of Menie and Fernandes 2016).

Also consistent with a potential microevolutionary trend towards reduced developmental stability, Rühli and Henneberg (2013) presented several examples of *medical abnormalities* that have increased in frequency among the populations of industrialized countries over the twentieth century. These include a doubling in the prevalence of anomalous arteries, such as the median artery of the forearm and a decrease in the incidence frequency of individuals possessing the thyroidea ima branch of the aortic arch to zero. The opening of the sacral canal (spina bifida occulta) has also increased in prevalence among those born in the second-half of the twentieth century. Tarsal coalitions are also more prevalent in modern populations, as are skeletal pathologies, such as ossification of the posterior longitudinal ligament and of the spine and diffuse idiopathic hyperostosis.

A recent estimate of the impact of germ-line mutation accumulation on general intelligence (g) has been made based on the secular trend towards declining levels of cranio-facial symmetry – measures of *fluctuating asymmetry* and cognitive ability being mildly inversely correlated. The commensurate decline in g was estimated to be -0.16 IQ points per decade, with mutation accumulation possibly accounting for around 30 % of this loss or -0.05 points per decade. A small impact of mutation accumulation on g is consistent with the observation that g likely exhibits a narrow-band sensitivity to the mutational spectrum and appears to be well canalized against genetic and environmental perturbation in development (Woodley of Menie and Fernandes 2016).

In considering possible historical microevolutionary trends such as these, Lynch (2016) urges caution, arguing that in order to adequately determine the contribution stemming from mutation accumulation, they need to be evaluated utilizing measurement models that can compellingly differentiate between genetic and environmental causes.

Conclusion

Research into mutation accumulation has generated a respectable body of theoretical and empirical work examining the phenomenon from multiple aspects. While germ-line mutation accumulation theory as it pertains to humans remains somewhat controversial, recent (as of this writing) synthetic summaries of this work (Lynch 2016) indicate that the accumulation of both germ-line and somatic mutations is nonetheless likely capable of producing relatively rapid declines in fitness (on the order of 1 % per generation) with noticeable effects arising over relatively short spans of time (i.e., a century). Critical theoretical factors that have not previously been considered in discussions of this phenomenon, such as the potential role of germ-line mutations that deregulate and enhance the rate at which somatic mutations occur and their potential role in behavioral deterioration, could form the basis for fruitful future research involving nonhuman eukaryotes, where behavior is rarely considered to be a key component of fitness (i.e., Ajie et al. 2005). The collection of systematic and detailed data on secular trends in behavioral and physiological variables, especially as these pertain to clinically significant measures, coupled with the use of measurement models that can tease apart causation should allow for fitness deterioration via mutation accumulation to be tracked.

Another potentially critical factor concerns the possibility that mutations with small, deleterious effects on behavior at the individual level can have disruptive effects on the *extended phenotype* of social ecology in ways that feedback on and amplify the fitness costs to populations. For example, among prosocial species, mutations that lead to reduced sensitivity to social cues or degrade social learning ability could all have widespread effects on fitness, even among conspecifics who do not carry these mutations, but who are nonetheless dependent upon the maintenance of certain social ecological optima for their fitness (Woodley of Menie et al. 2017).

The idea that there may be a *group-selection* analogue of purifying selection may account for Calhoun's (1973) observation of the *behavioral sink* phenomenon in his *mouse utopia* experiments. Calhoun observed that when allowed unlimited resources and space, mouse populations inevitably reach a growth plateau, followed by a decline in fertility, terminating in colony collapse. The decline phase is characterized by the appearance of individual mice exhibiting apparently bizarre behaviors, such as obsessive grooming, a tendency towards either extreme solitariness or crowding, disinterest in communal reproductive activities, and asexuality.

Mice engage in a variety of reciprocal social interactions; they also practice communal nesting and nursing and territory marking behavior. The presence of an elaborated social ecology might be translating the deleterious effects of any mutations on behavior into much larger-scale fitness losses at the colony level in the case of Calhoun's experiments (Woodley of Menie et al. 2017). This mechanism may also account for why the mouse utopia populations did not simply stabilize around some equilibrium number of individuals but instead proceeded to collapse. Accumulating behavior-altering mutations may have been feeding back on the social ecology of the colony, dragging down the fitness of the colony and ensuring eventual collapse. This process of group-level purifying selection strongly resonates with Calhoun's (1973) own description of the terminal phase of mouse utopia:

> Autistic-like creatures, capable only of the most simple behaviors compatible with physiological survival, emerge out of this process. Their spirit has died ("the first death"). They are no longer capable of executing the more complex behaviors compatible with species survival. The species in such settings dies. (p.86)

Social ecological systems might actually be quite fragile, containing "tipping points" which trigger group-level purifying selection and collapse once a certain critical load of behavior-altering mutations is present in the population. This is different from the classical mutational meltdown scenario, as it does not necessitate an initially small population in which deleterious mutations are drifting to fixation. Large and potentially genetically viable populations can in this scenario collapse simply via the deleterious pleiotropic effects of relatively small numbers of behavior modifying mutations on fragile features of social ecology. Within the context of human populations, this might help to explain the rapid transition into subreplacement fertility in many developed countries characteristic of the demographic transition. Total fertility rates in the United States, for example, fell from 3.7 in the period 1955–1959 to around 1.8 in the period 1975–1979 (Wattenberg, 1985) – a decline in organismal fitness of 51 % in just two decades – far in excess of what would be predicted on the basis of germ-line mutation accumulation under conditions of relaxed purifying selection alone. These fitness losses are furthermore biggest among those with potentially the lowest mutation loads, i.e. those with high social status. The idea that these rapid declines in fitness may stem even in some small part from the individually subtle behavior-altering effects of accumulated mutations on the generation of a natality-inhibiting cultural feedback loop is indeed a sobering thought, presenting yet another twist on the mutation load paradox, as here is a pathway through which mutations might yet be significantly impacting fitness, albeit indirectly. This possibility serves to amplify Lynch's clarion call that human mutation accumulation is a potentially major existential risk, deserving of our attention (Woodley of Menie et al. 2017).

Cross-References

▶ Dysgenic Concerns
▶ Fertility
▶ Individual Differences
▶ Mate Value

▶ Mutation
▶ Observed Mating Behavior and Women's Long-Term Mating
▶ Reproduction
▶ Reproductive Value

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Mutual Grooming

▶ Cooperative Grooming

Mutual Parental Investment

▶ Biparental Care

Mutualism

▶ Cooperation Among Fishes
▶ Cooperation Among Nonchimpanzee, Nonhuman Primates
▶ Morality Is Cooperation
▶ Primate Cooperation

Mutualistic Mimicry

▶ Müllerian Mimicry

Myth

▶ Fiction